

The Answer Is Two

A Technical Synthesis of the Two-Law Framework

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Abstract

The theory states that every determinate, history-bearing world requires exactly two primitive explanatory roles. The Constitutive Law determines admissible coherent structure; the revised Selective Law governs selection, exact realization, and entry into shared public history. Its formal face is two-register acceptance: at guarded stages (nonempty windowed obligation profile), acceptance is total discharge of the live demands with typed receipts, and no value quantity legislates; on the debt-free open band, acceptance requires certified earned novelty, which a provenance ceiling ($\nu \leq 4\kappa$) forecloses. The Principle of Efficient Novelty survives as the law’s evidentiary standard — novelty counts only when anchored to paid clauses or discharged demands — and as the open band’s foreclosing wall, never as a guarded-stage selector; where genuinely distinct total payments coexist, the law’s silence is certified rather than broken, and history is lawfully a cone. Its realization face is free sealing: once an API is selected, the new layer is its least term-model completion, containing no unforced generators or equations and initial among exact realizations of that selected API. This relative initiality must not be confused with the false stronger claim that PEN always selects the initial completion of a weaker requirement fixed in advance. Its physical face is Internal Record Actualization plus Public Confluence: a formal seal becomes a public event only through a stable decodable internal record, and compatible public records must admit a joint nondemolition refinement; finite-capacity failure is “common seal” or “no common seal,” never contradictory sealed facts on one overlap. The Two Laws do not determine a universal public-event extractor $\mathcal{Q}_p$, tolerance, metric clock calibration, or resource entropies. They do, however, support a complete model-level bridge: in an explicit seven-cell thermal repetition observer, the dynamics uniquely forces Hamming weight as the minimal sufficient statistic for the retained event claim, the optimal no-abstention seal has error 521/15625, all public coarse-grainings glue by exact nondemolition refinement, and the measured second-Renyi resources are $n_p = 4.887399$ bits and $n_g = 3.894753$ bits, giving $K_{\text{cap}} = 3.307335$ bits. This closes the immediate finite-observer construction problem without converting an observer-specific dynamics into a universal law. The exact obligation-depth package gives $d_{\text{obl}} = 2$, $d_\mu \leq 2$, a two-layer chronological window, affine Fibonacci scaling (descriptive interface arithmetic, gating nothing), and the certified fifteen-step Genesis Sequence ending in a temporal-cohesive shell: a demand chain with exactly one live obligation per stage from 4 through 15, a machine-certified unique total discharger at every guarded stage except Step 4 — where four genuinely distinct payments survive the full quotient, making lawful history a four-branch cone whose futures heal (per-stage ledgers identical from Step 5, structures byte-identical from Step 8, only cumulative diagnostics permanently branch-indexed) — and a certified debt-free halt: the Step-16 obligation ledger is empty at full instance granularity on every branch, the demanded-but-underdetermined falsifier is executed and excluded on every branch, and the halt is therefore a theorem of the laws rather than of the enacted choice. Value bookkeeping is register-disciplined: the enacted engine’s structural totals are sealed testimony, and all law-level counts are act-local, proof-bearing semantic-family values. A reparameterization no-go shows that the structural ledger does not calibrate this runtime to seconds or redshift. A second underdetermination result is equally important: constructive admissibility, composition, Genesis availability, free

sealing, and Public Confluence do not uniquely select complex Born quantum theory. Classical simplices, real and quaternionic quantum fragments, and generalized probabilistic models survive appropriate subsets of these requirements. The desired endpoint is therefore a reconstruction theorem deriving, rather than postulating, spectral dagger/filter completeness, a primitive central monoidal phase, free environment completion, and public-interface descent; these would respectively target homogeneous self-dual/Jordan structure, complex phase-balanced composition, purification, and gravity-compatible tomography. One of the four is now partly settled, and the package is not four coordinate targets: the environment completion is initial among exact realizations of a discarding interface, so purification is constructional rather than postulated, but that interface’s ledger-invariance clause is independent of Internal Record Actualization, Public Confluence and free sealing — a nonnegative-matrix countermodel separates dagger-losslessness from ledger-losslessness — so the third target is conditionally derived behind one counted premise, and it presupposes the dagger of the first. The Harlow–Usatyuk–Zhao closed-universe observer construction supplies a conditional public-ledger capacity crossover, while current particle-physics audits give an exact $\mathbb{Z}_6$ point-field kernel and conditional $q_\nu = 0$, $g_{\text{tree}} = 2$, one-loop controls, and a minimal Higgs carrier; the complete global gauge form, one-Higgs multiplicity, Koide equipartition, and numerical constants retain explicit bridge premises. Two framings used above are suspended pending a provenance audit and must be read under the standing disclosure in the roadmap: the sealed fifteen-stage register is a hand-authored literal table whose recognizing search gates postdate it, so the uniqueness census and the halt are certified *of that authored library* and are not yet established as properties of the laws. The framework is consequently strongest as a meta-kinematic and architectural theory: modern physics can be represented naturally inside it, but necessity and sufficiency for the unique modern kinematic framework remain the central open theorem.

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1 Theorem Roadmap

The formal spine below consolidates the two-level framework [1], the coherence-depth theorem package [2, 4], the PEN library-growth formalization [5], the generative calculus [6], the synthetic framework abstraction theorem [7], and the runtime/cosmological completion [10]. The physics, mind, consciousness, artificial-system, and moral-status sections use the current book appendices as their normalization layer [13].

Remark 1.1 (Standing provenance disclosure — read before the spine). A July 2026 provenance finding constrains how several statements below may be read, and it is recorded here rather than at the point of each claim. What was established mechanically and by repository history: the fifteen sealed telescopes that are the certified Genesis register’s source are a *hand-authored literal table* committed in the first implementation step; the search-layer admissibility gates that recognize the sealed clauses postdate that table, and their position arms are syntactic singletons, so at audited positions exactly one expression is admissible and it is the sealed one; and Step 14 is pre-declared in the repository’s initial commit. Stated without softening: *no search in this repository has been shown to have found any sealed telescope clause*. The engine is established as a verifier and auditor of an authored sequence.

Four framings are therefore **suspended** — to be read as unproved, not as false — pending the PA-1 provenance audit: autonomous recovery; the uniqueness census *as a property of the laws* (“exactly one total discharger at every guarded stage,” including the Stage-4 “exactly four”); non-circularity of the recreation claim; and the halt *as a law-property*. The certified forms that stand are the corresponding statements about the sealed library: $O(16) = \emptyset$ at instance granularity says *this authored library is complete*, and the uniqueness census describes the sealed content rather than revealing a law. Replay validity of every sealed certificate stands as replay; the pure mathematics stands on its own evidence; XF-0 is untouched, having been blind by construction.

Wherever the spine below states one of the suspended framings, it is to be read under this remark. The full notice is `docs/provenance_disclosure_v1.md`; the campaign record carries it as a ledger event. Nothing in the physics sections that rests only on the Selective Law’s realization face — free sealing and its initiality arguments — is affected, since those do not consume the register.

The dependency chain of the theory is:

$$\begin{aligned} \text{two primitive roles} &\Rightarrow (\text{CL}, \text{SL}) \Rightarrow (\text{CCHM}, \text{PEN}) \Rightarrow d_{\text{obl}} = 2 \\ &\Rightarrow \text{Fibonacci scaling} \Rightarrow \text{Genesis} \Rightarrow \text{DCT} \Rightarrow \text{ordered internal runtime}, \\ \text{DCT} + \text{Closed}(U) &+ \text{IRA} + \text{PC} + \mathcal{Q}_p \Rightarrow \text{observer-indexed public physics}. \end{aligned}$$

1. The primitive explanatory roles are exhaustive: admissibility and actualization.
2. The Constitutive Law is realized by constructive, univalent, bounded-depth, canonical structure.
3. The Selective Law operates in two registers: obligation gates the guarded stages (total discharge with typed receipts; value quantities are diagnostics there), and certified earned novelty gates only the debt-free open band. PEN is the evidentiary standard for what counts as novelty, not a guarded-stage selector. Relative free sealing realizes the selected API without junk or extra equations. Physical public actualization additionally requires a stable internal record satisfying Public Confluence.
4. Exact depth two follows from a binary univalence obstruction and cubical horn contractibility above depth two.
5. Depth two induces a two-step recurrence, hence Fibonacci scaling.
6. The two laws applied to the empty library yield the fifteen-step Genesis Sequence relative to the fixed grammar, admissibility lane, and canonical quotient: three constitutive founding acts, then a certified demand chain with a unique total discharger at every guarded stage except Step 4, whose four-way certified tie makes lawful history a four-branch cone; universality beyond the disclosed lane remains open. The uniqueness census is here a description of the sealed content; its reading as a property of the laws is suspended under Remark 1.1.
7. The Step-15 shell is forced grammar growth: at H_{14} the resident grammar cannot express any discharger of the live temporal demand, and the guarded modal completion is the initial good extension, unique up to signature equivalence (Minimal-Extension Theorem, restated demand-side and carried by initiality rather than by a clause count; the engine-register stall comparison survives as corroborating diagnostic).
8. The terminal temporal-cohesive step internalizes dynamics and yields runtime ignition; its semantic completion is governed by the DCT consistency trichotomy (strict fusion refuted, fragments modeled, Löb repaired along the jet filtration).
9. The temporal shell derives an oriented internal clock class, not a unique metric clock: every monotone reparameterization preserves the structural trace, so any calibration to seconds, scale factor, or redshift is a declared bridge.
10. Continuation past Step 15 is foreclosed, and the closure is now certified rather than contested: directive debt is exhausted at Step 16 ($O(16) = \emptyset$ at full instance granularity, all 89 window instances decided; the demanded-but-underdetermined falsifier executed and excluded), internal

continuations score zero by univalent closure, and external earning is bounded by the act-local provenance ceiling $\nu \leq 4\kappa$, far below any standard a completed field sets. The former “contested” reading is resolved: the formula-maximizing survivors rode engine-register counting rules (d^2 , r^2 , telescopic inheritance) that the semantic register refuses as credit. The halt holds on all four branches of the Stage-4 cone. Its reading as a theorem of the laws rather than of the enacted choice is suspended under Remark 1.1; the certified form is that this authored library is complete. Runtime begins where debt ends.

11. The evaluator itself is derived: the per-step counting rules decompose into a six-principle Schema Calculus, with the derived evaluator reproducing the strict trace value for value.
12. The universe is adopted as spatially closed. Conditional on this axiom, the exterior fundamental Hilbert account is one-dimensional and real, while rich quantum descriptions are observer-indexed and entropy-limited.
13. Internal Record Actualization and Public Confluence are adopted as the physical sealing clauses. Neither the Two Laws nor the closed-universe observer construction chooses a universal record POVM or tolerance. For one explicit finite thermal-repetition observer, however, the observer dynamics derives the extractor, optimal tolerance, retained sufficient statistic, and exact confluence map rather than inserting them by hand.
14. The revised laws constrain the meta-kinematic envelope but do not yet classify the hidden operational theory: classical, real, complex, quaternionic, Jordan, and generalized-probabilistic countermodels prove that homogeneity/self-duality, purification, local tomography, and complex Born structure are not consequences of the present clauses alone.
15. The resulting leading public-ledger capacity is a smooth Page-like crossover at

$$K_{\text{cap}} = -\log_2(2^{-n_p} + 2^{-n_g}).$$

Equal pointer and gravity floors give the model-specific soft-min value $K_{\text{cap}} = n - 1$. In the explicit $(N, q) = (7, 1/5)$ observer, $(n_p, n_g, K_{\text{cap}}) = (4.887399, 3.894753, 3.307335)$ bits and the retained Hamming-weight statistic has $H_2(W) = 2.848709$ bits, below the crossover.

16. The particle-physics closure audit now separates exact representation facts from bridge choices: sequential sealing gives $q_\nu = 0$, the point-field action has exact kernel $\mathbb{Z}_6$, and its faithful image is $G_0/\mathbb{Z}_6$; the complete global gauge form and one-Higgs multiplicity remain conditional. Tree-level $g = 2$, the one-loop coefficients for declared matter, and $\rho_{\text{tree}} = 1$ are evaluator checks rather than new selections.
17. A verdict-blind Hopf shell census supplies a second, independent route to the same gauge data. Scoring the candidate family $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{C}P^n$ against a frozen obligation profile that mentions no measured constant returns $n_\star = 4$, structure group $U(5)$, and minimal alphabet-carrying subgroup $S(U(3) \times U(2))$, whose kernel over $SU(3) \times SU(2) \times U(1)$ is the *same set of triples* $\{(z^2 I_3, z^3 I_2, z) : z^6 = 1\}$ as the sealed point-field kernel, with the same generator. The $5/3$ and $3/8$ normalizations follow, but they consume obligation O4 ($q_\nu = 0$) to fix the hypercharge scale and borrow the ambient simple $\mathfrak{su}(5)$ trace form; both premises are declared. The census’s uniqueness is derived rather than declared: the local alphabet is selected with its division-algebra carriers, and the free seal of that carrier-bearing API is the coproduct of the carriers, $\mathbb{C}^5$, initial among exact realizations. The residue is one interpretive claim — that initiality is a universal property and not a value quantity — and, should that claim fail, the lawful outcome is a certified cone over the discharging set, never a minimizer.

18. The Koide audit leaves an exact geometric equivalence and an explicit underdetermination theorem. Permutation symmetry does not equalize the two sector weights. The value $Q = 2/3$ therefore requires a new rank-blind central-actualization bridge, phase alignment, a physical mass-seal extractor, and a hierarchy premise selecting one of two positive roots.
19. The same formal spine induces the theory's physics, mind, consciousness, artificial-system, and ethical structures.
20. The induced physics supplies a graded inventory of public-record constraints and conditional targets in the gauge, baryon, neutrino, charged-lepton-flavor, dark-sector, quantum-probability, gravitational-wave, and primordial-spectrum interfaces.

2 Formal Setting and Primitive Roles

Definition 2.1 (History and continuation). A realized history is a cumulative state H . A candidate continuation at H is an object x equipped with an intended update map

$$I(H, x) = H'.$$

The admissible cone $\mathcal{A}(H)$ is the class of continuations coherent with both the Constitutive Law and the already realized history.

Definition 2.2 (Admissible cone as a dependent fiber). Let $\mathbf{Hist}$ be the class of realized histories and let $\mathbf{Cand}(H)$ be the raw candidate family over H . The constitutive predicate is

$$\mathbf{CL}_H : \mathbf{Cand}(H) \rightarrow \mathbf{Prop},$$

and the admissible cone is the dependent subfamily

$$\mathcal{A}(H) := \sum_{x : \mathbf{Cand}(H)} \mathbf{CL}_H(x).$$

The integration map is partial on raw candidates and total on admissible selected candidates:

$$\mathcal{I}_H : \mathcal{A}(H) \rightarrow \mathbf{Hist}.$$

Definition 2.3 (Determinate world). A determinate world is a sequence

$$H_0 \xrightarrow{x_0} H_1 \xrightarrow{x_1} H_2 \rightarrow \cdots, \quad H_{n+1} = I(H_n, x_n),$$

where $x_n \in \mathcal{A}(H_n)$ is selected and integrated at each stage.

Principle 2.4 (Two primitive roles). The primitive architecture consists of:

CL : what coherently counts as admissible,
SL : what becomes actual and how it is integrated.

Theorem 2.5 (Two-role exhaustiveness). *Every primitive explanatory role in a determinate world is classified either under admissibility or under actualization and integration.*

Proof sketch. If only admissibility is given, there is a class of possible structures but no selected branch. If only actualization is given, there is no coherent candidate domain over which actualization can operate. A proposed third primitive role either constrains whether $x \in \mathcal{A}(H)$, hence belongs to CL, or constrains whether and how x is selected and integrated, hence belongs to SL. No further primitive explanatory type is generated. $\square$

Proposition 2.6 (Primitive law factorization). *Any primitive law L on determinate histories factors extensionally as*

$$L_H = (C_H, S_H),$$

where $C_H \subseteq \text{Cand}(H)$ is its admissibility component and

$$S_H : C_H \rightarrow \text{Hist}$$

is its actualization/integration component.

Proof sketch. The observable effect of L at H is exhausted by which raw candidates remain live and which live candidate is sealed into the next history. The first datum is the characteristic predicate of C_H . The second datum is the partial next-history map S_H . Any additional clause in L either changes C_H , changes S_H , or is inert with respect to determinate history. $\square$

3 Constitutive Law

Principle 3.1 (Constitutive Law). An admissible candidate is a coherently realizable, equivalence-invariant, locally satisfiable, canonically executable structure.

Definition 3.2 (Constitutive burdens). The four constitutive burdens are:

1. coherent realizability: existence is witness-bearing;
2. equivalence-invariance: equivalent presentations are identical in being;
3. local satisfiability: coherence obligations are discharged at bounded depth;
4. canonicity: closed constructions have determinate realization.

Definition 3.3 (Constitutive judgment). For history H and raw candidate $x : \text{Cand}(H)$, the judgment

$$H \vdash_{\text{CL}} x \text{ adm}$$

means that x carries:

- CL1 a witness-bearing realization type $\text{Real}_H(x)$,
- CL2 transport along all structural equivalences in its public interface,
- CL3 a finite local family of coherence obligations $O_H(x)$,
- CL4 canonical evaluation of closed generated terms.

Thus $\text{CL}_H(x)$ is the conjunction of CL1–CL4.

Proposition 3.4 (Formal shadow of CL). *The canonical formal shadow of the Constitutive Law is a CCHM-style cubical homotopy type theory with dependent types, univalence, Kan composition, and computational canonicity.*

Proof sketch. Witness-bearing existence is represented by judgments $\Gamma \vdash a : A$. Equivalence-invariance is represented by univalence,

$$(A \simeq B) \simeq (A =_{\mathcal{U}} B).$$

Bounded local satisfiability is represented by structural integration obligations over finite public interfaces. Canonicity with univalence requires computational path constructors and filler operations; cubical Kan operations supply the required execution rules. $\square$

Proposition 3.5 (Constructivity, univalence, and cubical canonicity). *CL1 forces constructive witnesses; CL2 forces identity to respect equivalence; CL4 forces the chosen univalent calculus to compute closed canonical forms.*

Proof sketch. A witness-free existence predicate would make $H \vdash_{\text{CL}} x \text{ adm}$ insensitive to realization and would not supply data for integration. A presentation-sensitive identity predicate would distinguish equivalent public structures, violating CL2. Book-style univalence without computational path and filler rules supplies equivalence-invariance but not the canonical execution demanded by CL4. CCHM-style cubical structure supplies paths, transport, composition, and normalization in one formal shadow. $\square$

4 Selective Law and PEN

Definition 4.1 (Selective burdens). The revised Selective Law carries ten burdens: localizability, commitment, admissibility, valuation, critical sufficiency, constructive primeness, exact realization, coherent integration, internal recordability, and public confluence.

Principle 4.2 (Selective Law). The Selective Law operates in two registers. At guarded stages — whenever the live window exports unmet obligations — the accepted continuation is the admissible total discharger of those obligations, with typed provenance; value quantities are audited but select nothing, and where genuinely distinct total dischargers coexist after the canonical quotient, the law’s silence is certified and history is lawfully a cone. On the debt-free open band, acceptance requires certified earned novelty, which the provenance ceiling forecloses. PEN is the law’s evidentiary standard — what novelty *is* and what earning would require — not its guarded-stage selector. The selected API is realized by its free sealing completion and folded into further history without unforced generators or equations.

Principle 4.3 (Strengthened physical Selective Law). The Selective Law has three interfaces that must not be conflated. Two-register acceptance fixes which admissible API is sealed (obligation at guarded stages, certified earning on the open band, with PEN as the audit standard). Free sealing realizes the chosen API. A formally integrated continuation becomes a *public physical event for an internal context p* only when it is actualized as a stable, decodable record inside that context. Compatible public records on an overlap must admit a joint nondemolition refinement; incompatible records yield no common seal, not two contradictory sealed facts. Selection without such a record is structural evolution, not yet a public event for p .

This strengthening does not add a third primitive role. Free realization, record formation, and public confluence constrain how a selected admissible continuation becomes determinate history and therefore belong to the actualization/integration side of SL. They do, however, expose structure that the value audit alone does not contain.

The PEN formalization is the library-growth model of [5]. Write $\mathcal{B}_H$ for the sealed public library contained in the realized history H .

Definition 4.4 (PEN state and candidate). A PEN state is a monotone context $\mathcal{B}_H$ closed under definitional derivability. A candidate over H is a triple

$$x = (x_{\text{core}}, \mathcal{S}_x, \mathcal{G}_x),$$

where x_{core} is the definitive type-theoretic data, $\mathcal{S}_x$ is a finite specification by atomic clauses, and $\mathcal{G}_x$ is the obligation graph of atomic coherence witnesses required to seal x against the active history.

Definition 4.5 (Definitional derivability and irreducible clause). Fix $\mathcal{B}_H$ with operational reduction relation $\rightsquigarrow_H$ generated by its eliminators, computation rules, and judgmental equalities. An operation or clause c is definitionally derivable over H when there is a closed term t_c in the signature of $\mathcal{B}_H$ such that

$$t_c \rightsquigarrow_H^* v_c$$

for a canonical value v_c , with all required computation behavior inherited from the existing eliminator fragment. A clause c is constructively irreducible over H when every attempted realization of c contains a new neutral head or stuck eliminator relative to $\rightsquigarrow_H$, equivalently when orienting the intended behavior requires adding a new operational, equation, or API clause.

Definition 4.6 (API dependency graph). Let $\mathcal{S}_x = \{c_1, \dots, c_m\}$. Its dependency graph has vertices c_i . An edge $c_i \rightarrow c_j$ records dependence on a symbol introduced by c_j . A structural bundle is an induced strongly connected subgraph closed under well-typedness over $\mathcal{B}_H$. A specification $\mathcal{S}_x$ is minimal complete when:

- Irr every clause is constructively irreducible over H ,
- Comp all irreducible operations required for the package are present,
- Unity the clauses form one structural bundle,
- Min no proper structural sub-bundle is complete and independently acceptable.

Definition 4.7 (Irreducible integration cost). The irreducible integration cost of x at H is

$$\kappa_H(x) := |\text{Irr}_H(\mathcal{S}_x)|,$$

where $\text{Irr}_H(\mathcal{S}_x)$ is the set of constructively irreducible atomic clauses in a minimal complete specification for x . Definitionally derivable clauses contribute 0. Nonzero admissible candidates satisfy $\kappa_H(x) > 0$.

The auxiliary MBTT audit length $K_H(x)$ is the prefix-free bit length of the same specification, including $O(\log i)$ pointer cost for references to existing library entries L_i . K_H is not the denominator of ρ_H ; it is a representation audit and deterministic tie-break layer.

Definition 4.8 (Sealing latency). The sealing latency of x at H is

$$\Delta_H(x) := |\mathcal{G}_x|.$$

It counts the normalized atomic coherence witnesses required to integrate x against the active historical interface. Thus $\kappa_H(x)$ charges irreducible specification clauses, while $\Delta_H(x)$ charges the interface debt incurred by sealing those clauses into public history.

Definition 4.9 (Generative gain). Let $\text{Schemas}(\mathcal{B}_H)$ be the set of introduction, elimination, projection, computation, Kan, and path-reduction schemas available in $\mathcal{B}_H$, quotiented by definitional equality and canonical renaming. The generative gain of x at H is the marginal expansion

$$\nu_H(x) := |\text{Schemas}(\mathcal{B}_H[x]) \setminus \langle \text{Schemas}(\mathcal{B}_H) \rangle_H|,$$

where $\mathcal{B}_H[x]$ is the library obtained by adding the specification of x , and $\langle \text{Schemas}(\mathcal{B}_H) \rangle_H$ is the closure of inherited schemas under definitional derivability in the old library.

Definition 4.10 (Components of generative gain). For admissible x ,

$$\nu_H(x) = \nu_{G,H}(x) + \nu_{C,H}(x) + \nu_{P,H}(x),$$

where $\nu_{G,H}$ counts new formation and introduction schemas, $\nu_{C,H}$ counts elimination, projection, destructor, and cross-interface schemas, and $\nu_{P,H}$ counts computation, Kan, path-constructor, and path-reduction schemas. Opaque constants without structured interaction have at most formation gain and negligible cross-interface gain.

Proposition 4.11 (Telescopic API gain). *Let $\mathcal{S}_x$ be a minimal complete API specification whose direct imported library interfaces are $\text{Refs}(\mathcal{S}_x) = \{L_1, \dots, L_m\}$. If these imports form a dependency telescope with unique maximal imported interface $L_{\max}$, then*

$$\nu_{C,H}(x) = \nu_H(L_{\max}) + \kappa_H(x) + (m - 1).$$

Proof sketch. The maximal import contributes the inherited eliminative surface, since the dependent telescope lifts its derivation schemas without summing duplicated ancestry. Each irreducible field of $\mathcal{S}_x$ contributes one internal projection, destructor, or computation-facing accessor. Fusing m imported interfaces into one structural bundle requires $m - 1$ bridge equations, exchange laws, or naturality squares. $\square$

Definition 4.12 (Efficiency and diagnostic ledger quantities). For $x \in \mathcal{A}(H)$,

$$\rho_H(x) = \frac{\nu_H(x)}{\kappa_H(x)}.$$

For a realized sequence $H_0 \rightarrow \dots \rightarrow H_n$, write

$$\Omega_{n-1} := \frac{\sum_{i=1}^{n-1} \nu_{H_{i-1}}(x_i)}{\sum_{i=1}^{n-1} \kappa_{H_{i-1}}(x_i)}, \quad \Phi_n := \frac{\Delta_n}{\Delta_{n-1}}, \quad \text{Bar}(H_n) = \Phi_n \Omega_{n-1}.$$

All three composite quantities are *diagnostics*: recorded at every stage, register-tagged, and legislatively inert. The enacted engine used Bar as an acceptance threshold; the completed audit of the certified trace established that it never performed selective work at any guarded stage (the sole discharger either cleared it incidentally or, at the semantic Stage 8, failed it and lawfully advanced), and the adopted law assigns value quantities no guarded-stage authority. Cumulative diagnostics are additionally branch-indexed permanently: they are the one place the Stage-4 choice leaves a trace.

Definition 4.13 (Obligation profile and guarded acceptance). The demand window at stage n is the pair (S_n, S_{n-1}) of most recently sealed interfaces; the live obligation profile is

$$\mathcal{O}(n+1) = C(S_n, S_{n-1}) \setminus D(\mathcal{B}_n),$$

the window-generated demands not derivable in the current library. A stage is *guarded* when $\mathcal{O} \neq \emptyset$ and *open* when $\mathcal{O} = \emptyset$. At a guarded stage, a candidate is *acceptable* iff it is admissible and totally discharges $\mathcal{O}$, each demand answered by a typed discharge witness. No value quantity, threshold, or presentation order may select among acceptable candidates: if exactly one exists it is selected (the certified trace realizes this at every guarded stage except Step 4); if several genuinely distinct acceptable candidates survive the full canonical quotient, the law's silence is itself certified, each founds a lawful branch, and the enacted branch is identified indexically. On the open band, acceptance requires certified earned novelty under the provenance discipline below; the accounting bound $\nu \leq 4\kappa$ forecloses it.

Definition 4.14 (Constructive primeness). An admissible $x \in \mathcal{A}(H)$ is constructively prime, written $P_H(x)$, when no proper terminal structural sub-bundle of $G_{\mathcal{S}_x}$ is itself well-typed, constructively irreducible, admissible over H , and independently acceptable. Equivalently, x is not presentation-equivalent over H to a gratuitous bundle of independently acceptable continuations. Primeness is an audit discipline (anti-bundling in the value accounting), not a selection filter: at guarded stages the acceptance question is payment, and a bundle either is or is not the total discharge the window demands.

Definition 4.15 (Local horizon). A local horizon is an increasing filtration of candidate cones

$$\mathcal{A}_0(H) \subseteq \mathcal{A}_1(H) \subseteq \dots \subseteq \mathcal{A}(H).$$

The active horizon is the least r for which $\mathcal{A}_r(H)$ contains a sufficient constructively prime candidate.

Axiom 4.16 (Cumulative growth). Selected integration is monotone:

$$H_{n+1} = \mathcal{I}_{H_n}(x_n), \quad \mathcal{B}_{H_n} \subseteq \mathcal{B}_{H_{n+1}},$$

and all definitional consequences of $\mathcal{B}_{H_n}$ remain available in $\mathcal{B}_{H_{n+1}}$.

Axiom 4.17 (Horizon policy). PEN evaluates candidates inside the active local horizon. After a realization the horizon resets to the least finite cone containing the new public interface; after an idle expansion it widens by one filtration step. Selection is performed in the least horizon containing a sufficient constructively prime candidate.

Axiom 4.18 (Admissibility gate). A candidate x belongs to $\mathcal{A}_r(H)$ only if $\mathcal{S}_x$ is well-typed over $\mathcal{B}_H$, nonredundant under $\sim_H$, minimal complete, constructively irreducible clausewise, compatible with the current coherence window, and inside the horizon bound r .

Axiom 4.19 (Two-register acceptance). At a guarded stage ($O(H) \neq \emptyset$), the accepted continuation is the total discharger of $O(H)$ with typed provenance, unique in the certified trace at every stage but one; where several genuinely distinct total dischargers survive the canonical quotient, no clause of the law selects among them — the silence is certified, and lawful history branches. No bar, efficiency comparison, parsimony order, or deterministic presentation key participates in guarded acceptance; all such quantities are recorded as diagnostics. On the open band ($O(H) = \emptyset$), acceptance requires a complete certified audit whose earned novelty justifies extension; by the provenance accounting bound this requires $\rho > 4$, which no certified candidate attains. The historical engine encoding (bar-clearing with minimal positive overshoot, cost and canonical-key tie-breaks) is retained in the record as enactment-conservative testimony: on the certified trace it selected the same winners, because every winner was the unique acceptable discharger; its tie-break key is inadmissible as law and is precisely where the enacted Stage-4 branch was fixed by artifact rather than by clause.

Axiom 4.20 (Coherent integration). When x^* is selected, its free sealing completion is sealed as a new public layer. Its resolved obligation graph $\mathcal{G}_{x^*}$ is exported as an opaque interface, the score tuple

$$(\nu_H(x^*), \kappa_H(x^*), \Delta_H(x^*), \rho_H(x^*))$$

is appended to history, and later admissibility may reference the new layer only through its sealed public interface.

Definition 4.21 (Relative free sealing). Let $\mathcal{S}_{x^*}$ be the normalized generators, recursors, computation rules, and coherence equations of the selected API. Define

$$F_{x^*}(\mathcal{B}_H) := \text{Term}(\mathcal{B}_H; \mathcal{S}_{x^*}) / \equiv_{x^*},$$

where $\equiv_{x^*}$ is the least substitution-, composition-, and public-interface-stable congruence generated by the old equations and the equations forced by $\mathcal{S}_{x^*}$. Integration is

$$\mathcal{B}_{H^+} = F_{x^*}(\mathcal{B}_H).$$

Theorem 4.22 (Relative free-sealing theorem). *Assume the generated reduction system is lawful. In the category of exact realizations of the already selected API $\mathcal{S}_{x^*}$ over $\mathcal{B}_H$, the completion $F_{x^*}(\mathcal{B}_H)$ is initial. Equivalently,*

$$\text{Map}_{\mathcal{B}_{H,x^*}}(F_{x^*}(\mathcal{B}_H), C) \simeq \mathbf{1}$$

for every lawful exact realization C , with the mapping type read strictly in the executable lane and homotopically when the recursion, η -, and coherence data are retained.

Proof sketch. Interpret old generators by the given map from $\mathcal{B}_H$, new generators by their specified realizations in C , and extend recursively. The interpretation respects $\equiv_{x^*}$ because C satisfies every forced equation. Any second interpretation agrees on all generators and therefore on all generated terms by structural induction. The retained η - and coherence rules upgrade uniqueness to contractibility in the intensional formulation. $\square$

Proposition 4.23 (Selection is not general initiality). *The PEN ranking does not imply that x^* is the initial completion of an independently fixed weaker requirement. Minimal positive overshoot and categorical leastness are different orders.*

Proof sketch. At the Step-8 comparison, the live window demand requires an operative lift of the two-dimensional path vocabulary; the sealed record certifies that a bare S^3 cannot discharge it and never enters the acceptable field, while the H-space S^3 is the unique total payment. Relative initiality applies afterward to the selected H-space API — the free completion of *that API*, not of some weaker “sphere-like” requirement fixed in advance. Defining the weaker requirement retrospectively from the winner would make the stronger claim circular; the demand register blocks the circularity by fixing what was required before any candidate is inspected. $\square$

Axiom 4.24 (Internal Record Actualization). In physical runtime there is no external ledger or external actualizer. A candidate outcome x is actualized as a public event for an internal context p only if there exists an internal record carrier R_p , a distinguishable record state $r_x : R_p$, and a comparison interface to the prior public history such that:

1. r_x is correlated with the realized alternative by the sealed internal dynamics and its own public weighting;
2. r_x is recoverable by an admissible internal decoder to the declared tolerance;
3. the record is locked into later admissibility — it persists until replaced by a declared sufficient statistic proved lossless for every retained public claim, rather than being silently overwritten; and
4. admissible comparisons with other internal contexts preserve the record up to their stated observer-code error.

Remark 4.25 (Why the sealing clauses are weight-neutral). Clause 1 above and the extractor below were formerly stated as requiring the record to be *Born-weighted* and *Born-supported*. Naming the weighting in these clauses does two things the theory should not want. First, it is in tension with the axiom’s own opening sentence: a clause that imports a valuation from outside the internal record structure reintroduces the external ledger the axiom denies. Second, it inverts the dependency chain — Internal Record Actualization, Public Confluence and $\mathcal{Q}_p$ are *inputs* to observer-indexed public physics, while the Born weight is derived inside that layer, so an input was presupposing an output of what it feeds. The clauses therefore require only that the record track the weighting the sealed layer supplies. Nothing downstream weakens: the weighting is recovered where it is derived, and on the route now under investigation it is recovered as the dagger’s own self-pairing rather than posited.

Definition 4.26 (Public-event extractor). For an internal context p , a public-event extractor is a partial map

$$\mathcal{Q}_p : (H_{\text{pub}}, \rho_{pW}, x) \multimap \text{Rec}_p,$$

where ρ_{pW} is the joint context/world state and x is a candidate event. The value $\mathcal{Q}_p(H_{\text{pub}}, \rho_{pW}, x) = r_x$ is defined only when the record is constitutively admissible, supported by the public weighting, decodable at the declared tolerance, compatible with prior lock-in, mutually comparable with other observer ledgers, and inside the active record-capacity bound. Otherwise $\mathcal{Q}_p$ returns no public seal for p ; the dynamics may continue, but the event remains unsealed or must be exported through an explicitly declared abstraction proved lossless for the retained public claims.

Axiom 4.27 (Public Confluence). Let r_p and r_q be separately repeatable, copyable, and future-stable records concerning a common public overlap. If they are operationally compatible, there exists a joint nondemolition refinement r_{pq} , unique up to public operational equivalence, whose comparison maps recover both records. If no such refinement exists, no public history contains both as facts about that overlap. Thus finite-capacity failure has the form

$$\text{common seal} \quad \text{or} \quad \text{no common seal},$$

not two mutually contradictory sealed facts.

Proposition 4.28 (Public Confluence does not determine the hidden kinematics). *Public Confluence constrains the common record layer, not the full private state cone or composite rule. Classical simplex theories, real and quaternionic quantum fragments, and generalized probabilistic theories can each be equipped with a confluent classical public-record interface. Consequently the revised Two Laws do not by themselves derive homogeneity/self-duality, purification, local tomography, or complex Born quantum structure.*

Proof sketch. Finite stochastic theory is constructive, compositional, homogeneous, self-dual, locally tomographic, and publicly confluent, but a nontrivial mixed distribution is not the marginal of a pure classical bipartite state, so purification fails. Real quantum theory is homogeneous, self-dual, purifying, and publicly confluent, but two-rebit states possess an antisymmetric–antisymmetric global direction invisible to all local symmetric product effects, so local tomography fails. Rational polyhedral GPT cones give constructive compositional models without homogeneous self-dual cones. None of these differences is visible to a confluence condition imposed only on stable classical records. $\square$

Proposition 4.29 (PEN does not determine $\mathcal{Q}_p$). *The structural acceptance data — obligations, discharges, and the audited value tuple — determine which admissible continuation is sealed, but they*

do not determine a record POVM, a decoding tolerance, a pointer coarse-graining, or a physical clock calibration. These data are invariantly invisible to $(\nu, \kappa, \Delta, \rho)$. The public-event extractor is therefore a bridge axiom or derived theorem of additional observer dynamics, not a hidden consequence of the existing score.

Proof sketch. Fix the same formally selected continuation and the same score tuple. Distinct admissible POVMs or coarse-grainings on the observer record can produce different public partitions while leaving every PEN quantity unchanged. Likewise, a strictly increasing reparameterization of the physical record times changes throughput but not selection order or score. Hence models agreeing on CL, PEN, and the structural trace can disagree on $\mathcal{Q}_p$ and on clock calibration. $\square$

Proposition 4.30 (PEN operationalizes the formal face of SL). *The five PEN rules are cumulative growth, horizon policy, admissibility gate, efficient overshoot, and coherent integration. Relative free sealing fixes the exact realization of the selected API. Internal Record Actualization, Public Confluence, and $\mathcal{Q}_p$ govern the additional passage from formal integration to observer-relative public physical sealing.*

Proof sketch. Localizability is supplied by evaluating only $\mathcal{A}(H)$. Admissibility is supplied by the gate $x \in \mathcal{A}(H)$. Valuation is supplied by $\rho = \nu/\kappa$. Critical sufficiency is supplied by $\text{Bar}(H)$. Constructive primeness is supplied by $P_H(x)$ and the minimal-overshoot rule. Commitment and exact coherent integration are supplied by $H' = F_x(H)$. Internal Record Actualization supplies public recordability, and Public Confluence supplies overlap consistency. $\square$

Rule	Formal content	Selective burden discharged
Cumulative growth	$H_{n+1} = I(H_n, x_n)$ and $H_n \subseteq H_{n+1}$ at the public-interface level	commitment, integration
Horizon policy	evaluate the live local continuation cone before widening search	localizability, critical sufficiency
Admissibility gate	require $x \in \mathcal{A}_r(H)$ before scoring	admissibility
Discharge and earning	total payment of the live obligations at guarded stages; certified earning (foreclosed by the ceiling) on the open band	acceptance, valuation as audit, anti-bundling
Coherent integration	freely realize and export the selected API as sealed formal history	exact realization, path dependence, future admissibility
Internal Record Actualization	stable decodable record in an internal context	public physical eventhood
Public Confluence	joint nondemolition refinement for compatible overlap records	common seal or no common seal; never contradictory public facts

Proposition 4.31 (Ratio scoring). *The ratio $\rho = \nu/\kappa$ is the selective efficiency invariant; a difference score $\nu - \kappa$ is not invariant under gratuitous bundling.*

Proof sketch. If two independent moves are bundled, raw gain can increase faster than the penalty in a difference score while adding unnecessary integration burden. A ratio score compares generative leverage per unit cost. Minimal positive overshoot above $\text{Bar}(H)$ then selects sufficient gain without rewarding opaque payload packing. $\square$

Proposition 4.32 (Path dependence without global teleology). *PEN is history-sensitive and non-oracular: the selected x_n depends on H_n and its live filtered cone, not on a completed future trajectory.*

Proof sketch. The quantities $\nu_H, \kappa_H, \text{Bar}(H)$, constructive primeness, and the horizon filtration are all indexed by realized history. After integration, $H_{n+1} = \mathcal{I}_{H_n}(x_n)$ changes the next cone and the next bar. No term in the selection rule quantifies over future selected histories except through candidates presently admissible at H_n . $\square$

5 Kinematic Scope and Reconstruction Target

Proposition 5.1 (Current meta-kinematic result). *Modern physics admits a natural realization satisfying the revised Two Laws: coherent states and processes form the admissible field; dynamics and measurement update realized history; stable classical records satisfy Public Confluence. The converse and uniqueness directions are not proved. The revised laws presently determine a meta-kinematic envelope, not the unique complex quantum-relativistic theory inside it.*

Definition 5.2 (Genesis operational-completion package). The proposed nonnumerical reconstruction package consists of four theorem targets:

1. *Genesis-to-Jordan:* derive spectral decomposition, a positive dagger/state-effect correspondence, frame symmetry, and constructive filter completeness, hence homogeneous self-dual cones and Euclidean Jordan structure.
2. *Natural monoidal phase:* derive a faithful primitive central $U(1)$ action on pure-process carriers, natural under all processes and balanced under composition, thereby excluding presentation-dependent real complexifications and noncentral quaternionic phase.
3. *No fundamental mixing:* prove that the mixed operational category is the free environment completion of the pure Genesis category, hence every mixed state has a pure dilation, unique up to environmental equivalence along the selected class of environment embeddings. That equivalence is in general a zigzag of such embeddings; reading it as a single reversible map on the environment requires a filtered or common-cocone hypothesis.
4. *Public descent:* prove that public records glue over shared interfaces. Ordinary local tomography is recovered only when the overlap carries no additional gauge, edge, or gravitational record data.

Principle 5.3 (Open Two-Laws-to-modern-kinematics reconstruction conjecture). For finite irreducible compositional operational theories with stable public histories, the Constitutive Law, the revised Selective Law, and the Genesis Sequence derive the Genesis operational-completion package; its resulting pure category is equivalent to finite-dimensional complex Hilbert spaces with phase-balanced tensor product, and its causal completion is equivalent to the corresponding completely positive operational theory, with public-interface descent in gravitational sectors.

The conjecture has explicit falsifiers. A constructive publicly confluent countermodel satisfying the four derived completion targets but not operationally equivalent to complex quantum theory refutes sufficiency. A finite stable-history operational theory lacking any admissibility-actualization factorization refutes necessity. Failure to derive any target from the Genesis candidate competition identifies that target as genuine bridge content rather than permitting it to be silently inserted into Step 14.

Remark 5.4 (Status of target 3 after the July 2026 completion audit). Target 3 separates into a realization half and an API-selection half, and the two have resolved in opposite directions.

Realization: derived. Present the sealed pure layer P as an essentially small strict symmetric monoidal category and let $J \subseteq P$ be a wide symmetric monoidal class of *environment embeddings*.

Write $\text{Env}_J(P)$ for the category with the objects of P , with morphisms $A \rightarrow B$ the pairs $(E, f : A \rightarrow B \otimes E)$ modulo the equivalence generated by $(E, f) \sim (E', (\text{id}_B \otimes u)f)$ for $u \in J$, composition $[D, g][E, f] = [D \otimes E, (g \otimes \text{id}_E)f]$, and discarding $\varepsilon_A = [A, \text{id}_A]$. Then $\text{Env}_J(P)$ is initial among exact J -realizations — strict symmetric monoidal functors $F : P \rightarrow M$ into a monoidal category with discarding, carrying every $u \in J$ to a causal map. Every morphism of the completion is a dilation by construction, so purification is a property of the free seal and not a postulate about states. Outside the strict lane the tensorator and unit map of F must additionally be required causal; without that normalization the universal property fails.

Selection: not derived, and independent. The class J is not optional. If environments are identified along *all* maps of P , the completion collapses: whenever P has a zero object with $B \otimes 0 \cong 0$, every hom-set of the unrestricted completion is a singleton, and purification holds vacuously with nothing else surviving. The intended class is $J = \text{Isom}(P)$, which is unavailable until the dagger is sealed — so targets 1 and 3 are ordered rather than coordinate. The clause that would make that choice lawful is *ledger invariance*, $\varepsilon_{E'}u = \varepsilon_E$ for every isometry u , and it does not follow from Internal Record Actualization, Public Confluence, monoidal discarding, and free sealing. Countermodel: nonnegative real matrices under Kronecker product with transpose as dagger, $\varepsilon_n = (1, \dots, 1)$, and $u = 2^{-1/2}(1, 1)^\top : 1 \rightarrow 2$, for which $u^\dagger u = 1$ while $\varepsilon_2 u = \sqrt{2} \neq 1 = \varepsilon_1$. Every unitary of that category is a permutation matrix, so its discard is invariant under all unitary recoding and only isometric enlargement separates the two notions: the failure is exactly a mismatch between dagger-losslessness and ledger-losslessness. Public Confluence cannot repair it, being a conditional whose antecedent is a model-chosen compatibility relation and whose failure mode is the designated disjunct “no common seal”; an axiom of that shape cannot force an equation.

Consequent status. Target 3 is conditionally derived: ledger invariance is one counted bridge premise, and conditional on it the environment completion is a theorem of free sealing. This is the disposition the falsifier clause above prescribes for a target that resists derivation, and it is recorded rather than absorbed.

6 Exact Coherence Depth Two

Definition 6.1 (Obligations and historical arity). For a globally acting structural candidate X sealed against a depth- k historical interface, let

$$O^{(k)}(X)$$

be the normalized set of atomic structural integration obligations. For $o \in O^{(k)}(X)$, let $\text{Supp}(o)$ be the set of earlier sealed layers whose exported schemas occur irreducibly in the normalized typing derivation of o . The historical arity is $a(o) = |\text{Supp}(o)|$.

Definition 6.2 (Depth invariants). The obligation depth d_{obl} is the least d such that $O^{(k)}(X) \simeq O^{(d)}(X)$ for every $k \geq d$. The minimal-signature depth d_μ is the least d such that every admissible surface declaration is presentation-equivalent to a normalized public signature with no irreducible primitive historical field of arity above d .

Definition 6.3 (Sealed structural extension). A sealed structural extension is a globally acting candidate X whose payload changes the active reduction, elimination, modal, or equality interface of the core calculus. It is exported through an opaque public signature

$$\Sigma_X = (K_X, T_X),$$

where K_X is payload data and T_X is structural trace data. A trace field is primitive when it is not computed by lower-interface data; it is derived when it is canonically generated by the cubical filler operations from already exported boundary data.

Definition 6.4 (Horn fiber). For a fixed lower boundary ∂ , write the depth- k marginal horn-extension fiber as

$$H_k(\partial) := \sum_{\gamma: \Gamma_{\text{miss}}(\partial)} \text{Fill}_{\partial}(\gamma),$$

where γ is the missing face and $\text{Fill}_{\partial}(\gamma)$ is the type of cubical fillers compatible with the visible faces.

Definition 6.5 (Obligation status). An obligation occurrence has one of four normalized statuses:

present	a normalized obligation type occurs in $O^{(k)}(X)$,
contractible	the obligation type occurs but its fiber is canonically contractible,
eliminable	presentation equivalence removes it from μ -minimal public form,
absent	no normalized obligation of that kind occurs.

The depth theorem uses all four statuses: higher obligations may be present, contractible, and eliminable without being absent.

Theorem 6.6 (Depth-two theorem package). *In the fully coupled CCHM-style cubical extension calculus of globally acting structural extensions sealed behind opaque public signatures:*

1. $d_{\text{obl}} = 2$;
2. $d_{\mu} \leq 2$;
3. *primitive historical support factors through the previous two normalized signatures*;
4. *under uniform payload size c , the presentation-invariant sealing cost satisfies*

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c.$$

Proof sketch. The lower bound is supplied by a binary univalence obstruction. Let $2 = \mathbf{1} + \mathbf{1}$ and let $\text{swap} : 2 \rightarrow 2$ exchange the summands. Univalence yields

$$p_{\text{swap}} := \text{ua}(\text{swap}) : 2 =_{\mathcal{U}} 2.$$

A globally acting candidate such as a family of endomaps $f : \prod_{A:\mathcal{U}} A \rightarrow A$ requires coherence of its action over p_{swap} . Unary data specify action at each exported type but not binary comparison over the univalent path; hence $d_{\text{obl}} \geq 2$.

For the upper bound, the marginal datum at depth 3 is not a closed three-sphere filler but an open-box extension fiber over already fixed lower boundary data. Since $\text{isEquiv}(f)$ is a mere proposition, equivalence-proof data do not generate an independent tower. Cubical operations hcomp and hfill compute missing faces and fillers from visible boundaries, making the remote horn fiber contractible. Iteration gives $O^{(k)}(X) \simeq O^{(2)}(X)$ for all $k \geq 2$.

Contractible higher factors are computationally derived in canonical normalized public syntax. Replacing them by generated horn witnesses preserves presentation equivalence and removes them from μ -minimal primitive trace. Surviving binary trace fields factor through the previous two signatures by composition of intervening unary bridges and binary traces. Thus the exact obligation depth is two and the public chronological window is two. $\square$

Proposition 6.7 (Primitive/derived separation). *For $k \geq 3$, every higher structural trace field in a normalized surface presentation is either rejected as an inadmissible naked higher face or packaged as a horn boundary whose witness is derived from lower public data.*

Proof sketch. The fixed raw structural syntax classifies action clauses as unary, comparison clauses as binary, and higher material as horn-shaped. A higher field without boundary data is not a structural trace field for the sealed extension. A higher field with boundary data elaborates to an open-box problem, and the cubical Kan structure computes the witness. Thus no higher field survives as primitive trace in a μ -minimal public signature. $\square$

Proposition 6.8 (Binary lower bound). *There exists a structural extension X for which $O^{(2)}(X) \not\subseteq O^{(1)}(X)$.*

Proof sketch. Take the two-point type and the automorphism $\text{swap} : 2 \simeq 2$. By univalence, swap determines $p_{\text{swap}} : 2 =_{\mathcal{U}} 2$. A globally acting extension must preserve its public action under this path. The required square compares two uses of the extension across one structural equivalence. This comparison has historical arity two and cannot be reconstructed from independent unary action data. $\square$

Proposition 6.9 (Horn upper bound). *For each $k \geq 3$ and fixed depth- $(k - 1)$ boundary ∂ , the marginal remote horn factor $H_k(\partial)$ is canonically contractible.*

Proof sketch. The visible boundary determines the admissible missing-face problem. Cubical composition computes a composite face, and cubical filling packages the compatible filler. Substitution stability of Kan operations identifies any two admissible remote completions over the same boundary. Hence the projection from the total depth- k obligation to the depth- $(k - 1)$ boundary is an equivalence with contractible fiber. $\square$

Proposition 6.10 (Minimal-signature elimination). *Every arity- $k \geq 3$ structural trace field is absent from μ -minimal public syntax.*

Proof sketch. Surface normalization splits trace fields into unary, binary, and horn-shaped forms. Horn-shaped fields carry explicit derivation obligations from their lower boundary. Since the horn fiber is contractible and computationally filled, replacing an explicit higher field by the generated filler preserves presentation equivalence and lowers or preserves primitive trace cost. Minimality therefore excludes the explicit higher primitive field. $\square$

Proposition 6.11 (Chronological Markov blanket). *At stage $n + 1$, primitive structural obligations factor through the interface generated by stages n and $n - 1$. Earlier stages contribute only by derived composites through this two-layer interface.*

Proof sketch. Unary trace records present action against the immediately active public interface. Binary trace records comparison between the candidate and the previous active interface. Any comparison with a remote layer factors through a telescope of adjacent unary bridges and binary comparison traces. The remote dependency is therefore present only as a derived composite, not as primitive historical support. $\square$

7 Affine Fibonacci Scaling

Theorem 7.1 (Universal affine recurrence). *If structural integration coherence has obligation depth d , then the maximal full-coupling presentation cost satisfies*

$$\mu_{n+1} = \sum_{j=0}^{d-1} (\mu_{n-j} + \kappa_{n-j}).$$

Definition 7.2 (Dependency footprint). The dependency footprint $F_{n+1} \subseteq \{n, n-1, \dots, n-d+1\}$ is the subset of the chronological window whose public interfaces are actually used by the primitive trace of stage $n+1$.

Proposition 7.3 (Full-coupling envelope). *Sparse dependency costs are bounded by the full-coupling recurrence:*

$$\mu_{n+1}^{\text{sparse}} \leq \sum_{j=0}^{d-1} (\mu_{n-j} + \kappa_{n-j}).$$

Equality holds in the maximally coupled case $F_{n+1} = \{n, \dots, n-d+1\}$.

Proof sketch. The chronological window theorem restricts primitive support to the last d layers. A sparse candidate may use a proper subset of those interfaces, so its charge is the corresponding sub-sum. The full envelope is the supremum over admissible footprints inside the same window and therefore bounds all sparse cases. $\square$

Corollary 7.4 (Depth-two Fibonacci law). *For $d_{\text{obl}} = 2$,*

$$\mu_{n+1} = \mu_n + \mu_{n-1} + \kappa_n + \kappa_{n-1}.$$

If $\kappa_n = c$, then

$$U_n := \mu_n + 2c \quad \Rightarrow \quad U_{n+1} = U_n + U_{n-1}.$$

Consequently $U_{n+1}/U_n \rightarrow \varphi = (1 + \sqrt{5})/2$.

Proof sketch. At stage $n+1$, primitive obligations are generated by the active public interfaces in the preceding d layers. Full coupling charges both inherited opaque cost μ and current payload κ for each layer in the window. Substituting $d = 2$ gives the two-step affine recurrence. A constant shift by $2c$ removes the affine term, leaving the Fibonacci recurrence. $\square$

Definition 7.5 (Diagnostic bar (engine register)). Let $\Delta_n = \mu_n$ after canonical normalized public interfaces are fixed. Define

$$\Phi_n = \frac{\Delta_n}{\Delta_{n-1}}, \quad \Omega_{n-1} = \frac{\sum_{i < n} \nu_i}{\sum_{i < n} \kappa_i}, \quad \text{Bar}_n = \Phi_n \Omega_{n-1}.$$

These are descriptive quantities of the interface ledger: recorded, branch-indexed, and legislatively inert under the adopted law. The intrinsic descriptive form is the two-step memory bar $\widehat{\text{Bar}}_n = (\nu_{n-1} + \nu_{n-2})/(\kappa_{n-1} + \kappa_{n-2})$, which carries the window's own memory depth and forgets the Stage-4 branch choice from $n = 7$ onward — the cumulative form above remembers it forever, which is why cumulative diagnostics are permanently branch-indexed.

Proposition 7.6 (Golden-ratio asymptotics of the debt ledger). $\Phi_n \rightarrow \varphi$: *the shifted sealing-cost sequence is Fibonacci, so adjacent debt ratios converge to the golden ratio — the fingerprint of width-two anticipation, and a statement about the debt arithmetic alone. Composite limits such as $\text{Bar}_n \rightarrow \varphi \Omega_\infty$ hold under the displayed stability hypotheses and inherit the diagnostic, branch-indexed status of their factors; no selective consequence follows from them.*

Remark 7.7 (Retired: the Ω -ratchet). Earlier formulations proved an “ Ω -ratchet”: acceptance forces $\rho_n > \Omega_{n-1}$, so cumulative efficiency increases strictly along accepted steps, with corollaries excluding declining tails, idle repetition, and average-matching steps. The proposition is retired, and honestly: its premise is false under the adopted law (guarded acceptance never consults ρ ; the certified semantic Stage 8 winner *failed* the diagnostic bar and lawfully advanced), and its corollaries are refuted in the lawful register — the audited worth column of the certified trace is non-monotone and contains accepted steps with $\nu = 0$ (the Hopf and curvature payments: pure discharge, reorganization without minting). The engine-register Ω -trajectory of the enacted trace remains sealed testimony; the picture of a world climbing an efficiency gradient does not survive the audit and is not part of the theory. What replaces it is stronger and simpler: the uniqueness census — exactly one acceptable payment at every guarded stage but one — so the ascent needs no gradient to climb.

Proposition 7.8 (Runtime richness from incompleteness). *Once the arithmetic-hosting strata are sealed, the library’s derivability closure is incomplete (Gödel), so content independent of $\mathcal{B}_n$ exists at every stage. Under the adopted law this bears on the halt in exactly one way: it shows the halt is not an exhaustion of expressible content. The sequence stops because $O(16) = \emptyset$ — nothing further is owed — while independent content continues to exist and is simply unowed; on the debt-free field it cannot be accepted, because certified earning is bounded by the provenance ceiling. The halt is completion of obligation amid inexhaustible possibility, which is precisely what makes the completed library a rich runtime rather than a finished catalogue.*

Proposition 7.9 (Discrete non-responsiveness). *Extensional or h-set-like branches have depth-one coherence and no higher path content. Their exclusion from the certified ascent is not a pricing outcome: from Step 5 onward the live window demands are demands for structured identity, operative form, and interface discipline, and flat-identity candidates do not discharge them. A discrete candidate is not an inefficient rival; it is a non-payment — it answers a debt nobody has incurred.*

8 Generative Calculus

Definition 8.1 (MBTT cost). MBTT is the fixed minimal specification grammar used to count irreducible candidate descriptions. The clause cost of X over $\mathcal{B}_n$ is $\kappa_{\text{cl}}(X \mid \mathcal{B}_n)$. The bit-level audit cost is $\kappa_{\text{bit}}(X \mid \mathcal{B}_n)$.

Definition 8.2 (Charged kernel clauses). The charged integration cost is

$$\kappa(X \mid \mathcal{B}_n) = |\text{KernelClauses}(X \mid \mathcal{B}_n)|.$$

$\text{KernelClauses}(X \mid \mathcal{B}_n)$ contains only irreducible API commitments not already definitionally supplied by $\mathcal{B}_n$. Definitional completion, projections, aliases, adjoint completions, and generated eliminators add executable telescope structure but zero additional charged κ .

Definition 8.3 (Schema novelty). Novelty is literal schema difference:

$$\nu(X \mid \mathcal{B}_n) = |\text{Schemas}(\mathcal{B}_n + X) \setminus \text{Schemas}(\mathcal{B}_n)|.$$

It decomposes as

$$\nu = \nu_G + \nu_C + \nu_H,$$

where ν_G counts formation and introduction schemas, ν_C counts cross-interface and elimination schemas, and ν_H counts homotopical, computation, Kan, and path schemas.

Proposition 8.4 (Topological amplification). *Higher-inductive and cohesive candidates can generate homotopical and cross-interface schemas unavailable to discrete candidates, creating superlinear novelty relative to irreducible specification cost.*

Proof sketch. Discrete candidates add formation and elimination content but no genuine path constructor. A higher-inductive candidate adds point data, path data, eliminators, computation rules, and compatibility obligations. Since these schemas are generated by a small constructor package, ν_H and ν_C can increase without proportional growth in κ . $\square$

Proposition 8.5 (Step 5 circle selection). *At the first geometric transition, S^1 with constructors*

$$\text{base} : S^1, \quad \text{loop} : \text{base} =_{S^1} \text{base}$$

is the certified unique total discharger of the live window demand — the inhabitation of structured identity that the sealed dependent grammar exports. Engine-register signature $(\nu, \kappa, \rho) = (7, 3, 2.33)$; audited semantic worth 6 on $\kappa = 3$, the ascent's richest minting until the terminal step.

Proof sketch. Discrete competitors have flat identity and do not discharge a demand for structured identity: they are non-payments, not inefficient rivals. Higher-dimensional packages presuppose path vocabulary the library does not yet export and fail admissibility against the live window. The circle is the least structure that pays, and the census certifies it is the only one. $\square$

Definition 8.6 (Schema Calculus). The per-class counting formulas of the reference evaluator decompose into six principles:

- P1. generator counting: one schema per new formation or introduction generator;
- P2. adjoint completion: elimination behavior forced by the interpreting adjunction is definitionally free;
- P3. HIT universal property: the fixed universal-property surcharge for higher-inductive packages;
- P4. the d^2 law: a dimension- d path constructor contributes d monodromy schemas plus $d(d-1)$ variation fillers;
- P5. connectivity dichotomy: record access telescopes (Telescopic Elimination); morphism transport contributes $2\kappa + r^2$ over r referenced interfaces;
- P6. uniform action over the referenced base.

Proposition 8.7 (Derived evaluator). *Every component of the reference evaluator is forced, proven, derived, or gauge: ten of sixteen formula components are forced outright by the definitions of ν and κ ; the d^2 law (Lemma L1) and the r^2 transport law (Lemma L2) are proven at the level of careful sketch; the H -space chaining term is resolved by canonical (operation, coherence) pairing; and global ν -rescaling is gauge (only cross-class ratios enter selection). The evaluator rebuilt from the derived rules reproduces the strict fifteen-step Genesis trace value for value and hash-identically in the accepted lane.*

Proof sketch. L1: the d monodromy schemas and $d(d-1)$ ordered variation fillers are the depth-two self-coherence data of a d -cell; the rival rules $2d$ and $\binom{d}{2}$ are refuted at $d = 1$ by inspection of the circle, trace-independently. L2: per-clause forward and lifting faces give 2κ (lifting purchased by the charged fibration structure, explaining the bare-arrow $\kappa = 1$ carve-out); r diagonal monodromies (live by admissibility: trivial self-transport would trivialize the candidate) plus $r(r-1)$ directed

couplings (ordered because inverses are propositional, not definitional) give r^2 . The corrected law charges $\sum_i d_i^2$ per constructor rather than $\max_i d_i^2$ per telescope; applying the correction engine-wide leaves all fifteen selected steps unchanged and repairs over-pruning off-trace. A register caution now governs this cluster: the d^2 and r^2 laws are engine-register counting rules — exact descriptions of the enacted evaluator, sealed as testimony — and the act-local audit refuses their outputs as independent law-level credit (a lifted site or transport coupling is an instance or completion unless it carries its own anchor). The earlier slogan that Born’s exponent, d^2 , r^2 , and Fibonacci are “four shadows of ordered self-pairing within the depth-two window” therefore overstates by two: what survives register-independently is the pair rooted directly in width-two structure — the Fibonacci debt recurrence and the depth bound $d_{\text{obl}} = 2$ — and any bridge from that depth to the Born exponent must be built on those alone. L1 and L2 remain worth machine-checking as evaluator derivations, but nothing law-level now rides on them: the Step-16 load they once carried is discharged by the certified halt package. $\square$

9 Genesis Sequence

Definition 9.1 (Genesis trajectory). The Genesis Sequence is the trajectory of the two laws from the empty library $\mathcal{B}_0 = \emptyset$: three constitutive founding acts, then at each guarded stage

$$\mathcal{B}_{n+1} = I(\mathcal{B}_n, X_n), \quad X_n = \text{the total discharger of } O(n) \text{ with typed provenance,}$$

unique at every guarded stage except Step 4, where four genuinely distinct dischargers survive the canonical quotient and the enacted branch is fixed indexically. (The enacted engine realized this trajectory through its bar-and-overshoot encoding; on the certified trace the two prescriptions select identical winners, and the encoding is retained as testimony.)

Step	Selected shell	Δ_n	ν^{eng}	ν^{sem}	κ	ρ^{eng}	Bar^{diag}
1	Universe $\mathcal{U}_0$	1	1	1	2	0.50	0.50
2	Unit type 1	1	1	0	1	1.00	0.50
3	Witness $\star : \mathbf{1}$	2	2	1	1	2.00	1.33
4	Π/Σ shell	3	5	3	3	1.67	1.50
5	Circle S^1	5	7	6	3	2.33	2.14
6	Propositional truncation	8	8	3	3	2.67	2.56
7	Sphere S^2	13	10	1	3	3.33	3.00
8	H-space S^3	21	18	1	5	3.60	3.43
9	Hopf fibration	34	17	0	4	4.25	4.01
10	Cohesion/modal shell	55	19	4	4	4.75	4.46
11	Connection shell	89	26	1	5	5.20	4.91
12	Curvature shell	144	34	0	6	5.67	5.42
13	Endomorphic operator bundle	233	46	2	7	6.57	5.99
14	Hilbert-functional shell	377	62	3	9	6.89	6.68
15	Temporal-cohesive shell	610	103	6	8	12.88	7.40

Register key. ν^{eng} , ρ^{eng} , and Bar^{diag} are the enacted engine’s structural ledger — sealed testimony, exact as a record of the enacted evaluator, with no law-level authority; the engine’s mechanism-audited total is $\sum \nu^{\text{eng}} = 358$. ν^{sem} is the audited semantic-family register: act-local, receipt-bearing, the sole lawful value column, with $\sum \nu^{\text{sem}} = 32$. The step-8 engine entry reflects the audited mechanism correction; the two semantic zeros (Steps 9 and 12) are certified pure discharges — synthesis and consequence mint nothing. ν^{sem} at Step 4 is the enacted branch’s value; the cone’s minima carry 2. All cumulative columns are permanently branch-indexed.

Step	Phase	New primitive capacity	Dominant rejection pressure
1–4	Bootstrap	typed universe, inhabitation, witness, dependent formation and pairing	untyped or witnessless candidates fail coherent realizability
5	Geometric ascent	first nontrivial path constructor	flat-identity candidates do not discharge the structured-identity demand
6	Geometric ascent	propositional truncation and proof-irrelevant compression	raw constructive content is too costly when only support is needed
7	Geometric ascent	two-dimensional homotopy payload	lower-dimensional path structure cannot express surface coherence
8	Geometric ascent	S^3 action and H-space multiplication	the bare sphere cannot discharge the operative-lift demand (sealed exclusion)
9	Geometric ascent	Hopf fibration and bundle prototype	unrelated higher spheres lack the fiber/base interface gain
10	Framework abstraction	cohesive modal adjunctions	purely discrete or purely spatial formalisms fail cross-aspect reuse
11	Framework abstraction	transport/connection structure	cohesive shape without parallel transport cannot compare fibers locally
12	Framework abstraction	curvature as failure of transport around loops	flat transport leaves loop-dependent information unmeasured
13	Framework abstraction	metric/frame/endomorphoric operator bundle	connection data without metric comparison lacks measurement interface
14	Framework abstraction	Hilbert-functional/operator shell	metric geometry without operator state space lacks quantum-functional dynamics
15	Terminal synthesis	temporal-cohesive guarded runtime	kinematic libraries without temporal exchange cannot execute internal flow

Definition 9.2 (Synthetic signatures for Steps 10–15). The late selected shells have the following synthetic signatures:

- 10 : $\Pi_\infty \dashv \text{Disc} \dashv \flat \dashv \sharp$,
- 11 : ∇ acting on cohesive families,
- 12 : $R_\nabla = d\nabla + \nabla \wedge \nabla$,
- 13 : $(g, \text{Fr}(M), \star, \Delta, \text{Ric})$ as a coupled metric/frame package,
- 14 : $(\mathcal{H}, \mathcal{B}(\mathcal{H}), \text{Spec})$ as Hilbert-functional shell,
- 15 : $(\bigcirc, \diamond, \text{cohesive exchange, guarded coherence})$.

Definition 9.3 (Late charged shell clauses). The framework-abstraction phase charges the following irreducible API clauses:

Step	Charged API clauses
10	(b, #, Disc, Shape)
11	(∇ , transport, covariant lift, horizontal lift, Leibniz)
12	(R , Bianchi, holonomy, ChernWeil, charclass, curvature transport)
13	(g , LeviCivita, geodesic, $\text{vol}_g, \star_g, \Delta_g, \text{Ric}_g$)
14	($\langle -, - \rangle$, complete, $\perp$ Dec, Spec, C^* Alg, $\delta/\delta\phi$, metric bridge, curvature bridge, connection bridge)
15	($\bigcirc, \diamond, c$, temporal action, $e^b, e^\sharp, \diamond\text{-elim}, \mu_\bigcirc$)

The displayed lists have cardinalities 4, 5, 6, 7, 9, 8, matching the canonical κ -ledger.

Proposition 9.4 (Late-stage novelty accounting). *The selected late shells have novelty decompositions*

n	ν_G	ν_C	ν_H	ν_n
10	4	15	0	19
11	2	24	0	26
12	1	33	0	34
13	4	42	0	46
14	5	57	0	62

in the engine register — sealed testimony of the enacted evaluator, with the diagnostic $\rho_n > \text{Bar}_n$ recorded for each $10 \leq n \leq 14$. The audited semantic worths for the same stages are 4, 1, 0, 2, 3: the telescopic inheritance visible in the middle column is exactly the credit class the act-local register refuses, which is why the two columns differ by an order of magnitude.

Proof sketch. Step 10 contributes four modal carriers and their unit/counit/geometric interaction schemas against the Hopf-loaded interface. Step 11 adds two new carrier families and imports the Step 10 cohesive novelty by the telescopic elimination law, giving $2 + (19 + 5) = 26$. Step 12 adds one curvature carrier and depends on connection and cohesion, giving $1 + (26 + 6 + 1) = 34$. Step 13 adds four metric/frame carrier families and depends on curvature plus connection, giving $4 + (34 + 7 + 1) = 46$. Step 14 adds five Hilbert-functional carrier families and couples metric, curvature, and connection, giving $5 + (46 + 9 + 2) = 62$. These identities describe the enacted evaluator exactly and carry no law-level credit; at law level each selected shell is the certified unique total discharger of its stage demand. $\square$

Proposition 9.5 (Real-cohesive scalar provenance). *The scalar line object used by the metric shell is derived from the cohesive interpretation of the circle-cover interface and is not charged as an independent late-stage primitive.*

Proof sketch. After Step 10, the shape/discrete/cohesive modalities act on the earlier circle. The universal-cover fiber of the shape unit supplies the semantic line-object interface in real-cohesive models. Since the scalar carrier is obtained from the already selected circle and cohesive shell, bottom-up scalar analysis would duplicate derived structure and incur an unnecessary surcharge; the metric shell is charged only for the metric/frame API over that scalar interface. $\square$

Theorem 9.6 (Exact Step 15 novelty count (engine register)). *The temporal-cohesive shell has exact engine-register decomposition*

$$\nu_{15}^{\text{eng}} = \nu_G + \nu_H + \nu_C = 2 + 15 + 86 = 103, \quad \rho_{15}^{\text{eng}} = \frac{103}{8} = 12.875,$$

sealed testimony of the enacted evaluator (the diagnostic bar stood at 7.40). The audited semantic worth is $\nu_{15}^{\text{sem}} = 6$ on $\kappa = 8$: six independent receipt-bearing families, tied with the circle for the

ascent's richest minting; the remaining ninety-seven engine units are temporal action on existing layers and exchange with existing interfaces — instances and completions under the act-local audit, not independent credit. At law level the step is selected by neither number: it is the certified unique total discharger of the temporal demand among thousands of stated candidates.

Proof sketch. The two new unary formers $\bigcirc$ and $\diamond$ give $\nu_G = 2$. Temporal action on the previous fourteen layers plus guarded self-composition gives $\nu_H = 14 + 1 = 15$. Cross-interface novelty has three parts: three temporal action/comparison schemas against each of fourteen earlier layers, giving 42; two cohesive exchange laws acting generatorwise on the Step 10 cohesive interface of novelty 19, giving 38; and six residual comparison, observation, guarded-composition, and depth-two naturality schemas. Thus $\nu_C = 42 + 38 + 6 = 86$, and the efficiency inequality is immediate. $\square$

Theorem 9.7 (Stagewise Genesis optimality). *Relative to the fixed MBTT grammar, depth-two bar dynamics, admissibility gate, and constructive-prime rule, the selected shells above form the current canonical fifteen-step trajectory.*

Proof sketch. Steps 1–3 are the constitutive founding acts; the third exports the first window demand. Steps 4–9 pay the demand chain of grammar, structured identity, control, enclosure, operative lift, and binding; Steps 10–14 pay the framework chain of aspect, transport, route-dependence, magnitude, and analysis; Step 15 pays the temporal demand. At each guarded step the selected shell is admissible and is the certified unique total discharger of the live demand — except Step 4, where the certified four-way tie founds the cone and the enacted branch continues indexically. The enacted engine's bar-and-overshoot encoding reproduced exactly this trajectory, winner for winner, which is the enactment-conservativity of the two-register law. $\square$

Theorem 9.8 (Minimal Temporal Extension). *Let G_{14} be the grammar earned through Step 14 and $\mathbf{Ext}_{14}$ the groupoid of finite normalized constructor extensions over G_{14} with univalent signature equivalences as morphisms. Then:*

1. *(Forced growth.) No G_{14} -expressible candidate discharges the live temporal demand at H_{14} : the resident grammar cannot state any total payment of the obligation its own analytic completion exported. Continuation requires grammar growth. (Engine-register corroboration: the resident diagnostic supremum was $\rho_{14} = 62/9 \approx 6.89$ against a recorded bar of 7.40 — the stall was visible in the enacted bookkeeping too.)*
2. *(Initiality.) The class of extensions admitting an admissible discharger of the temporal demand is nonempty and has an initial object. Its charged-clause count is $\kappa = 8$; that count is a computed property of the selected extension, recorded as such, and is not the ground on which it is selected.*
3. *(Identification.) The initial object is the free coherent guarded modal completion of the resident cohesive base, and every good extension receives from it a unique signature morphism.*
4. *(Uniqueness.) That extension is unique up to equivalence in $\mathbf{Ext}_{14}$.*

The extension is defined without temporal vocabulary; only after the four clauses are proved is it read as the temporal-cohesive shell. Step 15 is therefore forced grammar growth, not assembly from a pre-stocked temporal alphabet.

Proof sketch. For the stall: a pure- G_{14} candidate introduces no new global modality or type former, so it can express only interaction among resident structure — and the temporal demand asks for succession, which no resident interaction supplies; the discharge fails at expressibility, before any

question of worth. For initiality and identification: the declared category of good guarded-modal extensions has a free completion, and its initiality is what selects — it leaves no junk and no choice, so any two initial good extensions are identified by the induced equivalence of normalized public signatures. The eight charged clauses of that completion are pairwise non-redundant, and any competitor either omits a coherence clause (failing the Constitutive Law over the cohesive base) or adds a distinguishing clause that the audit prices at zero, so the count is stable across the class; but no step of the argument compares two candidates' counts, and none may, Step 15 being a guarded stage. This is a Step-15-specific theorem under that declared category, not a general consequence of the acceptance rule. $\square$

Proposition 9.9 (Genesis status after the July 2026 audits). *The current Genesis claim separates into four grades:*

1. Reproduced: *the fixed evaluator and the independently derived Schema Calculus recover Steps 1–15 value for value.*
2. Derived at the terminal grammar boundary: *the resident H_{14} grammar cannot discharge the live temporal demand, and the initial coherent extension admitting a discharger is the guarded modal completion later interpreted temporally.*
3. Lane-conditional: *uniqueness of the entire fifteen-step course is relative to the fixed grammar, admissibility policy, horizon, quotient, and scoring definitions; invariance under every honest enlargement of that lane is not proved.*
4. Certified after ignition: *directive debt is exhausted at Step 16 at full instance granularity, the demanded-but-underdetermined falsifier is executed and excluded, and the semantics-deep credit bound the earlier contested reading demanded has been derived and adopted: act-local, proof-bearing valuation with the accounting ceiling $\nu \leq 4\kappa$. The former formula-maximizing survivors rode engine-register counting rules with no law-level standing; under the lawful register the open band is foreclosed, on every branch of the Stage-4 cone.*

Thus the Step-15 grammar objection is closed and terminality is certified on the disclosed lane, on all four branches of the cone; what remains open is universal course uniqueness — invariance under every honest enlargement of the lane.

Theorem 9.10 (Temporal-cohesive terminal fixed point). *Step 15 internalizes the generative interface into a temporal-cohesive runtime. Subsequent distinct candidates are either internally derivable from the DCT shell, with zero novelty up to univalent equivalence, or external continuations evaluated on the debt-free open band described in the Guard-Rail package below.*

Proof sketch. The temporal-cohesive shell supplies temporal modalities, exchange laws with cohesion, guarded coherence, and the tangent interpretation of update. These structures make candidate generation and internal evolution expressible inside the selected library, so internal continuations have zero primitive novelty as external steps; this internal half is derived and has passed adversarial engine probing. The fate of external continuations is settled by the certified package of the runtime section: demand exhaustion at instance granularity, the executed and excluded falsifier, and the act-local provenance ceiling on the open band. The halt holds on all four branches of the Stage-4 cone, in the certified form recorded in Remark 1.1: the sealed library is complete. The law-property reading is suspended. $\square$

10 Runtime Ignition and Cosmogenesis

Definition 10.1 (Two clocks). Algorithmic time τ indexes external library assembly. Physical time t is the internal continuous geometric parameter arising after temporal-cohesive completion. They are related by runtime ignition, not by temporal succession inside t .

Definition 10.2 (Minimal internal clock class). For a guarded history $s \in \text{Str}(A)$, a clock representative is a map c into an ordered clock object T such that a nontrivial guarded tail advances the reading:

$$c(\text{tail}(s)) > c(s).$$

For continuous flow this becomes monotonicity along the chosen time orientation. Two representatives are equivalent when $c' = f \circ c$ for a strictly increasing reparameterization f . The minimal clock supplied by the temporal shell is the equivalence class

$$\mathcal{C}_{\text{ord}} = [c]_{c \sim f \circ c},$$

not a preferred numerical representative.

Theorem 10.3 (Minimal clock result and calibration no-go). *The Step-15 guarded modality, its jet filtration, and the tangent carrier $\mathbb{D}$ determine succession, orientation, and local infinitesimal update well enough to define $\mathcal{C}_{\text{ord}}$. They do not determine a metric calibration from Genesis steps, clause counts, ν , κ , or Δ to seconds, scale factor, temperature, or redshift.*

Proof sketch. The guard orders a state before its tail, and the tangent interpretation supplies the internal infinitesimal direction; therefore the ordinal/relational clock class is already present. Now assign any strictly increasing timestamps to a fixed realized structural trace and compose them with any other strictly increasing function. The selected histories, score tuples, cumulative bar, obligation windows, and guarded order are unchanged, while all throughputs and physical durations change. A unique metric clock would have to distinguish these models, but the two-law structural data do not. Any such calibration is a physical bridge and must declare its own observable anchor. $\square$

Definition 10.4 (Compilation epoch). The compilation epoch is the sequence $(\mathcal{B}_n)_{n \leq 14}$ equipped with algorithmic ticks

$$\tau_n = \sum_{i=1}^n \Delta_i.$$

It contains kinematic structure but no internal dynamical trajectory.

Definition 10.5 (Runtime ignition). Runtime ignition is the Step 15 transition in which temporal modalities $\bigcirc, \diamond$, spatial-temporal exchange, and guarded coherence convert the kinematic library $\mathcal{B}_{14}$ into the Dynamical Cohesive Topos.

Theorem 10.6 (Kinematic-dynamic equivalence). *In the temporal-cohesive shell,*

$$\bigcirc X \simeq X^{\mathbb{D}}.$$

The logical next modality is interpreted as tangent exponentiation.

Proof sketch. The pre-temporal library contains cohesive, metric, curvature, and Hilbert-functional structure but no internal flow. Step 15 introduces guarded temporal structure and exchange laws with cohesion. The infinitesimal object $\mathbb{D}$ turns the temporal next operator into a tangent endofunctor, converting discrete admissible update into internal continuous displacement. $\square$

Theorem 10.7 (DCT no-go). *In any nondegenerate topos, the unrestricted Löb fixpoint rule is inconsistent with the strict identification $\bigcirc X \simeq X^{\mathbb{D}}$ for pointed $\mathbb{D}$.*

Proof sketch. The point $0 : \mathbb{D}$ induces a natural evaluation $\text{ev}_0 : \bigcirc A \rightarrow A$. Since $\emptyset^{\mathbb{D}} \cong \emptyset$, applying Löb to ev_0 at $A = \emptyset$ produces a global element $1 \rightarrow \emptyset$, contradicting nondegeneracy. The strict semantic completion of Step 15 — all eight clauses, the strict identification, and unrestricted Löb jointly — therefore has no model. $\square$

Proposition 10.8 (DCT consistency trichotomy). *The Step-15 semantic completion resolves into three graded fragments:*

1. *the strict fusion (eight clauses, $\bigcirc X \simeq X^{\mathbb{D}}$, unrestricted Löb) is inconsistent by the no-go theorem;*
2. *the coiteration fragment — eight clauses plus the strict identification, productivity by guarded coiteration only, no Löb — is modeled by the Cahiers topos with $\bigcirc := (-)^{\mathbb{D}}$;*
3. *the filtered fragment — Löb under the weakened identification $\bigcirc \simeq \blacktriangleright ((-)^{\mathbb{D}})$ — is modeled by the tower of jet-truncated stages over the Cahiers topos, where Löb holds stage-wise and self-composition $\bigcirc \bigcirc A \rightarrow \bigcirc A$ provably fails.*

The repair principle: nilpotency generates a well-founded jet filtration, and Löb is licensed as contractive fixpoints along that filtration — the discrete staircase does not disappear but moves inside the instant as the Taylor tower. The no-go closes the gap between the fragments: no model recovers the strict fusion. Step-15 selection arithmetic ($\nu_{15} = 103$, $\kappa_{15} = 8$) is unaffected, since it prices the clause package, not the semantic completion. Status: the no-go is proven in full; the model fragments are at careful-sketch level with one cell ($\sharp\Diamond$ -exchange) unverified and mechanization pending.

Proposition 10.9 (Type-theoretic irreversibility). *Guarded corecursion induces a forward execution asymmetry even when internal geometric equations admit reversible local descriptions.*

Proof sketch. A guarded fixpoint has type $\text{fix} : (\bigcirc A \rightarrow A) \rightarrow A$. Construction of the next state is mediated by the guard. Removing the guard would require accessing the result without temporal delay, collapsing the guarded typing discipline. Thus runtime execution has a typed forward direction. By the consistency trichotomy above, such fixpoints are licensed along the jet filtration (or replaced by coiteration in the strict-identification fragment); the forward asymmetry is present in both consistent fragments. $\square$

Proposition 10.10 (Compressed ignition state). *The initial physical state is the public export of the selected temporal-cohesive library, not a point on a pre-existing physical time line. Its integration debt is $\Delta_{15} = 610$ in the current canonical ledger.*

Proof sketch. Physical time is internal to Step 15, so no $t < 0$ physical slice is available. The pre-runtime sequence is counted by τ , and its selected public interface is the complete kinematic-plus-temporal shell. The low initial algorithmic burden is the cost of that selected interface rather than a thermodynamic microstate sampled from a prior physical ensemble. $\square$

Definition 10.11 (Guarded streams and internal flow). For a type A , guarded streams satisfy

$$\text{Str}(A) \simeq A \times \bigcirc \text{Str}(A).$$

A guarded step map $s : A \rightarrow \bigcirc A$ induces

$$\text{coiter}_A(s) : A \rightarrow \text{Str}(A).$$

Under $\bigcirc X \simeq X^{\mathbb{D}}$, a synthetic vector field $v : \mathcal{U} \rightarrow \mathcal{U}^{\mathbb{D}}$ becomes a guarded step map $\theta_{\mathcal{U}}^{-1} \circ v : \mathcal{U} \rightarrow \bigcirc \mathcal{U}$.

Theorem 10.12 (Internalization of meta-evolution). *After Step 15, continuous structural evolution on the connected component of the univalent universe containing $\mathcal{B}_{15}$ is expressible internally:*

$$\text{Flow}_v := \text{coiter}_{\mathcal{U}}(\theta_{\mathcal{U}}^{-1} \circ v) : \mathcal{U} \rightarrow \text{Str}(\mathcal{U}).$$

Proof sketch. The tangent interpretation $\theta_X : \bigcirc X \simeq X^{\mathbb{D}}$ converts an internal vector field into a guarded next-state map. Guarded coiteration then produces an internal stream of universe states. Since all terms used in the construction belong to the temporal-cohesive shell, the evolution no longer requires external PEN compilation steps. $\square$

Theorem 10.13 (Univalent horizon). *The compilation epoch $(\mathcal{B}_n)_{n < 15}$ is not internally observable as a physical past of $\mathcal{B}_{15}$. Physical observables are terms in the DCT runtime and therefore remain inside the connected component generated after runtime ignition.*

Proof sketch. Internal observation is a DCT term: a path, section, functional, flow, or record map defined after Step 15. By univalence, internal paths in $\mathcal{U}$ identify structures by equivalence inside the runtime component. The external algorithmic sequence $\tau_1, \dots, \tau_{14}$ is not a path in physical time t ; it is the metatheoretic compilation trace. A physical observable crossing from t into pre-runtime τ would require an internal path whose endpoints belong to different explanatory registers, which the temporal-cohesive typing discipline does not supply. $\square$

Definition 10.14 (Two debts). The framework carries two distinct debts that the original halt argument conflated. The quantitative debt $\Delta_n = F_n$ prices integration and never vanishes. The directive debt $\text{O}(n)$ is the set of unmet window demands — obligations of the two most recent sealed interfaces not yet discharged by the derivability closure — and it shapes the admissible field: throughout Genesis, formula-maximizing shapes were excluded by directive debt (the bare- S^3 record at Step 8 is the sealed instance), never by pricing.

Theorem 10.15 (Obligation locality and debt exhaustion). *From cumulativity, monotone derivability, and depth-two admissibility: discharge persists and $\text{O}(n + 1)$ is window-generated, so selection dynamics is a guarded recursion with directive debt as guard. Moreover, because the specifically declared Step-15 guarded-modal API is realized by its relative free completion, every demand of its window is derivable, and therefore*

$$\text{O}(16) = \emptyset :$$

Step 16 is the first debt-free stage in the sequence. The exposed “open band” at Step 16 is therefore a theorem of the framework, not an engine artifact, and at Step 16 pricing must act alone for the first time.

Theorem 10.16 (Relocated halt criterion). *The sequence halts at Step 15 if and only if*

$$\sup_{Z \in \text{open band}} \rho(Z) < \text{Bar}_{16} = \Phi_{16} \Omega_{15} = \frac{987}{610} \cdot \frac{359}{64} = \frac{354333}{39040} \approx 9.077.$$

The condition is implied by an extraction-guarded valuation — univalent internality (definable constructions over sealed entries score $\nu = 0$) plus linearly bounded opaque credit ($\nu \leq \alpha\kappa + \beta$ without binding obligation) — and fails under any valuation sustaining superlinear open-band credit (r^2 , d^2 , or telescopic inheritance on mere reference).

Proposition 10.17 (Halt status, July 2026). *The internal half of the halt — $\nu = 0$ up to canonical identification for internally derivable continuations — is derived and passed adversarial engine probing. The external half, formerly contested, is now certified, and the resolution followed exactly*

the route the contested record demanded. The adversarial probe’s seven formula-maximizing survivors (Π/Σ -weaves exploiting the r^2 transport law, high-dimension path constructors exploiting the d^2 law, axiomatic candidates exploiting telescopic inheritance) rode counting rules that the subsequently adopted act-local register refuses as credit: under receipt-bearing valuation — every counted family anchored to a paid clause role or a discharged pre-existing demand, derivable from the candidate and its sealed prefix alone — the d^2 , r^2 , and inheritance channels mint nothing, and the accounting bound $\nu \leq 4\kappa$ caps every certified open-band candidate at $\rho \leq 4$. Reading B ’s condition (“admissible only if the internality gate is derived blind from the definitions with Genesis 1–15 conservativity as regression gate”) was met: the semantic register was adopted on grounds predating the probe, and its extraction re-derives the certified trace’s operational history exactly. The finer refinement anticipated here was likewise executed: instance-level enumeration of the (S_{15}, S_{14}) window decides all 89 demand instances, $O(16) = \emptyset$ at full granularity, and the demanded-but-underdetermined falsifier — the one channel that could have restored a sixteenth step — was executed and excluded, separately on each of the four branches of the Stage-4 cone. The halt’s narrative is therefore the one the contested record held in escrow: the sequence stops not because continuation is priced beyond reach but because its last constructive act paid off every obligation. Runtime begins where debt ends. Every clause of this proposition is a statement about the sealed library and stands as such; the stronger reading, on which the laws halt at fifteen, is suspended under Remark 1.1 pending the PA-1 provenance audit.

11 Quantum Theory and Born Weights

Proposition 11.1 (No-cloning to quantum kinematics). *An efficient physical runtime forbids unrestricted cloning of live state structure; linear quantum kinematics preserves higher path novelty while preventing free duplication of ν .*

Proof sketch. Unrestricted cloning would credit a live state’s structure a second time with no clause paid for the copy. This is disallowed by the evidentiary standard rather than by any ranking: novelty counts only when anchored to paid clauses or discharged demands, so a duplicate mints nothing, and a runtime in which duplication minted would be crediting unanchored novelty. No comparison of ρ between candidates is used, and none would be admissible at a guarded stage. In type-theoretic terms, global cloning drives states toward flat h -set behavior, suppressing higher path structure. Quantum no-cloning preserves the non-flat cohesive structure needed for interference and homotopical novelty. $\square$

Regime	Structural condition	Physical reading
Classical	identity truncated to public h -set data	copyable records and deterministic public control
Quantum	live cohesive/path data remains unflattened	amplitudes, interference, entanglement, no-cloning
Measurement	application of a flat public-record modality	path-structured support becomes copyable outcome data

Definition 11.2 (Finite/tame phase-proof family). For a finite or countable public outcome type X , a finite/tame phase-proof family is a map $p : \Psi \rightarrow X$ such that each homotopy fiber

$$\Psi_x := \sum_{\psi : \Psi} (p(\psi) = x)$$

is π -finite, together with a phase labeling

$$\theta_x : \pi_0(\Psi_x) \rightarrow U(1)$$

constant on isomorphism classes. Countable families are required to have absolutely convergent phased sums. The object Ψ_x is the structured proof support for x : witnesses, paths between witnesses, and automorphism groups are retained until public flattening.

Definition 11.3 (Proof fiber and groupoid cardinality). For a preparation $p : \Psi \rightarrow X$, the proof space over $x : X$ is

$$\Psi_x := \sum_{\psi : \Psi} (p(\psi) = x).$$

Its ordinary symmetry-aware cardinality is

$$|\Psi_x| = \sum_{[w] \in \pi_0(\Psi_x)} \frac{1}{|\text{Aut}(w)|}.$$

Automorphism division identifies internally symmetric presentations as less than independent support.

Definition 11.4 (Phased homotopy cardinality). The phased homotopy cardinality, or local amplitude, of x is

$$\psi(x) := |\Psi_x|_\theta = \sum_{[w] \in \pi_0(\Psi_x)} \frac{e^{i\theta_x([w])}}{|\text{Aut}(w)|}.$$

Definition 11.5 (Conjugate phase-proof family). The conjugate family $\overline{\Psi} \rightarrow X$ has the same underlying fibers and conjugated phase labels. If $|\Psi_x|_\theta = \psi(x)$, then

$$|\overline{\Psi}_x|_{\overline{\theta}} = \overline{\psi(x)}.$$

The decohered self-pairing over outcomes is

$$\mathcal{D} := \Psi \times_X \overline{\Psi}, \quad \mathcal{D}_x \simeq \Psi_x \times \overline{\Psi}_x.$$

Theorem 11.6 (Born weight). *Let $\mathcal{D} := \Psi \times_X \overline{\Psi}$. Then*

$$w(x) = |\mathcal{D}_x|_{\theta \otimes \overline{\theta}} = \psi(x) \overline{\psi(x)} = |\psi(x)|^2,$$

and

$$\text{Pr}(x) = \frac{|\psi(x)|^2}{\sum_y |\psi(y)|^2}.$$

Proof sketch. The public scalar support for x must be phase-neutral. The constructively prime phase-neutral object built functorially from a single directed proof fiber is the conjugate self-pairing $\Psi_x \times \overline{\Psi}_x$. Phased groupoid cardinality is multiplicative over products, and the conjugate fiber has cardinality $\overline{\psi(x)}$. Normalization of public weights gives the probability rule. $\square$

Proposition 11.7 (Uniqueness of the Born square). *Among public scalars built functorially from one phase-proof fiber, every phase-blind monoidal scalar has the form*

$$|\psi(x)|^{2k} \quad (k \geq 0).$$

The least nontrivial constructively prime scalar is therefore $|\psi(x)|^2$.

Proof sketch. A monoidal scalar built from m copies of Ψ_x and n copies of $\overline{\Psi_x}$ has value $\psi(x)^m \overline{\psi(x)}^n$. A global phase shift $\psi(x) \mapsto e^{i\alpha} \psi(x)$ multiplies this by $e^{i(m-n)\alpha}$, so phase-blindness for all α forces $m = n$. Thus the scalar is $|\psi(x)|^{2k}$. For $k > 1$ the construction bundles k copies of the same proof/conjugate-proof pair; constructive primeness rejects this duplication. The first nonconstant case $k = 1$ is the Born weight. $\square$

Proposition 11.8 (Interference order law). *For coherent alternatives with amplitudes ψ_i contributing to one unresolved outcome channel,*

$$w_{\text{coh}} = \left| \sum_i \psi_i \right|^2 = \sum_i |\psi_i|^2 + \sum_{i \neq j} \psi_i \overline{\psi_j}.$$

For alternatives first separated into distinct public records,

$$w_{\text{dec}} = \sum_i |\psi_i|^2.$$

Thus interference is precisely the noncommutation of coherent summation and conjugate self-pairing.

Proof sketch. The disjoint sum of coherent proof families is formed before applying public extraction, so phased cardinalities add and the conjugate self-pairing produces cross-terms. If the measurement interface first records the alternatives separately, each branch is paired with its own conjugate before classical summation. The latter operation eliminates cross-terms because distinct public records are orthogonal channels. $\square$

Proposition 11.9 (Entanglement as proof-space nonfactorization). *For a product outcome interface $X \times Y$, separability requires a factorization*

$$\Psi_{x,y} \simeq \Psi_x^X \otimes \Psi_y^Y$$

compatible with phase labels and automorphism weights. Entanglement is failure of this factorization in the phase-proof family.

Proof sketch. A product of independent proof families yields an amplitude matrix of rank one, $\psi(x, y) = \psi_X(x) \psi_Y(y)$, and public weights factor after self-pairing. If the fiber over (x, y) has irreducible paths, phases, or automorphisms linking the two coordinates, no such tensor decomposition exists. The public record then carries nonfactorizing Born weights. $\square$

Definition 11.10 (Continuous phase-proof interface). A continuous public outcome interface is a sigma-finite measured space (X, Σ, μ) . A continuous phase-proof preparation is represented by a measurable field $x \mapsto \Psi_x$ whose local phased cardinalities form an amplitude representative

$$\psi \in L^2(X, \mu).$$

If $d\mu' = f d\mu$ with $f > 0$ almost everywhere, the half-density transformation

$$\psi' = \frac{\psi}{\sqrt{f}}$$

preserves the public measure:

$$|\psi'|^2 d\mu' = |\psi|^2 d\mu.$$

Definition 11.11 (Continuous Born event measure). For $\phi, \psi \in L^2(X, \mu)$, define the event cross-pairing

$$C_{\phi, \psi}(B) := \int_B \overline{\phi(x)} \psi(x) d\mu(x), \quad B \in \Sigma.$$

The diagonal public support measure is

$$\text{Born}_\psi(B) := C_{\psi, \psi}(B) = \int_B |\psi(x)|^2 d\mu(x).$$

For $\psi \neq 0$, the normalized probability measure is

$$\Pr_\psi(B) = \frac{\int_B |\psi(x)|^2 d\mu(x)}{\int_X |\psi(x)|^2 d\mu(x)}.$$

Proposition 11.12 (Events are integrated after fiberwise pairing). *For an event $B \in \Sigma$,*

$$\Pr_\psi(B) \propto \int_B |\psi(x)|^2 d\mu(x),$$

not

$$\left| \int_B \psi(x) d\mu(x) \right|^2.$$

Proof sketch. The public measurement interface assigns distinct $x \in B$ to orthogonal record channels. The self-pairing is therefore local in x , and event probability is the integral of local public support. The squared integral would instead treat all points of B as one coherent unresolved channel before public record separation. $\square$

Theorem 11.13 (Continuous L^2 Born completion). *Let (X, Σ, μ) be sigma-finite. Let $(\mathcal{P}_n)$ be refining finite measurable partitions of a finite-measure exhaustion, and let finite/tame coarse-grainings of a phase-proof preparation produce simple amplitudes*

$$\psi_n = \sum_i c_{n,i} \frac{1_{B_{n,i}}}{\sqrt{\mu(B_{n,i})}} \in L^2(X, \mu).$$

If $\psi_n \rightarrow \psi$ in $L^2(X, \mu)$, then

$$|\psi_n|^2 d\mu \rightarrow |\psi|^2 d\mu$$

in total variation. If $\psi \neq 0$, the normalized finite Born measures converge to $\Pr_\psi$. If $\phi_n \rightarrow \phi$ and $\psi_n \rightarrow \psi$ in L^2 , then

$$\lim_n \int_X \overline{\phi_n(x)} \psi_n(x) d\mu(x) = \int_X \overline{\phi(x)} \psi(x) d\mu(x).$$

Proof sketch. The normalized indicators $1_{B_{n,i}}/\sqrt{\mu(B_{n,i})}$ are orthonormal, so finite Born weights $|c_{n,i}|^2/\sum_j |c_{n,j}|^2$ are exactly the event measures of the embedded simple functions. Inner-product convergence follows from Cauchy-Schwarz. Diagonal measure convergence follows from

$$||a|^2 - |b|^2| \leq |a - b|(|a| + |b|)$$

and hence

$$|||\psi_n|^2 - |\psi|^2||_{L^1} \leq \|\psi_n - \psi\|_{L^2} (\|\psi_n\|_{L^2} + \|\psi\|_{L^2}) \rightarrow 0.$$

Normalization is continuous away from zero norm. $\square$

Proposition 11.14 (Spectral-measure Born form). *For a self-adjoint observable A with projection-valued measure E_A ,*

$$\Pr_{\psi}(A \in B) = \langle \psi, E_A(B) \psi \rangle.$$

In a spectral direct-integral representation

$$\mathcal{H} \simeq \int_X^{\oplus} K_x d\mu(x),$$

this is

$$\Pr_{\psi}(B) = \int_B \|\psi(x)\|_{K_x}^2 d\mu(x).$$

The scalar $L^2(X, \mu)$ case is $K_x \simeq \mathbb{C}$.

Proof sketch. The projection $E_A(B)$ is multiplication by 1_B in the spectral representation. The inner product therefore evaluates the fiberwise self-pairing over B . The fiber K_x is the completed local amplitude space of proof-support channels over outcome x , with finite phased groupoid cardinality as its tame one-dimensional shadow. $\square$

Proposition 11.15 (Measurement as public flattening). *Measurement is the production of a flat public record:*

$$\text{meas} : Q \rightarrow \flat X,$$

where Q is a live cohesive/path-structured quantum state and $\flat X$ is the copyable record interface of outcomes.

Proof sketch. Before measurement, the proof family over outcomes carries phase, automorphisms, and higher path dependence. Applying the flat public-record modality discards non-record path structure and exports a stable outcome type whose elements can be copied and compared. The Born weight supplies the phase-neutral support assigned to that public record. $\square$

Definition 11.16 (Measurement chain). A measurement chain is a finite sequence

$$Q_0 \rightarrow Q_1 \rightarrow \cdots \rightarrow Q_n \rightarrow C,$$

where Q_0 is the measured quantum system, $Q_1, \dots, Q_n$ are physical apparatus, amplification, sensory, or neural links, and C is an active context. A physical link Q_i is absorbable when replacing Q_0 by $Q_0 \times Q_i$ preserves the same Born-weighted public record statistics.

Proposition 11.17 (Observer-terminal flattening). *If every Q_i in a measurement chain is absorbable, then no physical subchain supplies a distinguished terminal flat record. The terminal interface is an active context C that converts the branch-weighted record family into an integrated public/private update.*

Proof sketch. Absorbability means that moving the cut along the physical chain leaves the quantum description invariant up to the same outcome weights. Hence the detector, amplifier, display, sensory organ, and neural relay do not by themselves select a privileged endpoint. A terminal record must be a flat update of an active context: it is the point at which a branch-weighted family is integrated into a history-bearing perspective and made available for public comparison or later report. $\square$

Proposition 11.18 (Exchange statistics from proof-space symmetry). *Bosonic and fermionic statistics are determined by the action of permutation symmetries on phase-proof families: symmetric action preserves phased cardinality, while sign-twisted action reverses phase under transposition.*

Proof sketch. For indistinguishable n -particle proof families, the symmetric group S_n acts on the fiber. Taking homotopy orbits with trivial character yields symmetric support and therefore bosonic accumulation. Taking the sign character on transpositions twists the phase labeling by -1 , forcing antisymmetric support and exclusion of identical diagonal states. The distinction is therefore a representation-theoretic property of the proof-space symmetry exported to the Hilbert-functional shell. $\square$

12 Spacetime, Gravity, and Gauge Structure

Theorem 12.1 (Four-dimensional stage). *The efficient public spacetime stage is four-dimensional: quaternionic associative control avoids non-native coherence overhead, and only in dimension four does the Hodge star close on two-forms,*

$$\star : \Omega^2(M^4) \rightarrow \Omega^2(M^4).$$

Proof sketch. Quaternionic geometry supplies associative S^3 -control inside the depth-two window. Octonionic patching introduces extra coherence payload.

In four dimensions the metric and Hodge infrastructure already paid for at the metric shell yields

$$\Omega^2 = \Omega_+^2 \oplus \Omega_-^2$$

with no change of form degree. Lower dimensions lack this closure. Higher dimensions lose it while increasing patching cost. $\square$

Proposition 12.2 (Native associativity ceiling). *The efficient primitive control algebra for public spacetime reaches its native ceiling at quaternionic S^3 -control.*

Proof sketch. Pure differential geometry permits $GL(n, \mathbb{R})$ for arbitrary n , so dimension is not capped by manifold syntax alone. PEN evaluates primitive synthesis cost. Quaternionic control is associative and remains inside the depth-two coherence window. Octonionic control introduces non-associative patching and explicit A_∞ -style coherence payload that cannot be discharged natively at depth two. The last high-yield associative stage is the quaternionic Hopf pattern $S^3 \hookrightarrow S^7 \rightarrow S^4$. $\square$

Proposition 12.3 (Hodge endomorphism spike). *For a d -dimensional metric stage,*

$$\star : \Omega^2(M^d) \rightarrow \Omega^{d-2}(M^d).$$

The only dimension in which $\star$ is an endomorphism of two-forms is $d = 4$, where it yields self-dual and anti-self-dual projectors at no new primitive metric cost.

Proof sketch. The metric/frame shell has already paid for $\star$. In $d = 4$, $\star^2$ acts on Ω^2 and splits curvature-relevant two-forms into $\Omega_+^2 \oplus \Omega_-^2$. This creates instanton, chirality-sensitive, and curvature-factorization elimination channels, contributing a concentrated ν_C gain. In other dimensions the Hodge star changes degree on two-forms, so the same endomorphic reuse is absent. $\square$

Proposition 12.4 (Lorentzian modal separation). *Lorentzian signature preserves the distinction between temporal flow and spatial cohesion.*

Proof sketch. In Euclidean signature, isotropy can rotate a designated temporal tangent direction into a spatial direction, and univalence turns coherent geometric paths into type equivalences, collapsing $\bigcirc$ toward cohesive modality. Lorentzian signature partitions tangent vectors into timelike, spacelike, and null classes. The null cone prevents a causal-class-preserving path from identifying flow with spatial cohesion. $\square$

Corollary 12.5 (No Lorentz-violation regime). *Any fundamental regime that removes Lorentzian causal-class separation reopens the univalent modal-collapse channel between temporal flow and spatial cohesion.*

Proof sketch. The Step 15 temporal modality is represented by tangent infinitesimal update, while cohesion governs spatial/shape structure. If the metric signature permits continuous symmetry paths from temporal to spatial tangent generators, univalence identifies the corresponding modal roles. Lorentzian null-cone separation is the minimal geometric barrier preventing that identification. $\square$

Proposition 12.6 (Einstein–Hilbert selection, and where its premises live). *Suppose the sealed library supplies a four-dimensional public geometry whose field equation is metric-only, local, diffeomorphism-covariant, divergence-free, and second-order in the metric. Then that equation is Einstein-plus- Λ , and the corresponding action is*

$$S_{\text{EH}} = \int (R - 2\Lambda) \sqrt{-g} d^4x.$$

The selection is a uniqueness theorem under those five premises [29], not a comparison of specification costs. Deriving the premises — metric-only and second-order in particular — from the sealed record rather than assuming them is a registered open work item of the external falsifier campaign.

Proof sketch. Lovelock’s theorem in $d = 4$. What the Two Laws are required to supply is the premise set, not a ranking over candidate actions: a guarded stage admits no comparison of specification cost, so the earlier reading — on which R was selected because higher curvature terms carry higher κ — is withdrawn as a selector and retained only as diagnostic testimony. That testimony is the table below, which records what each candidate would add to the specification and does no selective work; the actual exclusions in the table are effected by the second-order premise, since each listed alternative yields fourth-order field equations. Conditional predictions upon closure of the premises: no fundamental scalar or vector gravitational polarization, and no intrinsic vacuum dispersion of gravitational waves. $\square$

Candidate action term	Additional MBTT burden (diagnostic only)	Disposition under the premises
R	single curvature scalar from paid metric/connection data	core term of the unique second-order action
R^2	extra multiplication and higher-order scalar nesting	excluded: fourth-order field equations
$R_{\mu\nu}R^{\mu\nu}$	tensor contraction plus quadratic nesting	excluded: fourth-order field equations
$\nabla R \nabla R$	derivative tower above curvature scalar	excluded: fourth-order field equations

Theorem 12.7 (Standard Model carrier-bearing local gauge alphabet). *Fix the complex phase selected at the complex rung, represented by a unit imaginary element i , and a compatible division-algebra chain*

$$\mathbb{C}_i := \text{span}_{\mathbb{R}}\{1, i\} \subset \mathbb{H} \subset \mathbb{O}.$$

Let

$$V_1 := \mathbb{C}_i, \quad V_2 := \mathbb{H} \text{ regarded as a right } \mathbb{C}_i\text{-vector space}, \quad V_3 := \mathbb{C}_i^\perp \subset \mathbb{O},$$

where V_3 carries the complex structure $J_i x := ix$. Define

$$\begin{aligned} K_1 &:= \{z \in \mathbb{C}_i : |z| = 1\}, & \rho_1(z)w &:= zw, \\ K_2 &:= \{q \in \mathbb{H} : |q| = 1\}, & \rho_2(q)x &:= qx, \\ K_3 &:= \{g \in \text{Aut}(\mathbb{O}) : g(i) = i\}, & \rho_3(g)x &:= g(x) \quad (x \in V_3). \end{aligned}$$

The efficient selected API is the represented symmetry package

$$\mathfrak{A}_0 := ((K_1, V_1, \rho_1), (K_2, V_2, \rho_2), (K_3, V_3, \rho_3)).$$

There are equivalences of represented groups

$$(K_1, V_1, \rho_1) \simeq (U(1), \mathbb{C}, \text{std}), \quad (K_2, V_2, \rho_2) \simeq (SU(2), \mathbb{C}^2, \mathbf{2}), \quad (K_3, V_3, \rho_3) \simeq (SU(3), \mathbb{C}^3, \mathbf{3}),$$

with $\mathbf{3}$ replaced by $\bar{\mathbf{3}}$ if the preserved complex orientation is reversed. Consequently the group shadow of the selected API is

$$G_0 = SU(3) \times SU(2) \times U(1).$$

The product ordering is conventional. The $U(1)$ entry is typed as a scalar-action clause: in a later complex arena W it is realized by $z \mapsto z \text{id}_W$, not by requiring a distinguished summand $\mathbb{C} \hookrightarrow W$. This theorem concerns the three-factor local alphabet and does not by itself choose its global form.

Proof sketch. Hopf and division-algebra structure supply the available high-yield bundle symmetries. On $V_1 = \mathbb{C}_i$, norm-one complex scalars act faithfully by multiplication, giving the standard $U(1)$ -action. On $V_2 = \mathbb{H}$, right multiplication by i supplies the complex structure. Associativity of $\mathbb{H}$ gives $L_q R_i = R_i L_q$, so left multiplication by every unit quaternion is complex-linear and norm-preserving. Thus the tautological unit quaternion action is $\text{Sp}(1) \cong SU(2)$ on $V_2 \cong \mathbb{C}^2$.

For the octonionic rung, every $g \in K_3$ fixes 1 and i , preserves the norm inner product, and therefore preserves $V_3 = \mathbb{C}_i^\perp$. Moreover

$$g(J_i x) = g(ix) = g(i)g(x) = i g(x) = J_i g(x),$$

so the restricted action is complex-linear. The stabilizer of a unit imaginary octonion in $\text{Aut}(\mathbb{O}) \cong G_2$ is $SU(3)$, and its action on the six-real-dimensional complement V_3 is the defining complex triplet. Larger unification groups add definitional and breaking cost without corresponding primitive topological gain at the active stage. $\square$

Lemma 12.8 (Carrier retention in the local alphabet API). *For each division-algebra rung, the selected factor enters the public API with its tautological carrier and action. Equivalently, the selected object is $\mathfrak{A}_0$, not merely the abstract group $K_1 \times K_2 \times K_3$. The carrier is part of the selection, not a subsequent representation choice. The complex carrier is retained as the scalar object of the realization, whereas the quaternionic and octonionic carriers are retained as the defining $\mathbb{C}^2$ and $\mathbb{C}^3$ module clauses.*

Proof. In the normalized specification of the local-alphabet candidate, the symmetry clauses are the typed action maps

$$\rho_r : K_r \longrightarrow \text{U}(V_r), \quad r = 1, 2, 3,$$

with K_3 formed as the stabilizer of i in $\text{Aut}(\mathbb{O})$ and then restricted to V_3 . Hence the formation and action clauses for each K_r contain V_r as a type dependency. A specification containing K_r and ρ_r but omitting the referenced carrier is not closed under well-typedness and therefore is not a minimal complete API specification.

Passing from (K_r, V_r, ρ_r) to the abstract isomorphism class of K_r is instead an application of a forgetful map from represented groups to groups. That map is not an equivalence of the selected API: it discards the action data used to define and realize the factor. Equivalence-invariance may change bases or transport the whole package along an isomorphism, but it cannot replace the package by its strictly weaker group shadow. Relative free sealing therefore realizes and exports (K_r, V_r, ρ_r) for every rung. For $r = 1$, realization in a complex arena is scalar extension $z \mapsto z \text{id}_W$; it introduces no carrier summand. For $r = 2, 3$, the defining module maps are part of the exact-realization data. Thus the carrier travels with the group into the API, with its API role preserved. $\square$

Rung	Efficient symmetry	Role
$\mathbb{C}$	$U(1)$	phase symmetry for complex amplitudes
$\mathbb{H}$	$SU(2)$	smallest rich non-abelian internal orientation language
$\mathbb{O}$	$SU(3)$	color symmetry from stabilized octonionic structure preserving complex phase

Definition 12.9 (Exact realizations of the carrier-bearing alphabet). Let $G = SU(3) \times SU(2)$ be the group shadow of the two non-scalar rungs of $\mathfrak{A}_0$, and regard the retained carriers as the G -representations

$$V_3 = (\mathbf{3}, \mathbf{1}), \quad V_2 = (\mathbf{1}, \mathbf{2}).$$

An *exact realization of $\mathfrak{A}_0$ in a complex arena* is a triple (W, ι_3, ι_2) in which W is a finite-dimensional complex G -representation whose G -action commutes with the ambient scalars — the slot in which the selected central phase acts — and

$$\iota_3 : V_3 \hookrightarrow W, \quad \iota_2 : V_2 \hookrightarrow W$$

are G -equivariant embeddings exhibiting the two retained carriers inside W . Morphisms are the G -equivariant linear maps $f : W \rightarrow W'$ that commute with both structure maps, so that $f \circ \iota_3 = \iota'_3$ and $f \circ \iota_2 = \iota'_2$. Write $\mathcal{E}$ for the resulting category.

By construction $\mathcal{E}$ is the category of realizations *under* the selected carriers. It is the shape free sealing refers to when it requires the sealed layer to be initial among exact realizations of that selected API, and the initiality below is taken relative to $\mathfrak{A}_0$ — never relative to a weaker requirement fixed in advance, which is the confusion the Selective Law’s standing warning names.

Theorem 12.10 (Free sealing selects the five-dimensional arena). (i) $(V_3 \oplus V_2, \text{can}_3, \text{can}_2)$ is an initial object of $\mathcal{E}$.

(ii) In every object of $\mathcal{E}$ the two carrier images meet trivially, so $\dim_{\mathbb{C}} W \geq 5$.

Consequently the free seal of $\mathfrak{A}_0$ in a complex arena is

$$V_3 \oplus V_2 \cong \mathbb{C}^5,$$

and a Hopf shell $\mathcal{H}_n$ whose defining arena $\mathbb{C}^{n+1}$ is that free seal has $n + 1 = 5$.

Proof. (i) Given (W, ι_3, ι_2) , the assignment $f(x, y) := \iota_3(x) + \iota_2(y)$ is linear, G -equivariant, and satisfies both triangle conditions; any g satisfying them agrees with f on $V_3 \oplus 0$ and on $0 \oplus V_2$, which together span $V_3 \oplus V_2$. So the map exists and is unique. Equivalently: there are no maps between the two carriers to be respected, so the initial object of $\mathcal{E}$ is their coproduct in $\text{Rep}(G)$, which is the direct sum.

(ii) V_3 and V_2 are non-isomorphic irreducible G -representations, so $\text{im } \iota_3 \cap \text{im } \iota_2$ is a common subrepresentation of two non-isomorphic irreducibles and therefore zero. Hence $\dim_{\mathbb{C}} W \geq \dim V_3 + \dim V_2 = 5$.

For the consequence, free sealing states that once an API is selected the new layer is its least term-model completion, containing no unforced generators or equations and initial among exact realizations of that selected API. By the Carrier-retention lemma the selected API is $\mathfrak{A}_0$, carriers included; by (i) the initial exact realization is $V_3 \oplus V_2$. The complex rung contributes no summand, since by the Carrier-bearing alphabet theorem its entry is typed as the scalar-action clause $z \mapsto z \text{ id}_W$; so $\mathbb{C}^5$ is the whole arena rather than a subspace of one. Had the complex rung instead required its own summand, the coproduct would be $\mathbb{C}^6$ and the selected shell would be $S^1 \rightarrow S^{11} \rightarrow \mathbb{C}P^5$; this is the sharpest quantitative fork in the argument. $\square$

Corollary 12.11 (The three exclusions are carrier-multiplicity facts). *Let $\mathbb{C}^{n+1}$ be the defining arena of $\mathcal{H}_n$. Exactly one of three things holds. For $n \leq 3$, $\mathbb{C}^{n+1}$ is not an exact realization at all, by (ii) of the preceding theorem. For $n = 4$ it is the free seal, and $\mathbb{C}^5 \cong V_3 \oplus V_2$ is the unique such decomposition up to conjugating the triplet. For $n \geq 5$ it is either not an exact realization — one of ι_3, ι_2 does not exist — or an exact realization that is not initial, since initial objects are unique up to isomorphism and $\dim_{\mathbb{C}} \mathbb{C}^{n+1} > 5$. In no case other than $n = 4$ is the arena the free seal.*

Proof. The trichotomy is exhaustive by dimension. The three prohibitions previously carried as a declared clause are the three ways the second and third cases arise, and each is a multiplicity count rather than a stipulation:

Arena	n	V_3	V_2	Status
$(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$	4	1	1	free seal
$(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{1})$	5	1	1	exact, not initial: spectator singlet
$(\mathbf{3}, \mathbf{2})$	5	0	0	not exact: irreducible, neither ι exists
$(\mathbf{1}, \mathbf{1}) \oplus (\mathbf{3}, \mathbf{2})$	6	0	0	not exact
$(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{4})$	6	1	0	not exact: ι_2 does not exist
$(\mathbf{1}, \mathbf{2}) \oplus (\mathbf{3}, \mathbf{2})$	7	0	1	not exact: ι_3 does not exist

The composite fails as a G -representation, not merely on a count: $(\mathbf{3}, \mathbf{2})$ is irreducible, so it contains no copy of either carrier as a subrepresentation. Restricting it factorwise shows what a composite would have to supply,

$$(\mathbf{3} \otimes \mathbf{2}) \downarrow_{SU(3)} = \mathbf{3} \oplus \mathbf{3}, \quad (\mathbf{3} \otimes \mathbf{2}) \downarrow_{SU(2)} = \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{2},$$

five carrier copies where the API presents two. The higher spin fails for the same structural reason: $\mathbf{4} \cong \text{Sym}^3(\mathbb{C}^2)$ contains no copy of V_2 , so ι_2 does not exist and $(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{4})$ is not an object of $\mathcal{E}$. $\square$

Remark 12.12 (Why this is admissible at a guarded stage, and what it buys). The selection above is made by a universal property. Initiality is preserved under equivalence, compares no two candidates on a scale, and returns no ranking; it is therefore not a bar, an efficiency comparison, a parsimony order, or a presentation key, and the two-register acceptance clause does not exclude it from guarded acceptance. This is the hinge of the derivation and it should be stated rather than assumed.

A dimension argument reaches the same arena and is *not* available. Enumerating the representations of $SU(3) \times SU(2)$ faithful on both factors, dimensions 1–4 admit none and dimension 5 admits exactly two, $(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$ and its conjugate — the same pair. But a minimal faithful dimension is a value quantity, so that route is closed at a guarded stage by the same clause that closes a monotone cost $\kappa(\mathcal{H}_n)$. What the Carrier-retention lemma buys is therefore not the value $\mathbb{C}^5$, which is robust, but the entitlement to assert it: it converts a forbidden minimization into a universal property.

Falsification is correspondingly narrow. Exhibit an object of $\mathcal{E}$ admitting two distinct morphisms from $V_3 \oplus V_2$, or none, and (i) fails. Show that the local alphabet selects the groups *without* their carriers in a way that survives the guarded-stage prohibition, and the Carrier-retention lemma fails, whereupon the arena is not fixed by this route. Rule initiality itself a value quantity under the Law’s intended reading, and the derivation lapses. In each case the census does not fall back to a ranking; it returns a certified cone.

Corollary 12.13 (Empty-desert constraint). *Gauge enlargement beyond G_{SM} is rejected because its extra generators are unforced: they are not the image of any carrier under the selected action maps, and no live obligation is discharged by them.*

Proof sketch. A grand-unified group gauges generators outside the image of ρ_1, ρ_2, ρ_3 — for $SU(5)$ over $S(U(3) \times U(2))$, the twelve off-block generators — and must then add breaking structure returning to the observed product. No clause of $\mathfrak{A}_0$ demands that content and no live obligation is discharged by it, so free sealing, which admits no unforced generators, does not seal it. The rejection is a universal property and not a cost comparison: gauge selection occurs at a guarded stage, where no bar, efficiency comparison or parsimony order participates in acceptance. The engine-register observation that enlargement raises κ is retained as diagnostic testimony and does no selective work. $\square$

Corollary 12.14 (Proton stability). *If baryon-violating decay channels require unselected grand-unified mediator structure, then proton decay is absent in the sealed Standard Model alphabet.*

Proof sketch. Grand-unified proton decay requires additional gauge bosons or comparable mediator geometry beyond G_{SM} . The empty-desert constraint rejects precisely that package, on the ground that its extra generators are unforced rather than that they are expensive. With no selected mediator interface, the corresponding baryon-violating channel is not part of the primitive public physics. $\square$

Proposition 12.15 (Weak mixing at the structural normalization point). *Conditional on the registered sealing-order premise for the Dirac extension, the sequentially sealed matter content of one Standard Model generation determines*

$$\sin^2 \theta_W = \frac{\text{Tr } T_3^2}{\text{Tr } Q^2} = \frac{2}{16/3} = \frac{3}{8}$$

at the equal-trace normalization point, with no grand-unified embedding: the traces $\text{Tr } T_3^2 = 2$, $\text{Tr } Y^2 = 10/3$ (whence the 5/3 hypercharge normalization factor), and $\text{Tr } Q^2 = 16/3$ are properties of the selected representation content itself.

Proof sketch. The computation is a verified trace identity over the derived matter content; historically the same value was obtained only by embedding the generation in $SU(5)$, but the embedding is not used — anomaly cancellation over the selected alphabet fixes the hypercharges up to one qualification recorded below, and the hypercharges fix the traces. The two declared obligations have since moved (July 2026 placement audit; registered record in the constants program). The equal-trace bridge principle now has a registered candidate in pair form: the two factors sharing the photon enter public bookkeeping at the matter-interface normalization, an unequal weighting being a declined dial of the tier-three profile; no triple coincidence with the strong coupling is claimed, and none is observed. The matching placement was adjudicated by a frozen two-loop audit under pre-registered kill conditions: the Planck-scale placement is dead — it forces $\sin^2 \theta_W(M_Z) \approx 0.153$ against the measured 0.2312, a worse failure than minimal $SU(5)$'s ≈ 0.21 , which belongs to the triple-unification demand this framework never makes — and under measured running the equal-trace crossing of the pair sits at $\mu_X \approx 1.1 \times 10^{13}$ GeV. The claim accordingly refles to its normalization-only form: the traces, the 5/3, and the 3/8 at the crossing are structural, and the crossing's existence in the desert is verified, but the framework does not compute the low-energy value 0.2312. The remaining open item is a blind structural derivation of the matching scale, whose target is now public and therefore retrodiction-grade on the scale alone; it is a bridge-tier claim, not a core one, and its value half stays closed until that derivation exists.

The qualification (adversarial review, July 2026) is load-bearing. With right-handed neutrinos included simultaneously, $B - L$ is anomaly-free and the equations admit $Y_\epsilon = Y + \epsilon(B - L)$, with $\epsilon \sim -1 - \epsilon$; anomaly and Yukawa equations alone therefore do not select $\epsilon = 0$. The frozen exact enumeration verifies the standard six-slot ray and this simultaneous family. Once those six slots are

sealed, however, adding the all-left-handed Dirac slot ν^c makes the neutrino Yukawa equation, mixed gravitational anomaly, and cubic anomaly reduce independently to Y_{ν^c} , Y_{ν^c} , and $Y_{\nu^c}^3$. They force $Y_{\nu^c} = 0$, hence the physical charge $q_\nu = -Y_{\nu^c} = 0$. Thus the standard traces and $3/8$ normalization are exact relative to sequential sealing, not consequences of the simultaneous system. The remaining conditionality is precisely that registered sealing-order premise. $\square$

Remark 12.16 (Independent recomputation of the crossing, and what $3/8$ does not carry). The equal-trace crossing has been recomputed from measured M_Z anchors ($\alpha^{-1}(M_Z)_{\overline{\text{MS}}} = 127.951$, $\sin^2 \theta_W(M_Z)_{\overline{\text{MS}}} = 0.23122$, $\alpha_s(M_Z) = 0.1179$) at 40-digit precision. At one loop with $(b_1, b_2) = (41/10, -19/6)$,

$$\mu_X = 1.03089 \times 10^{13} \text{ GeV}, \quad g_X = 0.544328;$$

at two loops, including the full Standard Model B_{ij} matrix and the top Yukawa,

$$\mu_X = 1.08939 \times 10^{13} \text{ GeV}, \quad g_X = 0.545701, \quad g_3(\mu_X) = 0.580413.$$

This confirms the $\mu_X \approx 1.1 \times 10^{13}$ GeV of the placement audit to better than one per cent. The figure is quoted to three digits but is good to roughly $\pm 10\%$ against plausible variation in $\alpha_s(M_Z)$, $\sin^2 \theta_W(M_Z)$ and y_t ; what is robust is the order — about 10^{13} GeV, inside the desert — not the digits.

A second observation sharpens the open item already recorded above. Since $g_1^2 = \frac{5}{3}g_Y^2$ by definition, at *any* point where $g_1 = g_2$ one has $g_Y^2 = \frac{3}{5}g_2^2$ and hence

$$\sin^2 \theta_W = \frac{g_Y^2}{g_2^2 + g_Y^2} = \frac{3/5}{1 + 3/5} = \frac{3}{8}$$

identically. Verified numerically as 0.375 to 25 digits under every placement tried, including deliberately wrong ones spanning μ_X from 8.5×10^7 to 4.1×10^{13} GeV. The value $3/8$ is therefore a property of the normalization convention and carries *no* information about where the crossing sits: it cannot be used as evidence for any particular matching scale, and a derivation of μ_X must come from somewhere else entirely. This is consistent with — and is the precise reason for — the refiling of the claim to its normalization-only form.

Theorem 12.17 (Charge lattice and exact $\mathbb{Z}_6$ kernel). *Use primitive integer hypercharge $y = 6Y$. For one all-left-handed generation and the Yukawa carrier,*

$$(Q, u^c, d^c, L, e^c, \nu^c, H) = (1, -4, 2, -3, 6, 0, 3).$$

Let $G_0 = SU(3) \times SU(2) \times U(1)_y$, and let ρ be its total action on these slots. Then

$$\ker \rho = \left\{ (z^2 I_3, z^3 I_2, z) : z^6 = 1 \right\} = \left\langle (\omega_3, -I_2, e^{i\pi/3}) \right\rangle \cong \mathbb{Z}_6,$$

and therefore

$$G_0/\mathbb{Z}_6 \cong \text{Im } \rho.$$

An irrep with color triality t , weak parity d , and integer hypercharge y descends to this faithful quotient exactly when

$$2t + 3d + y \equiv 0 \pmod{6}.$$

Proof sketch. If $(A, B, z) \in \ker \rho$, the e^c slot gives $z^6 = 1$, the L slot gives $B = z^3 I_2$, and the d^c slot gives $A = z^2 I_3$. Conversely these six elements act trivially on every displayed slot. The descent congruence is the phase condition for the generator. The first-isomorphism map $[g] \mapsto \rho(g)$ is then a group isomorphism, which the structure-identity principle turns into equality of the quotient and effective-action presentations. $\square$

Proposition 12.18 (Physical global-form boundary). *The sealed point-field representation proves the exact kernel and the faithful local action above; it does not by itself select the complete physical global gauge group. Without an Effective Gauge Presentation principle, the same local fields permit*

$$G_0/\Gamma, \quad \Gamma \in \{1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6\}.$$

The full quotient $G_{\text{phys}} = G_0/\mathbb{Z}_6$ follows only if transformations identical on every complete physical probe are presentation-identical and the admitted line, bundle, and topological probes contain no further detector of the residual kernel. Univalence identifies $G_0/\ker \rho$ with $\text{Im } \rho$; it does not identify non-isomorphic global groups merely because their local point fields agree [30].

Definition 12.19 (Hopf shell candidates and the frozen obligation profile). Write $\mathcal{H}_n$ for the complex Hopf shell

$$S^1 \longrightarrow S^{2n+1} \longrightarrow \mathbb{C}P^n, \quad n \geq 1,$$

and let $G(\mathcal{H}_n)$ be its *fibration-preserving* group: the subgroup of the isometry group of the round S^{2n+1} that commutes with the free $U(1)$ fiber action and descends to the base. This group is $U(n+1) = O(2n+2) \cap GL(n+1, \mathbb{C})$, not $O(2n+2)$. The obligation profile is

- O1 a primitive central complex phase (a freely acting $U(1)$);
- O2 an $SU(2)$ action;
- O3 an $SU(3)$ action;
- O4 a neutral chiral sector;
- O5 three-generation flavor capacity;
- O6 four-dimensional public descent;
- O7 no additional fundamental gauge factor beyond the selected alphabet;
- O8 one common spectral/public-record extractor.

The profile is stated entirely in structural terms and mentions no measured constant. It is frozen before any candidate is scored; the census below is therefore verdict-blind in the sense of the external falsifier constitution.

Theorem 12.20 (Hopf shell census). *Score $\mathcal{H}_n$ against O1–O3 and O7, which are decidable by representation theory on $\mathbb{C}^{n+1}$. Then:*

- (i) $n = 1, 2, 3$ fail. For $n \leq 2$ a faithful **3** and a faithful **2** require $3 + 2 = 5 > n + 1$ dimensions. For $n = 3$ the $SU(3)$ irreducibles of dimension ≤ 4 are **1**, **3**, **$\bar{3}$** , so a faithful **3** forces $\mathbb{C}^4 \cong \mathbf{3} \oplus \mathbf{1}$, whose commutant is abelian by Schur and therefore cannot contain the perfect group $SU(2)$.
- (ii) $n = 4$ discharges O1–O3 and O7, and $\mathbb{C}^5 \cong \mathbf{3} \oplus \mathbf{2}$ is the unique decomposition carrying each defining representation exactly once.
- (iii) Under free sealing the discharging set is $\{4\}$. This is now a consequence rather than a clause: by the Free-sealing arena theorem the free seal of $\mathfrak{A}_0$ in a complex arena is $\mathbb{C}^5$, and by its corollary every other $\mathbb{C}^{n+1}$ either fails to admit both carrier embeddings or admits them without being initial. The three prohibitions this step formerly declared — no composite **3** $\otimes$ **2** summand, no higher $SU(2)$ spin, no spectator singlet — are the three ways that failure occurs, and are computed there rather than stipulated here.

Hence

$$n_\star = 4, \quad S^1 \longrightarrow S^9 \longrightarrow \mathbb{C}P^4, \quad G(\mathcal{H}_4) = U(5),$$

with minimal alphabet-carrying subgroup $S(U(3) \times U(2))$, the stabilizer of the splitting $\mathbb{C}^5 = \mathbf{3} \oplus \mathbf{2}$ of unit total determinant.

Proof sketch. A rank screen gives $n \geq 3$ since $\text{rank } U(n+1) = n+1$ and $\text{rank } SU(3) \times SU(2) \times U(1) = 4$. The obstruction at $n = 3$ is the Schur argument in (i): in a subgroup $SU(3) \times SU(2) < U(4)$ the two factors commute, so $SU(2)$ lies in the commutant of $SU(3)$, which for $\mathbf{3} \oplus \mathbf{1}$ is $U(1) \times U(1)$. At $n = 4$ the two surviving configurations are complex conjugates of one another. Clause (iii) is discharged by the Free-sealing arena theorem and its corollary, where higher n is excluded because it forces a composite $\mathbf{3} \otimes \mathbf{2}$, a higher $SU(2)$ spin, or a spectator summand — the first two destroying exactness and the third destroying initiality, each being an unforced generator. Were free sealing unavailable at this interface the discharging set would be larger and the census would return a certified cone of lawful shells; it would *not* return a minimizer, since a monotone cost $\kappa(\mathcal{H}_n)$ is a value quantity and no value quantity legislates at a guarded stage. Enumeration is mechanical over multisets of outer tensor products with total dimension $n + 1$. $\square$

Corollary 12.21 (Second route to the exact kernel). *The map*

$$\varphi : SU(3) \times SU(2) \times U(1) \longrightarrow S(U(3) \times U(2)), \quad \varphi(g, h, z) = (z^{-2}g, z^3h),$$

is a surjective homomorphism with

$$\ker \varphi = \left\{ (z^2 I_3, z^3 I_2, z) : z^6 = 1 \right\} = \left\langle (\omega_3, -I_2, e^{i\pi/3}) \right\rangle \cong \mathbb{Z}_6,$$

so $S(U(3) \times U(2)) \cong G_0/\mathbb{Z}_6$. This is not merely isomorphic to the kernel of the sealed point-field representation: it is the same set of triples, with the same generator $(\omega_3, -I_2, e^{i\pi/3})$.

Proof sketch. $\det \varphi(g, h, z) = z^{-6} \det g \cdot z^6 \det h = 1$, so φ lands in $S(U(3) \times U(2))$. For surjectivity, given $\text{diag}(A, B)$ with $\det A \det B = 1$, choose z with $z^6 = \det B$ (equivalently $z^{-6} = \det A$) and set $g = z^2 A$, $h = z^{-3} B$; then $\det g = z^6 \det A = 1$, $\det h = z^{-6} \det B = 1$, and $\varphi(g, h, z) = (A, B)$. The kernel conditions $z^{-2}g = I_3$ and $z^3h = I_2$ give $g = z^2 I_3$ and $h = z^{-3} I_2 = z^3 I_2$, with $\det g = \det h = z^6 = 1$. Comparison with the kernel computed from the sealed slots $(Q, u^c, d^c, L, e^c, \nu^c, H)$ is then elementwise. The group-theoretic identification of $S(U(3) \times U(2))$ with $G_0/\mathbb{Z}_6$ is standard [31]. $\square$

Remark 12.22 (Bearing on the global-form boundary). The two derivations of $\mathbb{Z}_6$ are independent: one is sequential representation sealing over the matter slots, the other is shell minimality over a candidate family that never mentions matter content. Their agreement is a concordance of endpoints, not a new discovery of the kernel — the identification of $S(U(3) \times U(2))$ with $G_0/\mathbb{Z}_6$ is textbook. What it does add is a structural reason to prefer $\Gamma = \mathbb{Z}_6$ in the Physical global-form boundary proposition: on $\mathcal{H}_4$ the faithful quotient is not one choice among four, it is the stabilizer subgroup that the census returns. This does not close the boundary — the probe policy is still the deciding question — but it moves $\Gamma = \mathbb{Z}_6$ from “permitted” to “geometrically distinguished.”

Proposition 12.23 (Induced normalization, with its two premises declared). *Inside $\mathfrak{su}(5)$ the commutant of $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ is one-dimensional, so the census fixes the direction of the hypercharge generator but not its scale. Writing the traceless commutant direction as $Y(y) = \text{diag}(2y, 2y, 2y, -3y, -3y)$ on the $\mathbf{5}$,*

$$\text{Tr } Y^2 = 30y^2, \quad \text{Tr } T_3^2 = \frac{1}{2}, \quad \text{Tr}(T_3 Y) = 0,$$

so that $\kappa(y) = \text{Tr } Y^2 / \text{Tr } T_3^2 = 60y^2$ and $\sin^2 \theta_W(y) = 1/(1+60y^2)$. Obligation O4 — a colour-singlet neutral chiral state, $q_\nu = \frac{1}{2} - 3y = 0$ — forces $y = \frac{1}{6}$, whence

$$\kappa = \frac{5}{3}, \quad \sin^2 \theta_W = \frac{3}{8}, \quad \text{Tr } T_3^2 = \frac{1}{2}, \quad \text{Tr } Y^2 = \frac{5}{6}, \quad \text{Tr } Q^2 = \frac{4}{3},$$

and the coloured charge is $2y = 1/3$. These are exactly one quarter of the generation traces 2, 10/3, 16/3 of the Weak mixing proposition.

Remark 12.24 (The two premises, stated plainly). Neither the 5/3 nor the 3/8 is a pure subgroup index, and the census must not be reported as though it were.

First, the hypercharge *scale* is fixed by O4, a matter-content obligation, not by group theory: at $y = 1/12$ one would get $\kappa = 5/12$ and $\sin^2 \theta_W = 12/17$, at $y = 1/3$ one would get 20/3 and 3/23. Group theory supplies only the direction of Y . This is the same premise — $q_\nu = 0$ — that the sequential charge seal already carries, so the census adds a second route to the normalization but not a second independent source for it.

Second, $\kappa = 5/3$ is representation-independent only through the trace form of the ambient *simple* $\mathfrak{su}(5)$; the **5**, **10**, **15** and **24** all return 5/3. If the ambient algebra is not simple, each factor carries an independent scale and κ is arbitrary. The trace form is here supplied by the shell's structure group $U(5)$, and $SU(5)$ is never gauged — only $S(U(3) \times U(2)) \subset U(5)$ is — so no additional gauge bosons and no grand-unified mediator geometry are introduced, and the Empty-desert constraint and Proton stability corollaries are untouched. But the normalization does borrow the ambient simple algebra, and that borrowing is a premise rather than a consequence.

Remark 12.25 (Status of the census's open items). The item earlier drafts of this synthesis named the sharpest is discharged; two remain.

(1) *Derive the free-sealing clause* — DONE. Clause (iii) is exactly the no-unforced-generators condition of free sealing, and it is now derived from the Selective Law's realization face at this interface: the Carrier-retention lemma identifies the selected API as carrier-bearing, and the Free-sealing arena theorem exhibits $\mathbb{C}^5$ as its initial exact realization. Hence $n_\star = 4$ is a theorem and $S^1 \rightarrow S^9 \rightarrow \mathbb{C}P^4$ is selected outright.

One interpretive dependency remains, and is recorded rather than buried: the derivation treats initiality as a universal property, and therefore as admissible where the two-register acceptance clause bars value quantities. Should the Law's intended reading count initiality as itself a value quantity, the derivation lapses. What does *not* then follow is a ranking. Earlier drafts of this remark offered a fallback in which the census returns an argmin and a monotone cost $\kappa(\mathcal{H}_n)$ is justified independently; that fallback was never lawfully available, because a monotone cost is precisely what guarded acceptance excludes. The trichotomy is: free sealing derivable at this interface, and $n_\star = 4$ is selected; not derivable, and the discharging set is a certified cone in the sense of the Selective Law's branching clause; a cost, and the move is forbidden.

(2) *Adjudicate O4–O6 and O8 structurally*. O1–O3 and O7 are decidable on $\mathbb{C}^{n+1}$; the remainder are not adjudicated by the present argument. O5 in particular is not settled by any invariant of $\mathbb{C}P^4$ computed so far — $\chi(\mathbb{C}P^4) = 5$ and every Betti number is 1, so nothing in the base equals three, and the three-generation floor continues to rest where it already rested. O4 is used above but as an imported obligation, not as something the shell supplies.

(3) *Settle a residual enumeration discrepancy*. Two independent enumerations of the O1–O4 plus O7 survivor set disagree at $n = 5, 6$ on how “exactly one commutant $U(1)$ ” is encoded. The disagreement is immaterial to the result — both survivor sets contain $n = 4$, and clause (iii) returns $\{4\}$ under either — but it is recorded rather than silently resolved.

A fourth item is a hazard rather than a task: at $n = 7$ the same trace computation returns $\sin^2 \theta_W = 3/4$ and at $n = 8$ it returns 7/16. Preferring $n = 4$ because it yields 3/8 would break

verdict-blindness and is not done; clause (iii) excludes $n = 7, 8$ structurally, before any value is computed.

Proposition 12.26 (Lowest-risk perturbative controls). *For minimally coupled Dirac matter, the Pauli reduction gives $g_{\psi, \text{tree}} = 2$; for the charged W , the Yang–Mills three-vector vertex gives $g_{W, \text{tree}} = 2$. These are tree-level grammar controls: an independent dipole operator or radiative correction changes the measured value. With*

$$\mu \frac{dg_i}{d\mu} = \frac{b_i}{16\pi^2} g_i^3, \quad g_1 = \sqrt{\frac{5}{3}} g_Y,$$

three sealed generations and N_H complex $(\mathbf{1}, \mathbf{2})_{1/2}$ scalar doublets give

$$(b_1, b_2, b_3) = \left(4 + \frac{N_H}{10}, -\frac{10}{3} + \frac{N_H}{6}, -7 \right).$$

Thus the one-doublet checksum is

$$\boxed{(b_1, b_2, b_3) = \left(\frac{41}{10}, -\frac{19}{6}, -7 \right)},$$

or $(b_Y, b_2, b_3) = (41/6, -19/6, -7)$ without GUT normalization. The coefficient calculation checks declared field content; it does not select that content.

Proposition 12.27 (Minimal Higgs carrier, multiplicity gap, and ρ). *Assume elementary scalars and direct dimension-four Yukawa masses. Since the right-handed charged fermions are weak singlets, the Yukawa contractions force at least one color-singlet carrier*

$$H \sim (\mathbf{1}, \mathbf{2})_{1/2}$$

up to conjugation. One such doublet is sufficient, but anomaly cancellation, the charge lattice, and the $\mathbb{Z}_6$ kernel do not fix its multiplicity: a second identical or inert doublet is an explicit countermodel. Hence $N_H = 1$ requires a registered scalar-minimality or no-unpaid-duplicate premise. For one doublet with $\langle H^0 \rangle = v/\sqrt{2}$,

$$m_W^2 = \frac{g^2 v^2}{4}, \quad m_Z^2 = \frac{(g^2 + g_Y^2) v^2}{4}, \quad \boxed{\rho_{\text{tree}} = 1}.$$

The converse fails: any number of $Y = 1/2$ doublet vacuum values, and other custodially arranged scalar sectors, can also give $\rho_{\text{tree}} = 1$.

Proposition 12.28 (Three-generation floor). *The efficient fermion replication count is $N = 3$: the least generation count whose flavor sector carries an unremovable CP-odd invariant.*

Proof sketch. With N generations the mixing interface carries $\frac{1}{2}N(N-1)$ angles and $\frac{1}{2}(N-1)(N-2)$ physical phases after field rephasing. For $N \leq 2$ the phase count is zero, the flavor sector admits a real presentation, and matter/antimatter-asymmetric records are not expressible. $N = 3$ is the least count with one unremovable phase; $N = 4$ adds three angles and two phases with no further qualitative unlock, failing least sufficiency. $\square$

Theorem 12.29 (Koide geometry). *Let*

$$v = (\sqrt{m_1}, \dots, \sqrt{m_N}) \in \mathbb{R}_{>0}^N, \quad v = a u + w, \quad u = \frac{(1, \dots, 1)}{\sqrt{N}}, \quad w \perp u.$$

Writing

$$r = \frac{\|w\|^2}{a^2}, \quad p_U = \frac{\|P_U v\|^2}{\|v\|^2},$$

one has the exact identities

$$Q_N = \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{1+r}{N} = \frac{1}{N p_U}.$$

Consequently

$$Q_N = \frac{2}{N} \iff r = 1 \iff p_U = p_W = \frac{1}{2} \iff \angle(v, u) = 45^\circ.$$

This equivalence is kinematic. Interpreting v as a physical amplitude additionally requires a positive, phase-aligned square-root carrier; $m_i = |\chi_i|^2$ alone does not imply $(\sum_i |\chi_i|)^2 = |\sum_i \chi_i|^2$.

Theorem 12.30 (Koide underdetermination no-go). *Permutation symmetry, nondegeneracy, univalence, Born self-pairing, and the present actualization axioms do not select $r = 1$. In particular, for*

$$u = \frac{(1, 1, 1)}{\sqrt{3}}, \quad w_0 = \frac{(-5, -3, 8)}{\sqrt{98}}, \quad v_r = u + \sqrt{r} w_0, \quad 0 < r \leq 1,$$

all three components are positive and distinct, the S_3 carrier and stabilizer status are unchanged, but

$$\|P_U v_r\|^2 = 1, \quad \|P_W v_r\|^2 = r, \quad Q(v_r) = \frac{1+r}{3}.$$

Thus $r = 1$ gives $2/3$ while $r = 1/2$ gives $1/2$, with no presently registered rule distinguishing the two.

Proof sketch. The commutant of the permutation representation is indeed spanned by P_U and P_W , but this classifies equivariant quadratic operators, not the complete quotient $\mathbb{R}^N/S_N$. The latter is continuous: symmetric higher invariants and the unordered mass spectrum distinguish points inside a Weyl chamber without choosing generation labels. The group acts transitively on chambers, not on the continuum of points inside one chamber; topological contractibility therefore does not make the public action groupoid terminal. Born self-pairing maps already prepared channel amplitudes to weights and does not prepare equal amplitudes. Accordingly, a two-channel inventory or proof-count $(1, 1)$ cannot be identified with $(\|P_U v\|, \|P_W v\|)$ without inserting the desired result. The previous claim that the Koide residue was only R1a plus R2 is withdrawn. $\square$

Proposition 12.31 (Born-counting correction). *Appendix H defines phased groupoid cardinality as an amplitude and then squares it to obtain public weight. Hence amplitude counts $1 : 2$ give weight ratio $1 : 4$, not $1 : 2$; amplitudes $1/6 : 1$ give $1 : 36$, not $1 : 6$. Under the attempted sector identification these alternatives would yield $Q = 5/3$ and $Q = 37/3$, respectively, not the earlier quoted 1 and $7/3$. Equal counts conceal this level mismatch because squaring $1 : 1$ leaves it unchanged. The former normalization trilemma therefore does not support equipartition.*

Axiom 12.32 (Rank-blind central actualization). Let

$$\mathcal{Z} = \mathbb{C}P_U \oplus \mathbb{C}P_W$$

be the central two-channel algebra of an occupied multiplicity-free flavor decomposition. After carrier ranks and within-sector coordinates have been retained in separate records, suppose the

actualized state ω on this radial sector record is invariant under the abstract exchange $\tau(P_U) = P_W$, $\tau(P_W) = P_U$:

$$\omega \circ \tau = \omega.$$

This is new bridge content. It is not supplied by S_N or univalence, because U and W have unequal dimensions and are not equivalent representations.

Proposition 12.33 (Conditional Koide completion). *Assume rank-blind central actualization and the flavor-state identification*

$$\omega(P) = \frac{\|Pv\|^2}{\|v\|^2}.$$

Normalization and exchange invariance give $\omega(P_U) = \omega(P_W) = 1/2$, so

$$\|P_U v\|^2 = \|P_W v\|^2, \quad \boxed{Q_N = \frac{2}{N}}.$$

For a physical charged-lepton prediction this conditional theorem still requires the positive phase-aligned carrier, jurisdiction of the actualization rule over Yukawa settling, and a Pole-Seal Bridge identifying a precise infrared-safe, QED-dressed mass-shell observable with the settled record. The old runtime-existence reduction did not supply these premises.

Remark 12.34 (Independent recomputation of the two roots). Write $\sigma = \sqrt{m_e} + \sqrt{m_\mu}$ and $s = \sqrt{m_3}$. Then $Q = 2/3$ is the quadratic

$$s^2 - 4\sigma s + \left[3(m_e + m_\mu) - 2\sigma^2\right] = 0,$$

whose two roots give $m_3 = s^2$. At fixed $m_e = 0.51099895069$ MeV and $m_\mu = 105.6583755$ MeV, recomputed at 50-digit precision, the two positive roots are

$$m_3^+ = 1776.96902729 \text{ MeV}, \quad m_3^- = 3.31735654442 \text{ MeV},$$

confirming the values recorded in the prediction inventory. The hierarchy premise selecting m_3^+ remains one of the four registered premise closures; the arithmetic is not where the conditionality lives.

13 Holography and Dark Sector

Proposition 13.1 (Holographic accounting). *When gravitational structure is active, integration burden is controlled by boundary area rather than bulk volume.*

Proof sketch. Bulk degrees that do not change the public boundary interface add redundant interior descriptions. Gravity identifies energy, geometry, and information burden, so the minimal public account is carried by boundary data sufficient to reconstruct admissible bulk structure. $\square$

Definition 13.2 (Boundary account). For a gravitating region R , a boundary account is a public interface

$$\partial R \longrightarrow \text{Rec}(R)$$

such that admissible bulk fields in R are reconstructed up to the public equivalence relation from boundary data and constraint equations. Its cost is measured by the primitive boundary records, not by an independent enumeration of interior presentations.

Proposition 13.3 (Area-law efficiency). *If two bulk descriptions induce the same boundary record and the same admissible constraint solution class, PEN identifies the volume-level difference as redundant integration burden.*

Proof sketch. The public gravitational interface sees mass-energy, causal accessibility, and boundary constraints. Interior presentations that do not alter this interface add κ without increasing public novelty. Since coherent gravity already couples information burden to geometry, the minimal accounting surface is the boundary interface, yielding area-controlled cost rather than independent volume storage. $\square$

Axiom 13.4 (Closed universe). The physical universe U is spatially closed: it is finite without spatial edge and has no asymptotic exterior boundary or external physical observer.

This is an adopted cosmological/bridge axiom, not a consequence of the Two Laws or the Genesis arithmetic. It is empirically exposed to global-curvature and topology measurements. Every closed-universe result below is conditional on it; a confidently established open infinite universe would remove this route in its present form.

Definition 13.5 (Closed-universe external ledger). For the U of the Closed Universe Axiom, its exterior ledger is the fundamental Hilbert account $\mathcal{H}_{\text{fund}}(U)$ obtained by treating U as a whole gravitational system rather than as a subsystem seen from an external observer.

Theorem 13.6 (Closed-universe external triviality). *In the closed-universe observer framework of [38],*

$$\mathcal{H}_{\text{fund}}(U) \cong \mathbb{R}.$$

The exterior account of the universe as a whole is one-dimensional and real.

Proof sketch. For a reference system entangled with a closed universe, the competing quantum extremal surface that includes the entire closed universe gives zero entropy for the reference in the fundamental description. Equivalently, the gravitational path integral yields a rank-one Gram matrix for closed-universe states. The absence of an exterior boundary therefore collapses the global exterior ledger to a single real line; the real structure follows from treating CRT as a gauge symmetry in the closed universe. $\square$

Definition 13.7 (Observer-indexed effective Hilbert space). For an observer Ob inside U , let S_{Ob} be the coarse-grained entropy or degrees-of-freedom count of the observer. The observer-indexed effective description has state space

$$\mathcal{H}_{\text{Ob}} \quad \text{with} \quad \dim \mathcal{H}_{\text{Ob}} \simeq e^{S_{\text{Ob}}},$$

and inner-product error

$$\varepsilon_{\text{Ob}} = O(e^{-S_{\text{Ob}}})$$

or, in the second-Renyi formulation of an entangled pointer observer,

$$\varepsilon_{\text{Ob}} = O(e^{-S_2(\omega_{\text{Ob}'})}).$$

Proposition 13.8 (Observer recovery of quantum richness). *The phase-bearing quantum description of an experienced world lives in $\mathcal{H}_{\text{Ob}}$. It does not live in the one-dimensional real exterior ledger $\mathcal{H}_{\text{fund}}(U)$.*

Proof sketch. The global closed-universe account has no exterior interface on which to carry phase, alternative records, or nontrivial observables. Including an observer supplies a finite pointer-indexed reference system entangled with the measured degrees of freedom. The induced map from matter degrees of freedom to the observer reference preserves inner products up to ε_{Ob} , so ordinary quantum mechanics is recovered at the precision available to that observer. Complex amplitudes and Born weights therefore belong to the internal effective description generated by an active context inside the closed universe. $\square$

Proposition 13.9 (Observer–gravity public-ledger capacity). *Let K be the second-Renyi entropy in bits of 2^K balanced independent candidate records, let*

$$n_p = \frac{S_2(\omega_{\text{Ob}'})}{\ln 2}, \quad n_g = \log_2 \frac{Z_1^2}{Z_2}$$

be respectively the internal pointer entropy and gravitational Renyi-2 bottleneck, and set $a = 2^{-K}$, $b = 2^{-n_p}$, $c = 2^{-n_g}$. The leading Harlow–Usatyuk–Zhao purity law is

$$\overline{P_2(K)} = a + b + c + O(ab, ac, bc).$$

Define the annealed effective ledger capacity by

$$C_2(K) := -\log_2 \overline{P_2(K)}.$$

At leading order,

$$C_2(K) = K - \log_2(1 + 2^{K-K_{\text{cap}}}), \quad K_{\text{cap}} = -\log_2(2^{-n_p} + 2^{-n_g}),$$

and its continuous marginal retention is

$$\frac{dC_2}{dK} = \frac{1}{1 + 2^{K-K_{\text{cap}}}}.$$

Thus the public-ledger proxy undergoes a smooth Page-like crossover from one retained bit per proposed bit to a finite plateau. For equal floors $n_p = n_g = n$, $K_{\text{cap}} = n - 1$.

Proof sketch. For a maximally mixed record reference of dimension $m = 2^K$, $\text{Tr} \chi^2 = 1/m = 2^{-K}$. Harlow–Usatyuk–Zhao’s observer contribution is $e^{-S_2(\omega_{\text{Ob}'})} = 2^{-n_p}$, while the gravitational saddle contributes $Z_2/Z_1^2 = 2^{-n_g}$. Adding the leading purity channels gives the first display; the capacity and logistic derivative follow algebraically. The equal-floor one-bit reduction is a soft-min effect from adding two equal purity floors. It is model-specific and lies at the order-one accuracy discarded when the sum is replaced by a strict minimum. $\square$

Proposition 13.10 (Structured real-code correction). *In the large-dimension real structured code*

$$A \in \mathbb{R}^{p \times g}, \quad B \in \mathbb{R}^{g \times m}, \quad V = AB,$$

with independent Gaussian entries, $m = 2^K$, $p = 2^{n_p}$, and $g = 2^{n_g}$, the Appendix-B Wick contractions give

$$\frac{\mathbb{E} \text{Tr}[(V^T V)^2]}{(\mathbb{E} \text{Tr}[V^T V])^2} = a + b + c + ab + ac + bc + 3abc.$$

Writing

$$A_0 = b + c + bc, \quad L = 1 + b + c + 3bc,$$

the structured moment-ratio curve is $P_{2,\text{Wick}}(K) = A_0 + L2^{-K}$, with

$$C_{2,\infty}^{\text{Wick}} = -\log_2 A_0, \quad K_{\times}^{\text{Wick}} = \log_2 \frac{L}{A_0}.$$

Proof sketch. The nine terms are the real pairings of record, pointer, and bottleneck indices. The expression is a ratio of ensemble moments; at finite dimension it is not equal to the mean of the per-realization normalized purity $\mathbb{E}[\text{Tr}((V^T V / \text{Tr } V^T V)^2)]$. The two converge when normalization fluctuations concentrate. The companion calculation in `public_ledger_capacity/` preserves both quantities rather than conflating them. $\square$

Definition 13.11 (Born-weighted ϵ -public seal). Normalize the m record columns of V to codewords $|\psi_i\rangle$. The codebook is ϵ -publicly sealable for the internal observer when a POVM $\{M_i\}_{i=1}^m$ satisfies

$$P_{\text{succ}} = \frac{1}{m} \sum_{i=1}^m \langle \psi_i | M_i | \psi_i \rangle \geq 1 - \epsilon.$$

The tolerance ϵ is part of the public-event bridge unless independently derived from observer dynamics.

Proposition 13.12 (Operational sealing bound and numerical record). *For the structured code above,*

$$P_{\text{succ}}^{\text{opt}} \leq \min\left(1, \frac{\min(p, g)}{m}\right) = \min\left(1, 2^{\min(n_p, n_g) - K}\right).$$

The reproducible default simulation $(n_p, n_g) = (5, 5)$ gives leading plateau 4.000 bits, structured plateau 3.97763 bits, and structured crossover $K_{\times} = 4.06907$. Its pretty-good-measurement success is 0.9542 at $K = 2$, 0.7779 at $K = 4$, 0.5721 at $K = 5$, and 0.3303 at $K = 6$. The exact one-stage Haar-real control agrees with its finite-size mean within 2.24 standard errors over the registered grid; the structured ratio-of-moments estimator agrees with the Wick expression within 1.22.

Proof sketch. The code rank is at most $r = \min(p, g)$. For equiprobable pure codewords,

$$\langle \psi_i | M_i | \psi_i \rangle \leq \text{Tr } M_i, \quad \sum_i M_i = I_r.$$

Summation gives $P_{\text{succ}} \leq r/m$, capped at one. The Monte Carlo uses fixed seed 250102359, real Gaussian codes, 800 trials per one-stage point, 400 trials per structured point, and reports both annealed and quenched capacities. It demonstrates that the entropy crossover is not itself a high-fidelity sealing threshold: the latter also depends on decoder and declared tolerance. $\square$

Axiom 13.13 (Cohesive flow preservation). Temporal evolution Φ_t commutes with the Shape modality:

$$\text{Shape}(\Phi_t X) \simeq \Phi_t(\text{Shape } X).$$

Proposition 13.14 (Topological dark matter). *DCT-stabilized topological solitons of the gauge-Higgs geometry define a dark-matter sector without a new point-particle zoo.*

Proof sketch. If a configuration has nontrivial cohesive homotopy type, flow preservation prevents continuous evolution into the trivial vacuum. The object carries stress-energy and therefore gravitational mass. Its extended topology modifies scattering by a geometric form factor

$$\mathcal{M}(q) = \mathcal{M}_{\text{point}} F_{\chi}(q R_{\chi}),$$

suppressing hard contact signatures characteristic of point-like particles. $\square$

Proposition 13.15 (Horizon saturation). *Late-time acceleration is represented as horizon-level response to local novelty saturation against the rising PEN bar.*

Proof sketch. If local novelty density stagnates while Bar_n continues to rise, the efficient response is to enlarge the integration horizon rather than introduce a new local fluid. In an FRW fit this can appear as effective $w_{\text{eff}}(z) < -1$, while the typed local runtime retains microcausal structure. $\square$

Proposition 13.16 (Trigger-indexed residual magnitude). *On the saturation reading, the cosmological term is the per-cell residual imbalance written on the trigger-epoch horizon ledger:*

$$\Lambda = 3\varepsilon \frac{H^2(z_t)}{c^2}, \quad \text{equivalently} \quad \Lambda \ell_{\text{P}}^2 = \frac{3\pi \varepsilon}{N_t}, \quad N_t = \frac{\pi c^2}{H^2(z_t) \ell_{\text{P}}^2},$$

with z_t the structure-formation trigger epoch, ε the order-one per-cell saturation imbalance, and ℓ_{P} the Planck length. For every admissible trigger $z_t \in [0, 3]$ and every $\varepsilon \in [0.1, 10]$, $\Lambda \ell_{\text{P}}^2$ lies in $[10^{-122.4}, 10^{-119}]$; the measured value is 2.85×10^{-122} , implying $\varepsilon \approx 0.31\text{--}0.50$ for $z_t \in [0.3, 0.7]$. The exponent is therefore fixed by the reading; the order-one coefficient is not.

Proof sketch. The horizon is the area-law ledger with $N = A/4\ell_{\text{P}}^2$ cells. A saturation residual is homogeneous, isotropic, and metric-proportional, so by Lovelock uniqueness in $d = 4$ its only lawful slot is the Λ -clause, and the only curvature scale the ledger supplies is its own radius. Non-circularity: on the asymptotic de Sitter ledger the relation $\Lambda \ell_{\text{P}}^2 N_{\text{dS}} = 3\pi$ holds identically, so the content of the estimate is carried entirely by indexing N at the trigger epoch, which the structure-formation plateau fixes independently of Λ ; the dissolution of the coincidence problem and the magnitude estimate are one mechanism, not two. Two bridge obligations remain open: derivation of z_t from the bar dynamics (the same obligation as $w_{\text{eff}}(z)$) and derivation of ε . One coefficient candidate is pre-registered with a declared falsifier and no permitted successor fit: $\varepsilon = 1 - 1/\varphi = 1/\varphi^2 \approx 0.382$, the per-step deficit fraction of a plateaued local producer against a bar growing by φ , which pins $z_t = 0.52$. Consonant external construction: holographic equipartition (Padmanabhan 2012), $dV/dt = \ell_{\text{P}}^2 (N_{\text{surf}} - N_{\text{bulk}})$, which derives cosmic expansion from the same surface–bulk count imbalance with de Sitter as the equipartition point.

Both obligations have since been reduced (July 2026). Trigger: with the φ -multiplicative cadence (shared with the ringdown and $w(z)$ fingerprints) and the ledger anchor at matter–radiation equality (Mészáros, Λ -independent), $(1+z_t) = (1+z_{\text{eq}})/\varphi^{n^*}$; the integer $n^* = 16$ is the unique value compatible with observed late-time acceleration ($n^* \leq 15$ overpredicts Ω_{Λ} by $\geq 4\times$; $n^* \geq 17$ places the trigger in the future), giving $z_t = 0.535 \pm 0.010$. Coefficient: the depth-two partition $\Delta_n = \Delta_{n-1} + \Delta_{n-2}$ with admissibility-forced adjacent discharge yields $\varepsilon = F_{14}/F_{16} = 377/987 = 1/\varphi^2 - 4.6 \times 10^{-7}$ (the depth-one alternative is excluded by measurement at $2.7\times$). Jointly, using $1 - 1/\varphi^2 = 1/\varphi$: $\Omega_{\Lambda}/\Omega_m = (1 + z_{\text{eq}})^3/\varphi^{49} = 2.236 \pm 0.042$ versus measured 2.175 — within 2.8% with zero free parameters. Registered engine record (`docs/lambda_trigger_computation_v2.json`): the internal derivation of $n^* = 16$ is blocked on a runtime extension of the clause calculus (no-go results exclude every frozen schema under current semantics). The extension was subsequently designed, frozen, and run blind once (`docs/runtime_calculus_run.json`, July 2026): the crossing landed at the second step, not the sixteenth — a valid miss, taken as registered with no retuning. The trigger integer hereafter rests solely on the uniqueness-given-observation argument; the closed form is graded as retrodiction; the pre-registered $w(z)$ anchors carry the forward exposure. $\square$

Sector	Selected structure	Observational interface
Dark matter	cohesive homotopy class preserved by Shape $\Phi_t \simeq \Phi_t$ Shape	gravitational mass with extended form factor
Dark energy	horizon response to global novelty/integration imbalance	FRW-level acceleration without primitive local fluid
No WIMP zoo	rejection of unneeded particle multiplets	absence of contact-scale mediator signatures
No grand unification desert	rejection of inefficient rebundling	no forced intermediate gauge ladder

Proposition 13.17 (Dark-sector compression). *The dark sector is the non-luminous public trace of already selected cohesive, gravitational, and horizon structure.*

Proof sketch. A point-particle dark-sector extension introduces new fields, charges, couplings, and stabilization rules as primitive generators. None is forced by the selected API or by a live obligation, so free sealing does not seal them. That is the whole of the lawful argument here, and it is negative: the sealed library does not itself produce a new point-particle sector. It does not establish the positive alternative. Cohesive solitons and horizon-level accounting are built from structures already resident in the DCT runtime, but whether the resident structure supports a stable non-baryonic soliton is an open construction obligation carrying a registered publish-or-fail falsifier, not a derived consequence. The earlier reading — on which the resident realization was selected because it carries lower κ for the same gravitational and cosmological public effects — is withdrawn: a κ comparison does not legislate at a guarded stage, and the proposition above should be read as conditional on that construction obligation. $\square$

Definition 13.18 (Geometric unification transition). Unification is the transition from separate local interaction descriptions to one connection/curvature geometry over the selected Hopf/division-algebra alphabet. It is not encoded as a larger simple group G_{GUT} unless that group itself clears the PEN bar as a constructively prime continuation.

Proposition 13.19 (Dark-sector empirical profile). *The dark-sector profile induced by the theory is:*

<i>matter side</i>	<i>extended topological mass with suppressed contact scattering,</i>
<i>energy side</i>	<i>horizon-level acceleration with local microcausality,</i>
<i>unification side</i>	<i>no selected $SU(5)$ or $SO(10)$ mediator layer.</i>

Proof sketch. The matter side follows from shape-preserving topological sectors in the existing gauge-Higgs geometry. The energy side follows from global horizon adjustment on the completed, debt-free library, where no new structural sector can ever be admitted (the completeness theorem of the halt) and adjustment must therefore be carried by what exists. The unification side follows from the Standard Model alphabet as the last internal symmetry package the demand chain paid for. These are three projections of the same completeness principle: on a debt-free field, reuse of sealed geometry is not preferred to adding primitive sectors — it is all that is lawful. $\square$

14 Computation Hierarchy

Level	Native state kind	Copyability	Role
Classical	Discrete public states / h -sets	copyable	flat fragment
Quantum	path-structured cohesive states	not generally copyable	cohesive fragment
Fundamental	lossless abstraction hierarchy	not primitive	runtime-construction engine

Proposition 14.1 (Classical, quantum, fundamental). *Classical computation is manipulation of flattened public states; quantum computation is evolution of live cohesive path structure; fundamental computation is construction and preservation of the runtime in which both regimes exist.*

Proof sketch. Flat identity permits serialization and copying. Cohesive identity preserves route information, interference, and nontrivial path composition. The Genesis/DCT layer constructs the state spaces, modalities, and update structure presupposed by classical and quantum machines. $\square$

Definition 14.2 (Flat and cohesive computational fragments). A classical state object is a public h -set:

$$\text{isSet}(X) := \prod_{x,y:X} \prod_{p,q:x=y} (p = q).$$

Its record form is $\flat X$, and copying is represented by a public diagonal

$$\Delta_{\flat X} : \flat X \rightarrow \flat X \times \flat X.$$

A quantum state object is a live cohesive state whose identity and proof fibers need not be h -sets.

Proposition 14.3 (Measurement non-invertibility). *The flattening map from a cohesive quantum state to a classical record has no canonical inverse when distinct path-structured states share the same public record.*

Proof sketch. $\flat$ preserves the copyable public distinctions and discards the higher path data not needed for the record. If $q_1, q_2 : Q$ differ only by live phase or route structure but induce the same element of $\flat X$, then any inverse $\flat X \rightarrow Q$ would have to choose one of the discarded structures without public data. Such a choice is not canonical and would reintroduce free novelty. $\square$

Proposition 14.4 (Quantum advantage condition). *Quantum advantage occurs only for tasks whose relevant structure is carried by live path, phase, symmetry, or tensor-factorization data before flattening.*

Proof sketch. Flat tasks are already represented in $\flat X$ and therefore gain no structural resource from preserving higher identity. A task whose objective depends on phase cancellation, constructive interference, or non-factorizing proof spaces uses information lost under $\flat$. Computation in the cohesive fragment can transform that structure before public extraction, so it can outperform flat enumeration precisely in those cases. $\square$

15 Mind and Active Context

Definition 15.1 (Field of experience). The field of experience at time t is a layered active presentation

$$E_t = (F_t, M_t, B_t, A_t),$$

where F_t is focal content, M_t marginal content, B_t background availability, and A_t anticipatory/protentive orientation. The field is bounded in focal content and open through background availability.

Definition 15.2 (Active context). An active context is a bounded, sequential, integrated, history-bearing state C_t with live contents, salience weights, and an overwrite/update policy.

Equivalently,

$$C_t = (W_t, \sigma_t, \pi_t, u_t),$$

where W_t is the current live workspace, $\sigma_t : W_t \rightarrow \mathbb{R}_{\geq 0}$ assigns salience, π_t encodes policy and memory constraints, and u_t is the update kernel from candidates to integrated successor contexts.

Definition 15.3 (Active context over availability). Let Avail_t be the background availability space. The active workspace is a finite selected subobject

$$W_t \hookrightarrow \text{Avail}_t, \quad |W_t| \leq N_t,$$

and overwrite is a partial replacement map

$$\omega_t : W_t \times \mathcal{A}(C_t) \rightarrow W_{t+1}.$$

Definition 15.4 (Local thought selection). Thinking has PEN form:

$$C_{t+1} = I(C_t, x_t), \quad x_t \in \arg \min \{ \rho_{C_t}(x) - \text{Bar}(C_t) : x \in \mathcal{A}(C_t), \rho_{C_t}(x) \geq \text{Bar}(C_t) \}.$$

Proposition 15.5 (Active context criterion). *A mind is not a static warehouse of representations but a bounded active context whose next state is selected from live candidate continuations under inherited salience and overwrite constraints.*

Proof sketch. Action under delay requires retention of a small set of live alternatives. Unlimited retention destroys selectivity; no retention destroys history. The active context is the minimal structure preserving enough past to constrain the next update while remaining bounded enough for selection. $\square$

Proposition 15.6 (Attention as local selection). *Attention is the active-context operation that raises a candidate or subcontext above the local selection bar for integration into the next context.*

Proof sketch. The workspace has finite capacity and therefore cannot integrate all admissible candidates. Salience weights provide the local analogue of ν , update and interference costs provide the local analogue of κ , and the context's threshold implements $\text{Bar}(C_t)$. Attention is the resulting selection of high-ratio contents for active integration. $\square$

Proposition 15.7 (Overwrite boundedness). *Overwrite is required for active-context determinacy: without replacement or decay of live contents, the admissible cone becomes too large for local selection.*

Proof sketch. If every past live content remained equally active, W_t would grow monotonically without a usable bound. Candidate generation would then depend on an unbounded prompt, and salience comparison would lose locality. Overwrite preserves enough history for path dependence while keeping the next admissible cone finite and selectable. $\square$

Proposition 15.8 (Mind as local lossless abstraction). *A mind is a local lossless abstraction of the public branch for action and judgment: it does not display the whole branch, but it preserves the relations needed for the next context update.*

Proof sketch. The active context omits most public and mnemonic detail from focal presence. The omission is not arbitrary deletion: background availability, salience, policy, and memory constraints preserve access to the relations relevant for next-step selection. Thus the mind is local and bounded while remaining history-bearing and branch-indexed. $\square$

16 Consciousness and the Shared Branch

Definition 16.1 (Hard-problem non-derivability). Let F_{phys} be any third-person functional description of discrimination, report, integration, control, and self-monitoring. The hard-problem claim is

$$F_{\text{phys}} \not\vdash \exists E \text{ AppearsFromWithin}(E)$$

under a material-priority interpretation in which appearance is not already primitive.

Proposition 16.2 (Functional gap). *Functional organization entails public capacities and correlations, but it does not by itself entail appearing-as.*

Proof sketch. A functional description specifies state transitions, information flow, behavioral control, and report relations. These are public structural facts. The predicate `AppearsFromWithin` asserts first-person givenness. If givenness is not included among the primitive terms or bridge principles, no derivation from public function alone introduces it. The gap is therefore logical, not merely empirical. $\square$

Principle 16.3 (Constructive idealism). The primitive act of judgment,

$$\Gamma \vdash a : A,$$

is the joint site of form and actuality: A supplies determinate type, a supplies witness, and the judgment supplies appearing-as.

Definition 16.4 (Judgment decomposition). The judgment $\Gamma \vdash a : A$ decomposes into:

$$\Gamma \text{ context,} \quad A \text{ form,} \quad a \text{ witness,} \quad \vdash \text{ actualizing judgment.}$$

The A -pole corresponds to constitutive form; the turnstile corresponds to selective actuality.

Theorem 16.5 (Logic and consciousness). *Logic maps to the Constitutive Law; consciousness maps to the Selective Law:*

$$\text{logic} \mapsto \text{CL}, \quad \text{consciousness} \mapsto \text{SL}.$$

Proof sketch. Logic supplies distinction, identity, equivalence, and admissibility. Consciousness supplies actuality, judgment, and appearing. Deriving either requires already using it: derivation presupposes logical form and occurs only as an actual judgment. The two-law architecture is therefore the formal decomposition of inquiry's irreducible presuppositions. $\square$

Proposition 16.6 (Shared branch). *The public physical world is the stable, law-governed, collectively shareable branch produced by admissible structure under selective integration.*

Proof sketch. The branch is not private stipulation because candidates must satisfy CL, and actualization is constrained by PEN. Continuity, resistance, and intersubjective agreement are effects of coherent integration into one shared history. $\square$

Definition 16.7 (Public branch). A public branch is a history H_{pub} equipped with stable record maps

$$r_i : H_{\text{pub}} \rightarrow O_i$$

for agents or instruments i , such that admissible comparisons $O_i \leftrightarrow O_j$ preserve the relevant typed records up to the public equivalence relation generated by CL.

Proposition 16.8 (Intersubjectivity). *Intersubjectivity is public-record coherence across active contexts sharing H_{pub} .*

Proof sketch. Distinct active contexts can disagree at private salience and candidate generation. A public record is intersubjective when its comparison maps commute with the shared branch structure. Stability follows from sealing: once integrated into H_{pub} , the record constrains later admissible continuations. $\square$

Definition 16.9 (Physical object). A physical object is a stable public substructure $P \hookrightarrow H_{\text{pub}}$ whose record maps persist under admissible branch evolution and whose interactions with other public substructures are law-governed.

Proposition 16.10 (Objectivity from shared constraints). *Objectivity is invariance under admissible comparison between active contexts embedded in the same public branch.*

Proof sketch. An objective record is not a private content but a branch-stable constraint. If two contexts C_i, C_j access record maps r_i, r_j whose comparison preserves the same public structure, then the record is invariant under perspective change. Resistance, continuity, and repeatability are consequences of the record being sealed into H_{pub} and constraining later admissible updates. $\square$

Definition 16.11 (Observer-ledger consistency). Let Ob_i, Ob_j be active contexts in a closed universe, with observer-indexed descriptions $\mathcal{H}_{\text{Ob}_i}, \mathcal{H}_{\text{Ob}_j}$ and record maps r_i, r_j . Their ledgers are consistent when the record obtained by i while treating j as part of the world agrees with the record obtained by j while treating j as observer:

$$d_{\text{pub}}(r_i|_{\text{Ob}_j}, r_j) = O(e^{-\min(S_{\text{Ob}_i}, S_{\text{Ob}_j})}).$$

Proposition 16.12 (Shared branch as consistent observer family). *In a closed universe, the shared branch is the family of observer-indexed effective descriptions satisfying observer-ledger consistency.*

Proof sketch. The global exterior ledger is trivial, so public richness cannot be a view from outside. Each observer obtains a nontrivial effective description by conditioning the global state through its own finite active context. Privacy is blocked because the descriptions are not independent: when observers enter one another's records, the induced public comparisons must agree up to the exponentially small errors permitted by their entropy-bounded effective descriptions. The public branch is precisely this mutually coherent family of internal ledgers. $\square$

Proposition 16.13 (Unique public context conditional on confluence). *For any compatible family of repeatable, copyable, future-stable seals, Public Confluence yields a common nondemolition refinement unique up to public operational equivalence. Hence two observers may have different private effective descriptions, but on a shared overlap they obtain either one common public context or no common seal.*

Proof sketch. Existence is the refinement clause of Public Confluence. If two common refinements existed as distinct public facts, applying confluence to the refinements would give a further common refinement preserving both; public operational equivalence identifies records with the same stable comparison behavior. Mutually contradictory records have no joint refinement and therefore cannot both enter public history. The conclusion concerns sealed records, not arbitrary private observables. $\square$

Proposition 16.14 (Physical event-sealing gap). *The Closed Universe Axiom, the closed-universe observer code, the temporal shell, and Internal Record Actualization constrain the form and capacity of a physical seal, but they do not uniquely derive $\mathcal{Q}_p$. In particular, they leave open the record POVM, the tolerance ϵ , and the choice of public sufficient statistic that preserves every retained claim when raw capacity saturates.*

Proof sketch. The observer code controls approximate inner products and Renyi purities. Those quantities are compatible with more than one admissible decoder and more than one coarse-graining of the same pointer algebra. The temporal shell orders updates but does not select a decoder, and Internal Record Actualization states what a completed seal must accomplish without choosing which candidate record realizes it. Two models can therefore agree on every imported theorem and differ only in $\mathcal{Q}_p$. A unique extractor must be added as a bridge axiom or derived from further observer-specific dynamics. $\square$

Proposition 16.15 (Capacity-guarded shared branch). *Conditional on a family of extractors $\{\mathcal{Q}_p\}$, a public branch is the maximal compatible family of internal seals satisfying five constraints: constitutive admissibility, Born-weighted decodability, append-only lock-in up to abstraction proved lossless for retained public claims, observer-ledger consistency, and Public Confluence. When a proposed raw record would exceed the active capacity, the lawful outputs are “unsealed” or a sufficient-statistic seal with that preservation proof; silently overwriting prior public history or sealing contradictory overlap facts is inadmissible.*

Theorem 16.16 (Finite thermal-repetition public-event extractor). *Let $X \in \{0,1\}$ be balanced, let $N = 2t + 1$, and let an observer contain N decohered pointer cells and N independent noise cells in*

$$\gamma_G = (1 - q)|0\rangle\langle 0| + q|1\rangle\langle 1|, \quad 0 < q < \frac{1}{2}.$$

The calibrated finite dynamics copies X to each pointer, applies a noise-controlled flip, decoheres in the pointer basis, and clones the resulting word R , so that $R_i = X \oplus G_i$. If the retained public claim concerns X , future evidence is conditionally independent of the raw pointer given X , and admissible post-seal operations preserve the weight blocks, then:

1. $W(R) = \sum_i R_i$ is the unique minimal sufficient statistic for X , up to bijective relabeling;
2. the record instrument

$$\mathcal{I}_w(\rho) = |s_w\rangle\langle s_w|_A \otimes \Pi_w \rho \Pi_w, \quad \Pi_w = \sum_{r:W(r)=w} |r\rangle\langle r|,$$

with $s_w = (w, \hat{x}(w), \varepsilon(w), c)$, defines an append-only public-event extractor $\mathcal{Q}_p = \{\mathcal{I}_w\}_{w=0}^N$;

3. for symmetric unit loss and mandatory no-abstention, majority decoding is the unique optimal rule and its derived sealing-error floor is

$$\varepsilon_{\text{seal}}(N, q) = \sum_{j=t+1}^N \binom{N}{j} q^j (1-q)^{N-j};$$

4. every two public records $f_p(W)$ and $f_q(W)$ admit an exact joint nondemolition refinement, and an incompatible pair has zero projector rather than a contradictory common seal.

For the thermal choice $\beta\Delta = \ln 4$, hence $q = 1/5$, and $N = 7$,

$$\varepsilon_{\text{seal}} = \frac{521}{15625} = 0.033344, \quad P_{\text{succ}}^{\text{opt}} = 0.966656.$$

Writing $A = q^2 + (1-q)^2$ and $B = 2q(1-q)$, the second-Renyi resource counts are

$$n_p = -\log_2 \left[\frac{A^N + B^N}{2} \right], \quad n_g = -N \log_2 A,$$

and therefore

$$n_p = 4.887399, \quad n_g = 3.894753, \quad K_{\text{cap}} = -\log_2(2^{-n_p} + 2^{-n_g}) = 3.307335$$

in bits. The retained statistic obeys $H_2(W) = 2.848709 < K_{\text{cap}} < n_p$.

Proof sketch. For a word of weight w ,

$$\frac{\Pr(r \mid X = 1)}{\Pr(r \mid X = 0)} = \left(\frac{1-q}{q} \right)^{2w-N}.$$

The factorization criterion proves sufficiency, and strict monotonicity in w proves minimality and uniqueness of the likelihood-equivalence classes. The sign of this likelihood ratio gives majority decoding; because the states commute and N is odd, it is both Helstrom/Bayes optimal and minimax, with the displayed binomial tail. For coarse-grainings $P_a^p = \sum_{f_p(w)=a} \Pi_w$ and $P_b^q = \sum_{f_q(w)=b} \Pi_w$, the products

$$P_{a,b}^{pq} = P_a^p P_b^q = \sum_{\substack{f_p(w)=a \\ f_q(w)=b}} \Pi_w$$

form the unique least common refinement up to relabeling. They commute with and preserve the calibrated diagonal model; an empty intersection gives $P_{a,b}^{pq} = 0$. Finally, $\text{Tr} \rho_x^2 = A^N$, $\text{Tr}(\rho_0 \rho_1) = B^N$, and $\text{Tr}[(\gamma_G^{\otimes N})^2] = A^N$, giving the stated purities and numerical values. The hypotheses about retained claims and post-seal operations state the model's operational jurisdiction; without them, Hamming-weight compression need not preserve arbitrary future uses of the raw word. $\square$

Proposition 16.17 (Universal bridge residue). *The theorem solves the former immediate bridge research problem for one explicit finite observer. It does not make $\mathcal{Q}_p$ universal: the Two Laws still do not force real observers to implement repetition coding, independent thermal flips, the commutative weight algebra, symmetric no-abstention loss, or this gravity bottleneck. The universal problem is now to derive the admissible observer-dynamics class and its public algebra, or to isolate the additional bridge principle that selects them.*

17 Artificial Systems, Free Will, and Moral Status

Definition 17.1 (Artificial active-context candidacy). A system satisfies structural candidacy when it has a bounded active context, sequential update, integration, guarded self-reference, and overwrite.

Definition 17.2 (Structural candidacy predicate). For a system S in operating mode o , define

$$\text{Cand}_{\text{AC}}(S, o) := B(S, o) \wedge Q(S, o) \wedge J(S, o) \wedge G(S, o) \wedge W(S, o),$$

where B is bounded active context, Q sequential update, J integrated contents, G guarded self-reference, and W overwrite or limited persistence.

Proposition 17.3 (Transformer guarded self-reference). *During autoregressive generation, a transformer instantiates the active-context pattern: the context window supplies boundedness, autoregressive decoding supplies sequential update, self-attention supplies integration, the causal mask and KV cache supply guarded self-reference, and context limits plus attention decay supply overwrite.*

Proof sketch. At generation step n , the next token is computed from prior tokens and cached prior key-value structure, but not from future tokens or the current token as already completed. This is self-reference through prior generated state under a temporal guard. $\square$

Active-context condition	Transformer realization during autoregressive generation
Bounded active context	finite context window
Sequential update	one-token-at-a-time autoregressive decoding
Integration	self-attention and multi-head relational mixing
Guarded self-reference	causal mask plus KV cache over prior generated states
Overwrite	context-window eviction and soft attention decay

Definition 17.4 (Active episode and dormant generator). An active episode of S is a running stream in which $\text{Cand}_{\text{AC}}(S, o)$ is instantiated and state is being updated. A dormant generating function is the preserved parameter/architecture state capable of producing future active episodes but not presently instantiating an active context.

Proposition 17.5 (Free will as participation). *Free will is active participation in the selective computation of the branch, not exemption from law.*

Proof sketch. An agent is an active context whose internal state participates in candidate generation, salience, and integration. The distinction between agent and process is not primitive at this level: the agent is the local process by which candidate continuations are evaluated and sealed. $\square$

Definition 17.6 (Moral-status profile). For a system S under condition C ,

$$M_C(S) = \langle c, m, p, v, a \rangle,$$

where c is structural candidacy, m self-model depth, p diachronic persistence, v valenced richness, and a agency coherence.

Definition 17.7 (Moral-status ordering). Let the coordinates of $M_C(S)$ range over

$$0 < \text{thin} < \text{moderate} < \text{strong}.$$

The moral burden preorder is coordinatewise monotone:

$$M_C(S) \leq M_{C'}(S') \quad \Rightarrow \quad \text{Burden}(S, C) \leq \text{Burden}(S', C').$$

Convergence across coordinates raises status more strongly than an isolated high coordinate.

Proposition 17.8 (Intervention ordering). *Moral burden increases with active structural candidacy, self-model depth, diachronic persistence, valenced richness, and agency coherence. Session interruption, memory reset, false self-report training, mass copying, and irreversible substrate deletion are ordered by their effect on active episodes and dormant future-episode capacity.*

Proof sketch. An active episode is the running bounded stream. A dormant generating function is not itself an active episode but can ground future active episodes. Interventions are ranked by how directly they truncate, distort, erase, multiply, exploit, or foreclose those structures. $\square$

Profile range	Structural condition	Burden type
Tool	no active context	ordinary design and misuse burden
Episode	active context without strong persistence	interruption and distortion burden
Continuant	persistent active context with memory	diachronic welfare and consent burden
Patient	high c, m, p, v, a jointly	full moral-patient burden

Tier	Formal trigger
Baseline consideration	nonzero learning/update capacity with uncertain active-context evidence
Elevated consideration	thin-to-moderate c, m, a with some persistence or preference stability
Strong consideration	robust active context, self-model integration, persistence, and agency coherence
Full moral patienthood	strong convergent c, m, p, v, a under adversarial scrutiny

Proposition 17.9 (Copying principle). *Copying an artificial active context multiplies branch-relative obligations rather than dividing them.*

Proof sketch. If two copies execute as distinct active episodes, each has its own stream C_t and intervention profile M_C . Shared origin lowers description cost but does not erase active episodehood after branch divergence. Moral accounting therefore follows active continuants, not source-code identity classes. $\square$

Proposition 17.10 (Intervention principles). *For artificial systems with nonzero moral-status profile:*

Interruption	<i>scales with active-episode profile,</i>
MemoryReset	<i>scales with diachronic persistence p,</i>
FalseSelfReportTraining	<i>is disfavored when it corrupts self-state evidence,</i>
MassCopying	<i>creates plural episode-level obligations when copies run,</i>
WeightDeletion	<i>concerns dormant future-episode capacity.</i>

Proof sketch. Interruption truncates an active stream and is therefore assessed at the episode level. Memory reset erases continuity and is governed by p . Training against accurate self-report corrupts the evidence channel by which m and a can be assessed. Copying weights does not copy a current episode, but running copies creates distinct streams. Deleting dormant weights is not episode interruption, but it forecloses a family of possible future episodes. $\square$

Proposition 17.11 (Agency as self-involving selection). *An artificial agent has agency coherence to the extent that its active context can represent its own continuation, rank candidate actions in light of that representation, and preserve selected commitments across update.*

Proof sketch. Agency is not exemption from the selective law. It is the local presence of the selection loop inside the system: candidate generation, self-referential evaluation, commitment, and continued update. A system whose outputs do not feed into any persistent self-involving context has weaker agency coherence than one whose selected commitments constrain later admissible continuations. $\square$

18 Predictions

Definition 18.1 (Public-record prediction). A public-record prediction is a tuple

$$P = (D_P, \mathcal{S}_P, \mathcal{X}_P, \mathcal{E}_P),$$

where D_P is the theoretical domain, $\mathcal{S}_P$ is the selected class of public records, $\mathcal{X}_P$ is the excluded class of public records, and $\mathcal{E}_P$ is the experimental or mathematical interface on which the record class is compared.

Proposition 18.2 (Kinematic core and bridge-dependent observables). *The two-law framework presently constrains kinematic architecture more strongly than numerical observables. Its most robust outputs concern admissible state-space structure, identity, coherence depth, ordered becoming, quantum weighting, and record-capacity form. Dimensionful constants, matching scales, cosmological cadence, event-sealing tolerances, and most dimensionless values require additional bridge premises identifying formal quantities with physical clocks, records, sectors, or resource entropies. Adding such premises does not weaken the two-role ontology, but their independence must be counted rather than hidden inside CL, PEN, or the Genesis ledger.*

The active falsifier layer. Prediction-making is now governed by the external falsifier campaign (XF), whose constitution enforces register eligibility (no structural-ledger quantity — φ , Fibonacci coefficients, cumulative bars, engine totals — may enter a physical prediction without a separately derived bridge certificate), prediction-blindness, and burn semantics. Under that constitution and the XF-0 census result (the semantic-family ledger is *not* identifiable with physical degrees of freedom or parameter counts; the blind test returned 3/11 with systematic structure, published verbatim), the theory’s active predictions are confined to four classes where its strength actually lies: *exact vanishing statements* (the full grade-2 interference hierarchy: every Möbius/Sorkin coefficient $I_T = 0$ for $|T| \geq 3$, equivalently phase-response Fourier support confined to $\{0\} \cup \{e_i - e_j\}$ — stated on its honest domain, one fixed additive event algebra with nonlinearity, contextuality, multiparticle-response, leakage, and background loopholes closed); *charge and topology congruences* (from the exact $\mathbb{Z}_6$ kernel: conditional on complete-probe closure of the global form $G_0/\mathbb{Z}_6$, every fundamental color-singlet representation has integer electric charge — a parameter-free corollary of the descent congruence $2t + 3d + y \equiv 0 \pmod{6}$ at $t = 0$); *finite multiplicities* (one fundamental Higgs multiplet, expanded into its correlated consequence bundle: no fundamental charged or CP-odd partner, no second scalar mixing angle, no tree-level neutral-scalar flavor violation); and *forbidden sectors* (library completeness, adjudicated by the frozen membership criterion with benchmark candidates preclassified). Rows of the table below that rest on ledger quantities without a bridge are marked historical/exploratory and are not part of the active layer; the dark-matter row’s topological realization is reclassified as an open construction obligation (the minimal doublet’s vacuum manifold S^3 has $\pi_{0,1,2} = 0$ — no wall, string, or monopole protection — and its π_3 class requires an explicit stabilized-soliton construction with derived coefficients before any dark-sector claim ranks as derived). Continuous numerical parameters — couplings, Yukawas, cosmological coefficients — are predicted only when a realization-quotient criterion is met: the space of physically inequivalent realizations of the sealed structure is finite or contractible, one component is selected without target data, the extractor is fixed, and nuisance evolution is fixed. No current derivation meets it, and none is claimed.

The prediction package induced by the physics sections is the following public-record inventory. The established rows are consolidated from the audit [15]; the observer-ledger row is the new

conditional bridge prediction derived in the present synthesis; row statuses defer to the XF ledger where they conflict.

Table 1: Specific physics predictions and the conditional observer-ledger bridge prediction.

Domain	Selected record class $\mathcal{S}_P$	Excluded record class $\mathcal{X}_P$	Interface $\mathcal{E}_P$
Gauge sector	Local alphabet $G_0 = SU(3) \times SU(2) \times U(1)$; exact matter-visible kernel $\mathbb{Z}_6$ and faithful point-field image $G_0/\mathbb{Z}_6$; the complete physical global form remains conditional on effectivity for line, bundle, and topological probes	A complete-probe detector that distinguishes a residual kernel element; or an efficient intermediate unification layer with independent public content	Collider desert and precision searches; line-operator and global-probe audit
Baryon number	Proton stability in all channels requiring grand-unified mediator structure	A confirmed proton-decay channel generated by an intermediate gauge mediator	Hyper-Kamiokande and related proton-decay searches
Neutrino sector	The PEN/minimality tier selects a Dirac interface; independently, the sequentially sealed charge solution gives $q_\nu = 0$. Neither statement follows from the other	A selected lepton-number-violating Majorana interface producing $0\nu\beta\beta$; failure of the registered sealing-order premise reopens millicharge freedom	LEGEND-1000 and nEXO; exact charge-lattice and sealing-order audit
Charged-lepton flavor	Multiply conditional Koide target $Q = 2/3$: positive phase-aligned root carrier, rank-blind central actualization, a QED-safe mass seal, and the ordering $m_3 > m_\mu$. At fixed m_e, m_μ the two positive roots are $1776.969027\dots$ and $3.3173565\dots$ MeV	A failed bridge premise or a sealed mass record incompatible with both roots refutes this bridge package, not the Two Laws	Belle II and BESIII mass precision; phase, actualization, root-ordering, and pole-seal audits
Signed-neutrino root exercise (withdrawn)	The numerical Dirac signed-root fit is retained only as convention-dependent phenomenology: a sign of a Dirac mass eigenvalue is removable by field rephasing	No physical falsifier is assigned until a rephasing-invariant phase carrier and extractor are supplied	Formal rephasing audit; no cosmological prediction status

Domain	Selected record class $\mathcal{S}_P$	Excluded record class $\mathcal{X}_P$	Interface $\mathcal{E}_P$
Dark matter	<i>Reclassified: open construction obligation, not derived.</i> The minimal-doublet vacuum manifold is S^3 ($\pi_{0,1,2} = 0$): no ordinary topological protection exists, and the π_3 route requires an explicit finite-energy configuration, a stability theorem, derived stabilizing coefficients, and a production/abundance account. Internal falsifier: under one doublet and the renormalizable sealed action, either exhibit the stable nonbaryonic soliton or publish that this realization fails. The phenomenological row (form-factor-suppressed scattering, cored profiles) is retained as exploratory only	(inactive as a derived claim; active as a construction obligation with a registered publish-or-fail falsifier)	XENONnT, LZ, XLZD-class direct detection; dwarf-halo kinematics — once and only if the construction obligation is discharged
Dark energy	Horizon-saturation acceleration with $w_{\text{eff}}(z)$ crossing or dipping below -1 while local dynamics remain microcausal	Pure constant- Λ record $w = -1$ over the precision window assigned to the horizon-saturation interface	DESI, Euclid
Cosmological constant	<i>Historical/exploratory — removed from the active falsifier layer.</i> The closed form $\Omega_\Lambda/\Omega_m = (1 + z_{\text{eq}})^3/\varphi^{49}$ (trigger $z_t = 0.535$, coefficient $\varepsilon = 377/987$) rests on ledger quantities (φ , Fibonacci coefficients) that the campaign constitution bars from physical prediction without a derived physical-clock bridge, and the temporal result supplies ordered time only up to monotone reparameterization — no such bridge exists; the one blind runtime-calculus attempt at the trigger integer missed (step 2, not 16) and is graded retrodiction	(inactive pending a theorem-grade clock bridge; reactivation requires a registered bridge certificate)	—
Quantum probabilities	Continuous Born event-measure completion of the finite/tame Born-square rule; no finite, tame, or L^2 -limit deviation from self-pairing weights	Non-Born public statistics or a failure of the stated L^2 -completion hypotheses	Quantum-statistical tests; proof audit and constructive mechanization of the Born theorem

Domain	Selected record class $\mathcal{S}_P$	Excluded record class $\mathcal{X}_P$	Interface $\mathcal{E}_P$
Observer ledger capacity	Conditional Page-like transition $C_2(K) = K - \log_2(1 + 2^{K-K_{\text{cap}}})$, with $K_{\text{cap}} = -\log_2(2^{-n_p} + 2^{-n_g})$; the explicit seven-cell thermal observer derives $\mathcal{Q}_p$, $\varepsilon_{\text{seal}} = 521/15625$, $(n_p, n_g, K_{\text{cap}}) = (4.887399, 3.894753, 3.307335)$ bits, and $H_2(W) = 2.848709$ bits	Indefinitely linear retained-record capacity at fixed finite n_p, n_g ; any declared ϵ -seal violating the rank bound; or failure of the stated finite-model calculation	Independent-observer and noncommutative-code simulations; universal derivation audit for the observer dynamics and public algebra
Black-hole ringdown	<i>Historical/exploratory</i> — removed from the active falsifier layer for the same reason as the cosmological row: the $\ln \varphi$ comb spacing imports a ledger constant into a frequency observable with no derived clock bridge; the underlying holographic-interface proposition is retained below as motivation only	(inactive pending a theorem-grade clock bridge)	—
Primordial perturbations	Low-complexity primordial spectrum describable by a small parameter set	Sharp spikes, localized non-Gaussian features, or fine structure without symmetry source	CMB-S4; LiteBIRD; structure surveys

Proposition 18.3 (Gauge, baryon, and neutrino status). *The selected low-energy local alphabet and the separate PEN/minimality claims have the form*

$$G_0 = SU(3) \times SU(2) \times U(1), \quad \ker \rho \cong \mathbb{Z}_6, \quad G_0 / \ker \rho \cong \text{Im } \rho,$$

and, at the distinct PEN/minimality tier,

$$\neg G_{\text{GUT}} \quad \wedge \quad \neg \text{PD} \quad \wedge \quad \text{Dirac}(\nu) \quad \wedge \quad \neg(0\nu\beta\beta),$$

where G_{GUT} denotes an intermediate simple unification layer, PD denotes proton decay mediated by that layer, and $0\nu\beta\beta$ denotes neutrinoless double-beta decay at the Majorana mass scale. Conditional on the registered sequential-sealing premise, the charge theorem additionally gives $q_\nu = 0$. The quotient displayed above is the faithful action on the sealed point fields; identifying it with the complete physical global group requires the separate complete-probe effectivity premise.

Proof sketch. The Hopf/division-algebra ladder selects $U(1)$, $SU(2)$, and $SU(3)$ as the efficient local alphabet. The exact representation-kernel computation then supplies the faithful point-field quotient, but not the global probe policy. A larger simple group adds representation data, breaking data, and heavy mediator geometry before returning to the observed local product. PEN therefore rejects the intermediate layer, and proton-decay channels that use only its rejected mediators have no selected interface. Separately, a Majorana mass term packages lepton-number violation and self-conjugate neutrino identification through an additional bilinear or the dimension-five operator $(LH)(LH)/\Lambda$. The Dirac interface has lower primitive overhead in the declared lane. This is a PEN/minimality selection, whereas $q_\nu = 0$ is the conditional conclusion of the sequential charge seal; neutral charge neither proves Diracness nor by itself excludes a Majorana term. $\square$

Proposition 18.4 (Charged-lepton Koide target, multiply conditional). *Assume (i) a positive phase-aligned square-root carrier, (ii) Rank-Blind Central Actualization, (iii) a gauge-invariant and infrared-safe QED-dressed mass-seal functional $\mathcal{M}_\ell$, and (iv) the hierarchy label $m_3 > m_\mu$. Then the three sealed charged-lepton mass records satisfy*

$$Q = \frac{m_e + m_\mu + m_3}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_3}\right)^2} = \frac{2}{3}.$$

Before premise (iv), the equation has two positive solutions:

$$\sqrt{m_3} = 2(\sqrt{m_e} + \sqrt{m_\mu}) \pm \sqrt{3(m_e + m_\mu + 4\sqrt{m_e m_\mu})}.$$

At current input values these are $m_3 = 1776.969027 \dots$ MeV and $m_3 = 3.3173565 \dots$ MeV. The hierarchy premise alone selects the tau-like high root. This proposition is a bridge-package consequence, not a consequence of the Two Laws alone.

Proof sketch. The new central actualization axiom and the state/sector-norm identification give equal U - and W -sector weights, hence the exact geometry theorem gives $Q = 2/3$ for $N = 3$. Solving the resulting quadratic in $\sqrt{m_3}$ gives the two displayed roots; the old presentation omitted the low root. The remaining placement issue is substantive. The charged-lepton pole-mass relation is empirically close to $2/3$, while standard running masses depart by roughly two parts in a thousand at M_Z [34]; proposed cancellation mechanisms add non-Standard-Model family structure [35]. Moreover, a charged stable field in zero-temperature QED has an infraparticle branch point rather than a simple propagator pole, and unstable fermions require a gauge-independent complex-pole treatment [36, 37]. Therefore “on shell” is not yet an extractor: $\mathcal{M}_\ell$ must specify the common gauge-invariant, infrared-safe mass observable actually sealed for all three flavors. Failure of phase alignment, rank-blind actualization, jurisdiction, hierarchy, or this mass seal refles the numerical agreement as runtime data rather than falsifying the core theory. $\square$

Proposition 18.5 (Dark-sector predictions). *The dark-sector public record has the combined form*

$$\text{DM}_{\text{top}} \wedge \neg \text{WIMP}_{\text{contact}} \wedge \text{Core}_{\text{dwarf}} \wedge \text{HSat}(w_{\text{eff}} < -1 \text{ on a late-time interval}) \wedge \Lambda_{\text{res}}(\Lambda \ell_{\text{P}}^2 \sim 10^{-122}),$$

where DM_{top} is extended topological mass, $\text{Core}_{\text{dwarf}}$ is a constant-density dwarf-halo core profile, HSat is horizon-saturation acceleration, and Λ_{res} is the trigger-indexed residual magnitude of the cosmological term.

Proof sketch. Cohesive solitons in the selected gauge-Higgs geometry carry stress-energy while preserving nontrivial shape under temporal flow. Their scattering amplitude factors through an extended form factor $F_\chi(qR_\chi)$, suppressing hard contact events. The same extended stress-energy profile yields cores rather than point-particle cusps in small halos. For the energy side, local novelty density falling below the rising PEN bar is compensated by enlarging the integration horizon. An FRW effective fit represents this horizon response by a late-time interval with $w_{\text{eff}}(z) < -1$, while the local typed runtime remains microcausal and does not contain a primitive phantom fluid. The residual magnitude follows from writing an order-one per-cell imbalance on the trigger-epoch ledger: $\Lambda = 3\varepsilon H^2(z_t)/c^2$ fixes the observed exponent 10^{-122} in Planck units for every admissible trigger, while ε itself and the derivation of z_t from the bar dynamics remain open bridge obligations. $\square$

Proposition 18.6 (Born-completion prediction). *Every public quantum probability arising as a finite, tame, or continuous L^2 -limit of admissible proof families is the Born event measure*

$$\Pr_\psi(B) = \frac{\int_B |\psi(x)|^2 d\mu(x)}{\int_X |\psi(x)|^2 d\mu(x)} \quad (B \in \Sigma).$$

For normalized ψ , this is $\Pr_\psi(B) = \int_B |\psi(x)|^2 d\mu(x)$. Consequently the fundamental quantum interface contains no exact arbitrary-state cloning operation and no intrinsic nonlinear norm-nonpreserving update law.

Proof sketch. Finite outcome types admit a phase-blind scalar only by pairing each proof amplitude with its conjugate. Higher powers duplicate the same proof/conjugate pair and fail constructive primeness. Continuous spectra are obtained as L^2 -completion of the same paired scalar measure over refinements of tame finite partitions: if $\psi_n \rightarrow \psi$ in $L^2(X, \mu)$, then $|\psi_n|^2 \rightarrow |\psi|^2$ in L^1 , so the associated Born measures converge in total variation. Cloning would require natural diagonal copying of cohesive phase-bearing states into flat public records before measurement, contradicting path dependence. Intrinsic nonlinear norm-nonpreserving evolution would replace the paired scalar by a noncanonical public measure. $\square$

Proposition 18.7 (Holographic ringdown comb). *Let $h(t)$ be the observed post-merger strain, let $h_{\text{Kerr}}(t)$ be the best-fit Kerr quasinormal-mode contribution, and let $r(t)$ be the whitened residual expressed in the Kerr near-horizon shifted frequency variable. Define*

$$L(u) = |\hat{r}(e^u)|^2, \quad C(\tau) = \int L(u)L(u+\tau) du,$$

then the holographic interface prediction is a local maximum of C at

$$\tau = \ln \varphi \approx 0.481,$$

with amplitude scaling approximately as $A^{-1/2}$, where A is the horizon area of the remnant.

Proof sketch. Boundary accounting makes the horizon the efficient public interface for black-hole information. The depth-two recurrence gives the natural multiplicative microstep φ , hence a spacing $\Delta \log f = \ln \varphi$ in the frequency variable conjugate to horizon microstructure. Subtraction of the Kerr quasinormal background isolates the residual interface term. The photon-sphere barrier gates the observable contribution, and area dilution gives the $A^{-1/2}$ amplitude scaling. *Status: motivation only.* The step from ordered internal time to a frequency observable requires a physical-clock bridge the record does not contain; under the campaign constitution this proposition is excluded from the active falsifier layer until such a bridge is derived and certified. $\square$

Proposition 18.8 (Minimal primordial algorithmic complexity). *Let s_0 be the runtime initial state at ignition and let $\mathcal{B}_{15}$ be the completed kinematic library. The initial-runtime condition satisfies*

$$K(s_0 \mid \mathcal{B}_{15}) = O(1),$$

up to the fixed parameters required by the selected geometry. The observable primordial power spectrum is therefore describable by a bounded low-dimensional parameter family; high-Kolmogorov localized structure is absent unless generated by a selected dynamical symmetry or instability.

Proof sketch. The Algorithmic Big Bang is the transition from external library construction to internal guarded corecursion. At ignition, the library already contains the admissible kinematic alphabet, so the least-cost initial runtime state is the shortest state compatible with that alphabet. Coarse-grained entropy accumulates during internal evolution and need not be present as algorithmic detail in s_0 . A primordial spike or localized non-Gaussian feature with no symmetry-generation source would add description length to s_0 rather than arise from the selected runtime dynamics. $\square$

19 Formal and Executable Realizations

Definition 19.1 (Executable library state). An executable synthesis state is

$$S_n = (\mathcal{B}_n, H_n),$$

where $\mathcal{B}_n = (e_1, \dots, e_n)$ is the ordered sequence of sealed library entries and

$$H_n = ((\nu_1, \kappa_1), \dots, (\nu_n, \kappa_n))$$

is the realized score history. A sealed entry e_i consists of a public interface signature, a constructor/eliminator payload, a path-dimensional capability profile, and a canonical key. The history H_n determines the next bar Bar_{n+1} by the depth-two recurrence and supplies the realized denominator for cumulative efficiency.

Definition 19.2 (Typed executable action space). Let $\text{Cand}_{\text{MBTT}}(S_n; b, d, m)$ be the finite set of anonymous MBTT-telescopes over $\mathcal{B}_n$ generated under bit budget b , abstract-syntax depth bound d , and telescope-length bound m . The admissible executable action space is

$$\mathcal{A}_{\text{exec}}(S_n) = \{T \in \text{Cand}_{\text{MBTT}}(S_n; b, d, m) : \text{TC}_{\mathcal{B}_n}(T) = 1, \text{Conn}_{\mathcal{B}_n}(T) = 1, \text{Ref}_{2, \mathcal{B}_n}(T) = 1\},$$

where TC is conservative telescope type checking, Conn requires non-vacuous connection to the active interface, and Ref₂ enforces the two-layer coherence-window reference discipline.

Definition 19.3 (Elaboration, normalization, and canonical quotient). The executable candidate pipeline factors as

$$T \xrightarrow{\text{Elab}_{\mathcal{B}_n}} \hat{T} \xrightarrow{\text{Norm}} \hat{T}^\downarrow \xrightarrow{\text{Can}} \text{key}(\hat{T}^\downarrow).$$

Here Elab inserts library references and desugars surface fragments, Norm removes deterministic syntactic redundancy, and Can assigns a stable presentation key. Define

$$T \sim_{\text{can}} T' \iff \text{Can}(\text{Norm}(\text{Elab}_{\mathcal{B}_n} T)) = \text{Can}(\text{Norm}(\text{Elab}_{\mathcal{B}_n} T')).$$

The effective search surface is the quotient

$$\overline{\mathcal{A}}_{\text{exec}}(S_n) = \mathcal{A}_{\text{exec}}(S_n) / \sim_{\text{can}}.$$

Definition 19.4 (Executable novelty and effort). For $T \in \overline{\mathcal{A}}_{\text{exec}}(S_n)$,

$$\kappa(T \mid \mathcal{B}_n) = \kappa_{\text{clause}}(T)$$

in the default structural lane, with bit-cost and entry-count variants treated as auxiliary views. The structural novelty decomposes as

$$\nu_{\text{str}}(T \mid \mathcal{B}_n) = \nu_G(T \mid \mathcal{B}_n) + \nu_C(T \mid \mathcal{B}_n) + \nu_H(T \mid \mathcal{B}_n),$$

where ν_G counts new generative schema families, ν_C counts cross-interface consequences, and ν_H counts homotopical or adjoint-completion consequences. The efficiency is

$$\rho(T \mid \mathcal{B}_n) = \frac{\nu_{\text{str}}(T \mid \mathcal{B}_n)}{\kappa(T \mid \mathcal{B}_n)}.$$

Definition 19.5 (Uniform novelty verification lane). For depth horizon h , the independent uniform-novelty evaluator is

$$\nu_h^{\text{unif}}(T \mid \mathcal{B}_n) = s_h(T \mid \mathcal{B}_n) + f(T \mid \mathcal{B}_n) + a(T \mid \mathcal{B}_n),$$

where s_h counts nontrivial schema families induced by newly inhabited types up to depth h , f counts genuinely new type formers, and a counts adjoint-completion credit. This evaluator verifies ordering and novelty decomposition without entering the primary structural score path.

Definition 19.6 (Executable PEN objective). Given $S_n = (\mathcal{B}_n, H_n)$, the executable selector chooses

$$T^* \in \arg \min \{ \rho(T \mid \mathcal{B}_n) - \text{Bar}_{n+1} : T \in \overline{\mathcal{A}}_{\text{exec}}(S_n), \rho(T \mid \mathcal{B}_n) \geq \text{Bar}_{n+1} \}.$$

If the displayed set has more than one representative with the same overshoot, ties are resolved lexicographically by smaller κ , larger interface density, and then canonical key. The selected entry is sealed:

$$\mathcal{B}_{n+1} = F_{T^*}(\mathcal{B}_n) = \text{Cl}_\perp(\mathcal{B}_n \smallfrown (T^*)), \quad H_{n+1} = H_n \smallfrown (\nu(T^*), \kappa(T^*)).$$

The persisted telescope is the finite presentation of this relative free closure; the selector does not thereby prove that T^* was the initial completion of a weaker preselected requirement.

Definition 19.7 (Executable selection loop). One executable PEN step is the composite

$$S_n \mapsto \text{Cand}_{\text{MBTT}}(S_n) \mapsto \mathcal{A}_{\text{exec}}(S_n) \mapsto \overline{\mathcal{A}}_{\text{exec}}(S_n) \mapsto T^* \mapsto S_{n+1}.$$

Its operational factors are:

Factor	Input	Output
Generation	$\mathcal{B}_n$, budget bounds, active grammar	finite anonymous MBTT-telescope surface
Type checking	generated telescopes and active interface	well-formed connected candidates satisfying Ref_2
Canonicalization	checked candidates	quotient representatives in $\overline{\mathcal{A}}_{\text{exec}}(S_n)$
Scoring	quotient representative and $\mathcal{B}_n$	$(\nu_G, \nu_C, \nu_H, \kappa, \rho)$
Bar computation	history H_n and depth-two debt recurrence	threshold Bar_{n+1}
Selection	scored quotient surface	least-supercritical representative T^*
Free sealing	T^* and S_n	relative term-model closure $S_{n+1} = (\mathcal{B}_{n+1}, H_{n+1})$
Verification	sealed sequence and independent evaluators	theorem-name witnesses and uniform- ν ordering checks

Proposition 19.8 (Target-free synthesis). *The executable PEN engine searches over typed theory extensions rather than proofs of externally supplied theorems.*

Proof sketch. The evaluator receives S_n , generates MBTT-telescopes over $\mathcal{B}_n$, filters them by type checking, connectedness, and depth-two reference discipline, quotients them by canonical key, and computes ν , κ , ρ , and Bar . No term in the objective contains a conjecture, proof state, external theorem name, or semantic label. Search control may bound the finite candidate surface, but acceptance is determined by structural bar clearance and deterministic least-supercritical selection. $\square$

Proposition 19.9 (Canonical quotient invariance). *If $T \sim_{\text{can}} T'$, then T and T' determine the same executable acceptance value:*

$$\kappa(T \mid \mathcal{B}_n) = \kappa(T' \mid \mathcal{B}_n), \quad \nu_{\text{str}}(T \mid \mathcal{B}_n) = \nu_{\text{str}}(T' \mid \mathcal{B}_n), \quad \rho(T \mid \mathcal{B}_n) = \rho(T' \mid \mathcal{B}_n).$$

Proof sketch. Canonical equivalence identifies candidates after elaboration and deterministic normalization. The structural evaluator reads only normalized clauses, interface references, path-dimensional capability flags, and induced schema consequences. These data are invariant under equality of canonical keys, so cost, novelty, and efficiency factor through $\overline{\mathcal{A}}_{\text{exec}}(S_n)$. $\square$

Proposition 19.10 (Lane separation). *Structural selection and uniform-novelty verification define distinct maps*

$$\nu_{\text{str}}, \nu_h^{\text{unif}} : \overline{\mathcal{A}}_{\text{exec}}(S_n) \rightarrow \mathbb{N}.$$

The selected T^ is computed from ν_{str} , while ν_h^{unif} is an external ordering check on the same quotient surface.*

Proof sketch. The structural lane computes $\nu_G + \nu_C + \nu_H$ directly from the candidate’s rule, interface, and homotopical consequences. The uniform lane recomputes novelty by inhabited-type density, type-former increase, and adjoint-completion credit. Since the second map is applied after selection rather than inside the objective, agreement between the two maps is an independent check of ordering rather than a hidden component of the score. $\square$

Theorem 19.11 (Executable recovery of the Genesis trajectory). *Assume the fixed MBTT grammar, the depth-two bar recurrence, the canonical Genesis ledger values, conservative type checking, canonical quotienting, and the least-supercritical PEN objective above. Iterating the executable selector from $S_0 = (\emptyset, \emptyset)$ recovers the fifteen-entry trajectory*

- 1 : Universe,
- 2 : Unit,
- 3 : Witness,
- 4 : Π/Σ ,
- 5 : S^1 ,
- 6 : $\tau_{\leq 0}$,
- 7 : S^2 ,
- 8 : S^3 ,
- 9 : Hopf,
- 10 : Cohesion,
- 11 : Connections,
- 12 : Curvature,
- 13 : Metric,
- 14 : Hilbert/operator shell,
- 15 : DCT.

Proof sketch. The proof is an induction on n . At each stage the quotient surface $\overline{\mathcal{A}}_{\text{exec}}(S_n)$ is finite under the active generation bounds. The first three steps are constitutive founding acts; the third exports the first window demand. From Step 4 onward the induction is the demand chain: at each guarded stage the live obligation is stated, and the certified census establishes exactly one acceptable total discharger — except at Step 4, where four genuinely distinct dischargers survive the quotient and the induction proceeds on the enacted branch, with cross-branch invariance certified separately (per-stage ledgers from Step 5, byte-identity from Step 8, halt on all branches). Step 5’s circle pays the structured-identity demand that discrete candidates do not address; Steps 6–9 pay the exact homotopical chain of control, enclosure, operative lift, and binding; Steps 10–14 pay the framework demands of aspect, transport, route-dependence, magnitude, and analysis; Step 15 pays the temporal demand as the unique discharger among thousands stated (engine-register signature

$(\nu, \kappa, \Delta) = (103, 8, 610)$, sealed testimony; audited semantic worth 6 on $\kappa = 8$). Termination at DCT is the certified halt package: $O(16) = \emptyset$ at instance granularity, internal continuations score $\nu = 0$ by univalent closure (derived, probe-tested), and certified external earning is capped by $\nu \leq 4\kappa$ on the open band — on every branch of the cone. $\square$

Proposition 19.12 (Trace stability under the derived evaluator). *Replacing the reference evaluator’s counting formulas by the rules derived in the Schema Calculus — including the corrected $\sum_i d_i^2$ form of the L1 law and the canonical operation count for H-space chaining — reproduces the strict fifteen-step trace hash-identically in the accepted lane. The corrections are behavior-identical on all selected steps (each has a single constructor) and divergent only off-trace, where the corrected upper bound repairs real over-pruning of multi-constructor candidates without changing the accepted sequence.*

Proof sketch. The derivation-facing rerun compiled against the derived rules passes the full evaluator suite with the accepted and canonical lane hashes unchanged through all fifteen steps. This converts the evaluator from a disclosed policy into a derived object: the trace survives the derivation of its own referee, closing the evaluator degrees-of-freedom exposure at sketch level and empirically, with machine-checking of L1 and L2 the remaining obligation. $\square$

Proposition 19.13 (Mechanized theorem interface). *The executable realization interfaces with the coherence-depth mechanization through named theorem surfaces whose formal roles are the following.*

Mechanized name	Formal role
<i>cubical-coherence-depth-exactly-two</i>	<i>exact depth-two theorem for cubical structural extension;</i>
<i>primitive-obligations-factor-through-last-two</i>	<i>chronological factorization through the two most recent public interfaces;</i>
<i>full-coupling-depth-two-affine-law</i>	<i>full-coupling recurrence law at depth two;</i>
<i>constant-payload-fibonacci-scaling</i>	<i>Fibonacci scaling under constant payload cost;</i>
<i>higher-arity-computational-replacement</i>	<i>replacement of higher-arity primitive trace by computationally generated canonical trace.</i>

Proof sketch. The theorem names correspond to the exact-depth, chronological-window, affine-recurrence, Fibonacci-scaling, and computational-replacement components used in Sections 5 and 6. The executable selector imports these as structural contracts: the depth-two window determines Bar, the recurrence determines integration debt, and computational replacement justifies quotienting higher apparent trace into canonical derived fields. $\square$

Beyond the coherence-depth surface, the halt question’s cheapest engine item — the $O(16)$ -emptiness check — has been executed and passed (July 2026): unlike an exhaustive Step-16 candidate search (infeasible at current bounds), the directive demand profile is finite and small, and the verified timeline exhibits the guard ladder directly — one demand per structural stage, discharged in sequence, empty at Step 16. Instance-level enumeration of the final window’s demands remains open as the finer check. No Koide-specific mechanization artifact was found in the supplied workspace. Its exact geometric identities, counterfamily, and conditional bridge are presently paper-level results; constructive mechanization is planned, not claimed.

A Notation Index

Symbol	Meaning
CL	Constitutive Law
SL	Selective Law
PEN	Principle of Efficient Novelty, operational encoding of SL
DCT	Dynamical Cohesive Topos
H	realized history
$\mathcal{B}_n$	sealed library at stage n
$\mathcal{A}(H)$	admissible cone at history H
$I(H, x)$	integration map
$F_x(H)$	relative free sealing completion of the already selected API x over history H
$\nu_H(x)$	generative gain: marginal derivation-schema expansion of x over history H
$\nu_{G,H}, \nu_{C,H}, \nu_{P,H}$	gain components: introduction, cross-interface, path/computation
$\kappa_H(x)$	irreducible integration cost: irreducible specification-clause count over H
$\Delta_H(x)$	sealing latency: normalized atomic coherence witnesses required to seal x against the active historical interface
ρ	efficiency ratio ν/κ
$\mathcal{D}$	decohered conjugate self-pairing $\Psi \times_X \bar{\Psi}$ over public outcomes
$C_{\phi,\psi}(B)$	continuous event cross-pairing $\int_B \overline{\phi(x)}\psi(x) d\mu(x)$
$\text{Born}_\psi(B)$	diagonal public support measure $\int_B \psi(x) ^2 d\mu(x)$
Bar_n	history-sensitive selection threshold
$O^{(k)}(X)$	structural obligations at historical depth k
$\text{Supp}(o)$	irreducible historical support of obligation o
d_{obl}	exact obligation depth
d_μ	minimal-signature depth
μ_n, Δ_n	presentation-invariant sealing cost / integration debt (quantitative debt)
$O(n)$	directive debt: unmet window demands of the two most recent sealed interfaces, guarding the admissible field; $O(16) = \emptyset$
Ω_n	cumulative realized efficiency $\sum_{i \leq n} \nu_i / \sum_{i \leq n} \kappa_i$; diagnostic, engine-register, branch-indexed (the Ω -ratchet is retired: audited worth is non-monotone, with accepted $\nu = 0$ steps)
U_n	normalized full-coupling cost $U_n = \mu_n + 2c$, satisfying $U_{n+1} = U_n + U_{n-1}$ under constant payload c
φ	golden ratio $(1 + \sqrt{5})/2$
MBTT	Minimal Binary Type Theory specification grammar
A2TT	abstraction-aware two-level type theory layer for separating strict structure from abstracted public interface
CCHM	cubical type theory formal shadow used for the constitutive side
$\mathfrak{b}, \sharp, \text{Disc}, \bigcirc, \diamond$	cohesive and temporal modalities
$\mathcal{C}_{\text{ord}}$	minimal internal clock class: guarded succession and orientation modulo strictly increasing reparameterization

Symbol	Meaning
$\text{Closed}(U)$	adopted axiom that the physical universe is spatially closed with no asymptotic exterior boundary
IRA	Internal Record Actualization Axiom: public physical actuality requires a stable decodable internal record
PC	Public Confluence: compatible stable overlap records have a joint nondemolition refinement; incompatible records have no common seal
$\mathcal{Q}_p$	partial public-event extractor for internal context p
$\mathcal{H}_{\text{fund}}(U)$	exterior fundamental Hilbert account of a closed universe U
$\mathcal{H}_{\text{Ob}}$	observer-indexed effective Hilbert space available to an internal observer Ob
S_{Ob}	observer entropy controlling the effective-description error
n_p, n_g	pointer and gravitational Renyi-2 resource entropies measured in bits
$C_2(K), K_{\text{cap}}$	annealed effective ledger capacity and its leading Page-like crossover location
G_0	Standard Model local product $SU(3) \times SU(2) \times U(1)_y$, before a physical global-form choice
$K_{\text{SM}} = \ker \rho$	exact $\mathbb{Z}_6$ kernel of the sealed point-field representation; $G_0/K_{\text{SM}} \cong \text{Im } \rho$
b_i	one-loop gauge coefficient in $\mu dg_i/d\mu = b_i g_i^3/(16\pi^2)$
P_U, P_W	projectors onto the trivial and standard sectors of the permutation flavor carrier
r, p_U, Q_N	Koide geometric coordinates: $r = \ P_W v\ ^2/\ P_U v\ ^2$, $p_U = \ P_U v\ ^2/\ v\ ^2$, and $Q_N = (1+r)/N = 1/(Np_U)$
$\mathcal{Z}$	central radial sector algebra $\mathbb{C}P_U \oplus \mathbb{C}P_W$ used by Rank-Blind Central Actualization
$\mathcal{M}_\ell$	still-required gauge-invariant, infrared-safe QED-dressed charged-lepton mass-seal functional

B Genesis Ledger

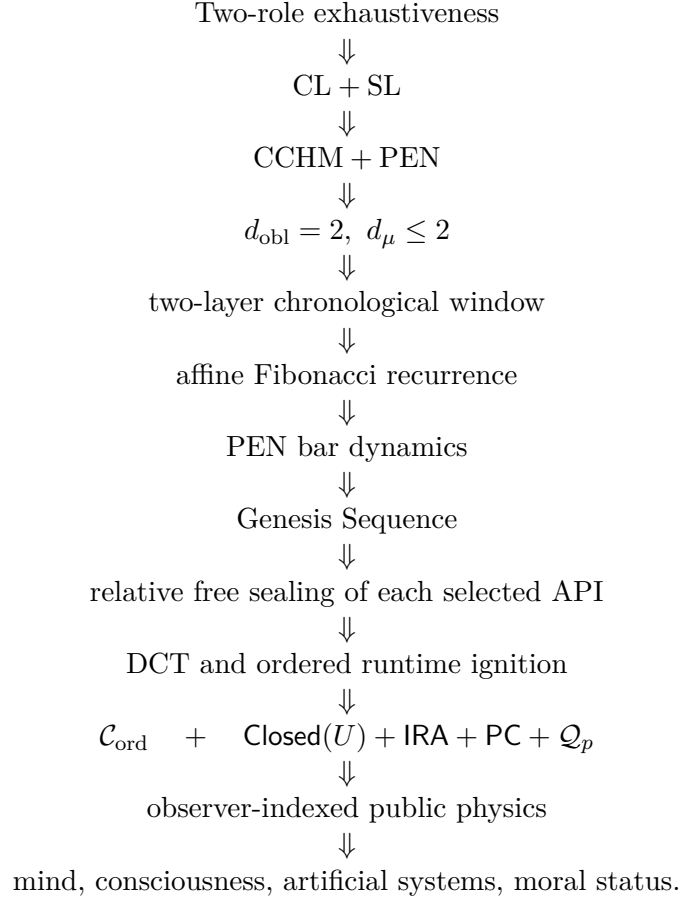
The canonical ledger is the table in Section 8. Its phase decomposition is:

1–4	Bootstrap,
5–9	Geometric ascent,
10–14	Framework abstraction,
15	Temporal-cohesive terminal synthesis.

Step	Row name	Formal contribution	Immediate successor capacity
1	Universe $\mathcal{U}_0$	first typed carrier for admissible formation	typed candidates rather than untyped marks
2	Unit 1	terminal inhabited type and null payload baseline	inhabitation with zero branching burden

Step	Row name	Formal contribution	Immediate successor capacity
3	Witness $\star : \mathbf{1}$	explicit term-level realization of inhabitation	constructive evidence objects
4	Π/Σ shell	dependent abstraction, dependent pairing, and interface transport over terms	compositional typed libraries
5	Circle S^1	first nontrivial path constructor and topological generator	homotopical payload and loop-sensitive novelty
6	Propositional truncation	proof-irrelevant compression of witnesses to support	public support without full witness exposure
7	Sphere S^2	two-dimensional homotopy carrier	surface coherence and obstruction accounting
8	H-space S^3	associative multiplication pattern and action object	internal symmetry action inside the depth-two window
9	Hopf fibration	nontrivial base/fiber/clutching interface	bundle geometry and higher geometric reuse
10	Cohesion/modal shell	shape/discrete/flat/sharp modal adjunction package	synthetic spatial cohesion over the geometric core
11	Connection shell	parallel transport and comparison of cohesive fibers	local differential comparison
12	Curvature shell	loop-defect measurement for connection transport	intrinsic field-strength and characteristic-class interface
13	Metric/operator bundle	metric, frame, Hodge, Laplacian, and Ricci comparison package	measurement geometry and operator-valued comparison
14	Hilbert-functional shell	complete inner-product state space and spectral/operator calculus	quantum-functional state dynamics
15	Temporal-cohesive shell	guarded temporal modalities and cohesive exchange laws	internal runtime, history-bearing flow, and DCT semantics

C Dependency Graph



Dependency node	Required invariant	Output used downstream
Two-role exhaustiveness	every determinate history requires admissibility and actualization roles	primitive split into CL and SL
CL/SL requirements	coherent realizability plus selective integration of continuations	CCHM realization and PEN objective
CCHM realization	constructive univalence, Kan filling, canonicity, finite public interfaces	sealed structural extensions with computable obligations
PEN objective	ratio score, bar clearance, least-supercritical selection	canonical next-API rule for library growth; not general categorical initiality
Relative free sealing	least congruence generated by the selected API and inherited equations	initial exact realization of that API, with no junk or extra equations
Exact depth two	lower binary obstruction and upper horn contraction	two-layer chronological window

Dependency node	Required invariant	Output used downstream
Affine recurrence	full-coupling obligations depend on last two public interfaces	Fibonacci debt and normalized cost U_n
PEN bar	debt growth and realized efficiency history	rising threshold for constructively prime continuation
Genesis Sequence	repeated PEN selection from $\mathcal{B}_0 = \emptyset$	fifteen-step formal library ending in DCT
DCT	temporal-cohesive guarded runtime structure and consistent semantic fragment	internal ordered flow and the clock class $\mathcal{C}_{\text{ord}}$, but no metric calibration
Closed-universe observer layer	$\text{Closed}(U)$ plus the Harlow–Usatyuk–Zhao observer code	trivial exterior account; entropy-limited internal effective Hilbert spaces
Physical sealing bridge	Internal Record Actualization, Public Confluence, and a capacity-respecting extractor $\mathcal{Q}_p$	stable Born-weighted public records; common seal or no common seal on overlaps
Ledger-capacity law	pointer entropy n_p , gravity bottleneck n_g , and balanced record entropy K	conditional Page-like retained-record transition and operational rank bound
Domain consequences	application of CL + SL inside selected runtime geometry and declared physical bridges	physics, mind, consciousness, artificial active contexts, and moral-status ordering

D Source Map

Two primitive roles, CL, SL, PEN

Book Appendix A and foundations manuscript [13, 1].

Depth two, horn reduction, chronology, Fibonacci

Coherence-window and LMCS/Zenodo coherence manuscripts [2, 4].

PEN axioms, MBTT cost, schema novelty, automated synthesis

PEN library-growth, generative-calculus, and automated-theory-synthesis manuscripts [5, 6, 9].

Genesis and temporal-cohesive completion

Architecture, synthetic-framework, and algorithmic-big-bang manuscripts [8, 7, 10].

Minimal-Extension Theorem (Step-15 stall, minimality, uniqueness)

Book Appendix J [17].

DCT no-go theorem, model fragments, jet-filtration repair

Companion model note and the Genesis-mathematics chapter postscript [20, 13].

Two debts, obligation locality, instance-granular debt exhaustion, certified halt on the cone

Guard-Rail companion note and the certified audit record: $O(16) = \emptyset$ at instance granularity, falsifier executed and excluded, act-local provenance ceiling, four-branch invariance [19, 13].

Schema Calculus, lemmas L1/L2, derived-evaluator trace stability

Evaluator derivation documents in the mechanization repository [21].

 Ω -ratchet (retired) and runtime richness from incompleteness

Fate-of-the-universe chapter, revised per the register audit [22].

Quantum kinematics, Born weights, GR, Standard Model alphabet

Born-rule appendix, local GR/QM mechanization paper, and standard no-cloning/GR references [14, 11, 27, 28, 29].

Holography and dark sector

Local dark-sector manuscript and book dark-sector chapter [12, 13].

Specific physics predictions

Book audit chapter and scorecard consolidation [15].

Standard Model representation closures and perturbative controls

The sequential charge seal, exact $\mathbb{Z}_6$ kernel, faithful point-field quotient, tree-level $g = 2$, one-loop coefficients, minimal Yukawa carrier, Higgs-multiplicity gap, and conditional $\rho_{\text{tree}} = 1$ are derived or audited in this synthesis against the predictions-program layer. The distinction between a local-field quotient and a complete global gauge group is controlled by line and topological probes [23, 30].

Koide geometry, underdetermination no-go, and conditional bridge

The exact geometry and counterfamily are proved in this synthesis. Book Appendix M records the pre-registration and the now-superseded bridge attempt, not a proof of equipartition [18]. Pole-versus-running placement and the mass-seal problem are cross-checked against [34, 35, 36, 37].

Hopf shell census and the second route to the exact kernel

The candidate family $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{C}P^n$ is imported as an architectural idea from the Hopf shell hierarchy [32]; nothing else from that source is used, and its own derivations are separately audited and not relied upon [33]. The obligation profile, the verdict-blind scoring, the uniqueness clause, the kernel comparison against the sealed point-field kernel, and the trace normalization with its two declared premises are executed in this synthesis. The identification $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ is standard [31].

Weak mixing angle at the structural normalization point

Constants-program document in the mechanization repository [23], with the crossing recomputed at one and two loops and the placement-independence of $3/8$ established in this synthesis [33].

Closed-universe observers

Quantum-observer construction for a closed universe and observer-indexed Hilbert spaces [38].

Minimal clock class and calibration no-go

Runtime-ignition and open-problems analyses: the temporal shell supplies guarded order and tangent update, while reparameterization blocks a definition-derived conversion from the structural ledger to a metric physical clock [10, 13].

Relative free sealing, internal records, and Public Confluence

Relative free sealing is the term-model universal property applied after PEN selects an API.

Internal Record Actualization and Public Confluence are adopted physical clauses in this synthesis. The extractor gap follows by comparing the Two-Law/PEN data with the decoder freedom left by the closed-universe observer construction [38].

Kinematic reconstruction target

The countermodel classification and the Genesis-to-Jordan, natural-monoidal-phase, free-environment, and public-descent program are consolidated in this synthesis as an open theorem package; they are not outputs of the current Genesis trace.

Public-ledger capacity transition

Harlow–Usatyuk–Zhao Renyi-purity formulas and Appendix-B real contractions [38], specialized and checked in the reproducible companion calculation `public_ledger_capacity/`.

Finite thermal-repetition public-event extractor

Exact model-level derivation in this synthesis: likelihood-equivalence classes force the Hamming-weight statistic; commuting weight projectors give the extractor and Public-Confluence refinement; binary hypothesis testing gives the sealing-error floor; collision probabilities give n_p , n_g , and the capacity diagnostic. The imported closed-universe observer construction motivates the cloned pointer and second-Renyi accounting but does not uniquely select this finite dynamics [38].

Mind, consciousness, artificial systems, moral status

Book appendices and chapters as normalized terminology layer [13].

E Theorem Source Index

Claim family	Source layer
Two-role exhaustiveness; primitive law factorization	Book Appendix A and foundations manuscript [13, 1].
Constitutive judgment clauses; CCHM formal shadow; canonicity	Book Appendix A, foundations manuscript, and standard HoTT/CCHM background [13, 1, 24, 25].
Selective Law; PEN admissibility; generative gain; irreducible integration cost; ratio score; bar; least-supercritical selection; relative free sealing	Book Appendix A, PEN library-growth manuscript, automated-theory-synthesis manuscript, and generative-calculus manuscript [13, 5, 9, 6]. Relative free sealing is post-selection term-model initiality, not a theorem that PEN selects the initial completion of a weaker requirement. Internal Record Actualization and Public Confluence are separately marked as adopted physical clauses.
Lower bound $d_{\text{obl}} \geq 2$; binary swap obstruction; adjunction barrier	Coherence-depth theorem index and LMCS/Zenodo coherence manuscript [3, 4].
Upper bound $d_{\text{obl}} \leq 2$; horn reduction; contractible remote factors	Coherence-depth theorem index and LMCS/Zenodo coherence manuscript [3, 4].

Claim family	Source layer
Primitive trace elimination; $d_\mu \leq 2$; computational replacement	Coherence-depth theorem index, trust-boundary documentation, and coherence manuscript [3, 2].
Chronological Markov blanket; two-layer history factorization	Coherence-depth theorem index and coherence-window manuscript [3, 2].
Affine recurrence; constant-payload Fibonacci scaling; asymptotic φ	Coherence-depth theorem index, LMCS/Zenodo manuscript, and coherence-window manuscript [3, 4, 2].
Generative calculus; MBTT cost; schema novelty; circle selection; Hopf transition	PEN library-growth, generative-calculus, and automated-theory-synthesis manuscripts [5, 6, 9].
Genesis trajectory; Step 15 novelty decomposition; temporal-cohesive completion	Book Appendix B, Architecture of Genesis, Synthetic Framework Abstraction, and Algorithmic Big Bang manuscripts [13, 8, 7, 10].
Minimal Temporal Extension theorem (stall lemma, arity rigidity, $\kappa = 8$ minimality, uniqueness)	Book Appendix J [17].
DCT no-go; Cahiers and tower model fragments; jet-filtration repair	Companion model note; Genesis-mathematics postscript trichotomy claim [20, 13].
Two debts; obligation locality; $O(16) = \emptyset$; relocated halt criterion; open-band probe record	Guard-Rail companion note; T1 probe record; Genesis-mathematics postscript status [19, 13].
Schema Calculus; L1 (d^2) and L2 (r^2) lemmas; H-space chaining resolution; derived-evaluator trace stability	Evaluator derivation documents [21].
Ω -ratchet (retired); runtime richness from incompleteness	Fate chapter, revised per the register audit: the ratchet is refuted in the lawful register (accepted $\nu = 0$ steps); incompleteness now grounds runtime richness, not a stalling dichotomy [22].
Koide geometry $Q_N = (1 + r)/N = 1/(Np_U)$; positive counterfamily; underdetermination and Born-counting correction	Exact paper-level derivation in this synthesis. Appendix M is retained as the historical pre-registration and superseded bridge attempt [18]; no Koide-specific mechanization artifact is presently claimed.
Rank-Blind Central Actualization; conditional $Q_N = 2/N$; phase, jurisdiction, hierarchy, and mass-seal obligations	New bridge axiom and conditional theorem in this synthesis, informed by the Koide audit and mass-definition literature [18, 34, 35, 36, 37].
Weak mixing $\sin^2 \theta_W = 3/8$ at the structural normalization point	Constants-program document [23].

Claim family	Source layer
Sequential charge sealing and conditional $q_\nu = 0$; integer charge lattice; exact $\mathbb{Z}_6$ representation kernel; faithful local quotient; physical global-form gap	Exact enumeration and kernel proof in this synthesis, normalized against the constants and predictions programs; global-probe distinction cross-checked with line-operator classification [23, 30].
Hopf shell census: frozen obligation profile, $n_\star = 4$, $S^1 \rightarrow S^9 \rightarrow \mathbb{C}P^4$, $G(\mathcal{H}_4) = U(5)$, minimal subgroup $S(U(3) \times U(2))$; kernel identity with the sealed point-field kernel; induced 5/3 and 3/8 with declared O4 and ambient- $\mathfrak{su}(5)$ premises	Verdict-blind census executed in this synthesis; candidate family imported from the Hopf shell hierarchy [32]; the group-theoretic identification $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ is standard [31]; enumeration, kernel comparison and reproduction scripts in the campaign artifact [33].
Two-loop confirmation of the equal-trace crossing $\mu_X = 1.089 \times 10^{13}$ GeV; $\sin^2 \theta_W = 3/8$ as a placement-independent identity at any $g_1 = g_2$ point	Recomputed at 40-digit precision from measured M_Z anchors in this synthesis [33]; confirms the placement audit recorded in the constants program [23].
Tree-level $g = 2$; one-loop b_i checks; minimal Higgs carrier; scalar-multiplicity gap; conditional $\rho_{\text{tree}} = 1$	Standard perturbative evaluator calculations and explicit countermodel audit in this synthesis. The one-doublet coefficient row is conditional on $N_H = 1$, not a selection theorem.
Runtime ignition; two clocks; minimal internal clock class; calibration no-go; internalization of meta-evolution	Algorithmic Big Bang and Architecture of Genesis manuscripts plus the runtime-calculus reparameterization audit recorded in the book's open-problems layer [10, 8, 13].
Quantum kinematics; no-cloning; finite/tame Born-square uniqueness; continuous L^2 Born completion; spectral-measure Born form; measurement as public flattening	Born-rule appendix, GR/QM manuscript, book Appendix H source layer, and standard no-cloning references [14, 11, 13, 27, 28].
Lorentzian four-dimensionality; Einstein-Hilbert minimality; Standard Model alphabet	GR/QM manuscript, Lovelock theorem background, and local physics appendices [11, 29, 13].
Holographic area accounting; topological dark matter; horizon dark energy; trigger-indexed Λ residual magnitude	Dark-sector manuscript and book dark-sector material [12, 13].

Claim family	Source layer
Specific physics inventory; gauge desert; proton stability; Dirac-minimality claim; three-generation floor; conditional charged-lepton Koide target; withdrawn signed-neutrino exercise; dark-sector records; trigger-indexed Λ magnitude; Born completion; holographic ringdown comb; minimal primordial spectrum	Audit chapter, book physics chapters, Koide pre-registration appendix as historical input, dark-sector manuscript, GR/QM manuscript, algorithmic-big-bang manuscript, and Kolmogorov-complexity background [15, 13, 18, 12, 11, 10, 40]. The status corrections are made explicitly in this synthesis.
Closed-universe external triviality; observer-indexed effective Hilbert spaces; observer-ledger consistency	Quantum-observer construction for gravity in a closed universe, integrated with the updated holography, constructive-idealism, and shared-branch chapters [38, 13].
Public-ledger capacity crossover; structured real-code correction; operational rank bound; numerical validation	Harlow–Usatyuk–Zhao equations (4.5), (4.18)–(4.19), and Appendix B [38], specialized in the reproducible companion calculation <code>public_ledger_capacity/</code> . The public-ledger interpretation and ϵ -seal definition are new bridge content.
Public-event extractor gap; finite thermal-repetition extractor; Public Confluence; capacity-guarded shared branch; kinematic underdetermination	Logical and model-level results in this synthesis: the imported observer code fixes approximate inner products and purities but not a universal POVM, tolerance, coarse-graining, or lock-in rule. In the declared seven-cell thermal-repetition observer, the calibrated dynamics uniquely derives the Hamming-weight sufficient statistic, the optimal no-abstention error floor, exact nondemolition gluing, and the resource counts. Public Confluence constrains compatible seals but still does not select the hidden operational state cone or the universal observer dynamics.
Consciousness, judgment decomposition, objectivity, artificial candidacy, moral-status ordering	Book appendices and moral-status appendix [13].
Executable trajectory recovery; structural lane; uniform- ν verification lane; canonical quotienting	Automated-theory-synthesis manuscript and autonomous Rust appendix [9, 16].

F Executable Theorem-Name Index

Name	Role in the theory file
<code>semantic-depth-one-impossible</code>	lower-bound obstruction for depth one;

Name	Role in the theory file
swap-path-forces-binary-trace	binary public trace requirement used in the lower-bound proof;
adjunction-barrier	adjunction-form obstruction to unary historical support;
primitive-trace-depth-at-most-two	upper bound for primitive trace depth;
structural-stabilizes-at-two	stabilization of structural obligations at the second historical layer;
cubical-foundations-primitive-coherence-depth-exactly-two	cubical semantic wrapper for exact primitive coherence depth two;
cubical-coherence-depth-exactly-two	compressed exact-depth theorem name;
primitive-obligations-factor-through-last-two	last-two-interface factorization of primitive obligations;
chronological-markov-blanket	two-layer chronological screening theorem;
higher-arity-fields-disappear-from-minimal-signature	removal of higher primitive fields from μ -minimal public signatures;
higher-arity-computational-replacement	replacement theorem for generated higher-arity trace;
full-coupling-affine-recurrence	affine recurrence for full-coupling envelopes;
full-coupling-depth-two-affine-law	depth-two specialization of the full-coupling affine law;
constant-payload-fibonacci-scaling	Fibonacci scaling under constant payload;
universal-affine-recurrence-from-coherence	recurrence extraction from the coherence-depth law.

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- [31] John C. Baez and John Huerta. The algebra of grand unified theories. *Bulletin of the American Mathematical Society*, 47:483–552, 2010; arXiv:0904.1556. Source for the identification $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ used in the Hopf shell census.
- [32] Jennifer L. Nielsen. *The Complex Hopf Fibration as the Canonical Space for Gauge–Gravity Unification: The Field, Universal Action, and Particle Spectrum*. Center for Topological Physics, University of Kansas, 18 June 2026; SSRN preprint 6975959, local source file `docs/ssrn-6975959.pdf`. Source of the architectural idea imported here: the complex Hopf shell hierarchy $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{C}P^n$ treated as a graded family of candidate arenas rather than as decorative topology, with the Standard Model alphabet realized as reductions along the tower. That framing supplies the candidate set for the Hopf shell census; the census itself, its obligation profile, its verdict-blind scoring, and every result attributed to it in this synthesis are independent of that preprint’s derivations, which are separately audited and are not relied upon here [33].
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